1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

J113 MLB SCHEMATIC

10/03/14

ALIASES RESOLVED

PRODUCT SAFETY REQUIREMENTS:
PCB, UL RECOGNIZED, MIN. 130-C TEMP. RATING AND V-0 FLAME RATING PER UL 796 & UL 94.
PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP. RATING AND V-0 FLAME RATING.

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00385	1	SCHEM,MLB,J43A	SCH	CRITICAL	
820-00165	1	PCBF,MLB,J43	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED=Fri Oct 3 11:36:00 2014

DRAWING TITLE
<PART_DESCRIPTION>

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

BOM Groups

BOM GROUP	BOM OPTIONS
MLB_COMMON	ALTERNATE,COMMON,MLB_MISC,MLB_DEBUG:PVT,MLB_PROGPARTS
MLB_MISC	PP5V5_DCIN:NO,TBTHV:P15V,EDP,CAM_XTAL:NO,CAM_WAKE:NO,APCLKRQ:ISOL,TPAD_INTWAKE:SHARED,USB_PWR:S3,SD_ON_MLB,VCORE_FETS,SSD_LPSR:S3
MLB_DEVEL:ENG	ALTERNATE,BKLT:ENG,XDP_CONN,DDRVREF_DAC,S0PGOOD_ISL,DBGLED,ISNS:ENG
MLB_DEVEL:PVT	XDP_CONN
MLB_DEBUG:ENG	XDP,SAMCONN
MLB_DEBUG:PVT	BKLT:PROD,XDP,SAMCONN,ISNS:ENG,DBGLED,XDP_CONN
MLB_DEBUG:PROD	BKLT:PROD,SAMCONN,XDP,ISNS:PROD

Current Sensor Configuration

BOM GROUP	BOM OPTIONS
ISNS:ENG	CPU_HS_ISNS:YES,CPUVR_ISNS:YES,DRAM_ISNS:YES,P1V05_ISNS:YES,AIRPORT_ISNS:YES,SSD_ISNS:YES,LCDBKLT_ISNS:YES,P3V3S5_ISNS:YES,3V3S0_ISNS:YES,OTHER_HS_ISNS:YES,CAM_ISNS:YES,CPUDDR_ISNS:YES,PANEL_ISNS:YES
ISNS:PROD	CPU_HS_ISNS:YES,CPUVR_ISNS:YES,DRAM_ISNS:YES,P1V05_ISNS:NO,AIRPORT_ISNS:NO,SSD_ISNS:YES,LCDBKLT_ISNS:NO,P3V3S5_ISNS:NO,3V3S0_ISNS:NO,OTHER_HS_ISNS:NO,CAM_ISNS:NO,CPUDDR_ISNS:NO,PANEL_ISNS:NO

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DDR3:HYNIX_4GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:HYNIX_4GB
DDR3:HYNIX_8GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:HYNIX_8GB
DDR3:SAMSUNG_4GB	RAMCFG0:L,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:SAMSUNG_4GB
DDR3:SAMSUNG_8GB	RAMCFG0:L,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:SAMSUNG_8GB
DDR3:ELPIDA_4GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:ELPIDA_4GB
DDR3:ELPIDA_8GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:ELPIDA_8GB
DDR3:MICRON_4GB	RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:MICRON_4GB
DDR3:MICRON_8GB	RAMCFG0:H,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:MICRON_8GB
DDR3:HYNIX_16GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:HYNIX_16GB
DDR3:SAMSUNG_16GB	RAMCFG0:L,RAMCFG1:H,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:SAMSUNG_16GB
DDR3:ELPIDA_16GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:ELPIDA_16GB
DDR3:MICRON_16GB	RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:MICRON_16GB

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0915	1	EEPROM,4MBIT,SPI,50MHZ,1.8V,USON8	U2890	CRITICAL	TBTROM:BLANK
341S00159	1	T29,EEPROM,FALCON RIDGE(V27.1), PROtO 0,J110/J113	U2890	CRITICAL	TBTROM:PROG
338S1214	1	IC,SMC12-B1,40MHZ/50DMIPS MCU,157BGA	U5000	CRITICAL	SMC:BLANK
335S00006	1	IC,SERIAL FLASH,64 MBIT 3V,WSON,QE=1	U6100	CRITICAL	BOOTROM_MAC:BLANK
335S00007	1	IC,SERIAL FLASH,64 MBIT 3V,WSON,QE=1	U6100	CRITICAL	BOOTROM_NUM:BLANK
341S00153	1	IC,EFI ROM(V0108)), PROTO 0,J110/J113	U6100	CRITICAL	BOOTROM:PROG

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00029	1	BDW,QGH9,D0,1.8,15W,2+2,0.7,4M,B1168	U0500	CRITICAL	CPU:2.1GHZ
337S00073	1	BDW,QGHB,D0,1.6,15W,2+2,0.6,4M,B1168	U0500	CRITICAL	CPU:1.6GHZ
338S00069	1	IC,TBT,FR-2C,288,12x12 ,FC-CSP,TRAY	U2800	CRITICAL	
338S1264	1	IC,BCM15700A2KFEB4G,S2 CMRA,8X8,208FCBGA	U3900	CRITICAL	
607-6811	1	ASSEMBLY,SUBASSY,PCBA,HALL EFFECT,K99	J6955	CRITICAL	J113_MLB
946-5477	1	UV GLUE,MLB,J41_J43	GLUE	CRITICAL	
825-7987	1	LABEL,MLB,J41/J43	NEW_LABEL		
376S00036	2	MOSFET,N-CH,30V,52A,5.9M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:REN
376S00037	2	MOSFET,N-CH,30V,64A,3.5M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:REN
376S1194	2	MOSFET,N-CH,30V,15.3A,12M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET,N-CH,30V,22A,6.0M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:VSHY
900-0090	1		SOLDERPASTE	CRITICAL	
825-7670	1	LABEL,TEXT,MLB,K21/K78	LABEL		

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0677	4	IC,SDRAM,8Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0681	4	IC,SDRAM,16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S00001	4	IC,SDRAM,23NM,8GB,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S00003	4	IC,SDRAM,23NM,16GB,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S0793	4	IC,SDRAM,8Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0791	4	IC,SDRAM,16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:ELPIDA_8GB
333S0793	4	IC,SDRAM,8Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0791	4	IC,SDRAM,16GB,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:MICRON_8GB
333S0789	4	IC,SDRAM,25nm,32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	DRAM_TYPE:ELPIDA_16GB

CPU DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

SIZE	CFG 3	CFG 2
4GB	0	0
8GB	0	1
16GB	1	0
RSVD	1	1

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1032	376S0855		ALL	Toshiba alt for Diodes dual
376S1129	376S0855		ALL	NXP alt for Diodes dual
376S1089	376S1128		ALL	NXP alt for Diodes single
138S0684	138S0660		ALL	Murata alt to Taiyo Yuden
138S0703	138S0648		ALL	Murata alt to Taiyo Yuden
152S0586	152S1301		ALL	Dale/Vishay alt to Cyntec
372S0186	372S0185		ALL	NXP alt to Diodes
197S0479	197S0478		ALL	200uW Epson alt to NDK
376S1053	376S0604		ALL	Diodes alt to Fairchild
371S0713	371S0558		ALL	Diodes alt to ST Micro
128S0371	128S0376		ALL	Kemet alt to Sanyo
152S1821	152S1757		ALL	Cyntec alt to NEC
197S0480	197S0343		ALL	NDK crystal alt to TXC
197S0481	197S0343		ALL	Epson crystal alt to TXC
107S0254	107S0241		ALL	Cyntec sense R alt to TFT
353S3452	353S1286		ALL	Maxim alt to Microchip
128S0386	128S0284		ALL	Kemet alt to Sanyo
128S0397	128S0325		ALL	Kemet alt to Sanyo
377S0155	377S0104		ALL	OnSemi alt to Infineon
128S0398	128S0220		ALL	Kemet alt to Sanyo
197S0542	197S0544		ALL	NDK alt to TXC
197S0545	197S0544		ALL	Epson alt to TXC
138S0681	138S0638		ALL	Taiyo alt to Samsung
138S0841	138S0638		ALL	Murata alt to Samsung
376S00014	376S0761		ALL	Renesas alt to Vishay
152S1876	152S1804		ALL	TDK alt to Toko
107S0255	107S0240		ALL	Cyntec alt to TFT
107S0250	107S0248		ALL	Cyntec alt to TFT
870-5074	870-1938		ALL	ALT POGO PIN W_O CAP
860-3428	860-1327		ALL	ALT STANDOFF W_O MYLAR
333S0787	333S0677	DRAM_TYPE:HYNIX_4GB	ALL	ALT STANDOFF W_O MYLAR
860-3690	860-1328		ALL	ALT STANDOFF W_O MYLAR
333S0785	333S0681	DRAM_TYPE:HYNIX_8GB	ALL	ALT STANDOFF W_O MYLAR
353S3814	353S3812		ALL	ALT TBT PORT MUX
311S00008	311S0271		ALL	ALT AND GATE
311S00007	311S0426		ALL	ALT SNGL BUFFER
311S00015	311S0450		ALL	ALT 2-INPT AND
311S00013	311S0508		ALL	ALT SNGL BUFFER
311S00014	311S0515		ALL	ALT DUAL BUFFER
353S00133	353S2741		ALL	ALT PWR DIST SW

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-00623	PCBA,MLB,BEST,HY-4GB,X430	MLB_CMNPTS,CPU:2.1GHZ,DDR3:HYNIX_4GB,ALTERNATE
639-00624	PCBA,MLB,BEST,HY-8GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:HYNIX_8GB,ALTERNATE
639-00625	PCBA,MLB,BEST,HY-16GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:HYNIX_16GB
639-00626	PCBA,MLB,BEST,SM-4GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:SAMSUNG_4GB,ALTERNATE
639-00627	PCBA,MLB,BEST,SM-8GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:SAMSUNG_8GB,ALTERNATE
639-00628	PCBA,MLB,BEST,MI-4GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:MICRON_4GB
639-00629	PCBA,MLB,BEST,MI-8GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:MICRON_8GB
639-00630	PCBA,MLB,BEST,MI-16GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:MICRON_16GB
639-00631	PCBA,MLB,BEST,EL-4GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:ELPIDA_4GB
639-00632	PCBA,MLB,BEST,EL-8GB,X433	MLB_CMNPTS,CPU:2.1GHZ,DDR3:ELPIDA_8GB
639-00633	PCBA,MLB,BETTER,HY-4GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:HYNIX_4GB,ALTERNATE
639-00634	PCBA,MLB,BETTER,HY-8GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:HYNIX_8GB,ALTERNATE
639-00635	PCBA,MLB,BETTER,HY-16GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:HYNIX_16GB
639-00636	PCBA,MLB,BETTER,SM-4GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:SAMSUNG_4GB,ALTERNATE
639-00637	PCBA,MLB,BETTER,SM-8GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:SAMSUNG_8GB,ALTERNATE
639-00638	PCBA,MLB,BETTER,MI-4GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:MICRON_4GB
639-00639	PCBA,MLB,BETTER,MI-8GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:MICRON_8GB
639-00640	PCBA,MLB,BETTER,MI-16GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:MICRON_16GB
639-00641	PCBA,MLB,BETTER,EL-4GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:ELPIDA_4GB
639-00642	PCBA,MLB,BETTER,EL-8GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:ELPIDA_8GB
685-00046	CMN PTS,PCBA,MLB,X433	MLB_COMMON,J113_MLB
685-00047	VCORE FET,REN,X433	VCORE_FET:REN
685-00048	VCORE FET,VSHY,X433	VCORE_FET:VSHY
639-00697	PCBA,MLB,BETTER,EL-16GB,X433	MLB_CMNPTS,CPU:1.6GHZ,DDR3:ELPIDA_16GB

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00047	685-00048		ALL	Renesas alt to Vishay
333S0704	333S0700		ALL	Elpida CAM DRAM alt to Hynix

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S1246	1	IC,GL3219,USB3 SD CARD READER,46P,LQFN	U4500	CRITICAL	

BOM Groups

BOM GROUP	BOM OPTIONS
MLB_PROGPARTS	BOOTROM:PROG,SMC:PROG,TBTROM:PROG

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S00148	1	IC,SMC-B1,EXT(Vxxxx),PROTO 0,J113	U5000	CRITICAL	SMC:PROG

Sub-BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00046	1	CMN PTS,PCBA,MLB,J113	CMNPTS	CRITICAL	MLB_CMNPTS
685-00048	1	VCORE FET,VSHY,J113	VCOREFETS	CRITICAL	VCORE_FETS

SYNC_MASTER=MASTER SYNC_DATE=MASTER
PAGE TITLE
BOM Variants
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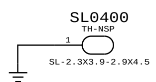
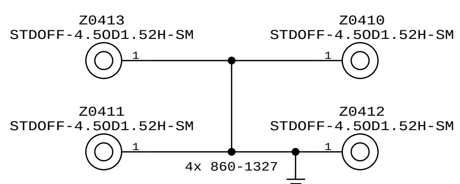
PD Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-5107	1	CAN,TOPSIDE,ALT,J41/J43	TBTTOPSIDE_2P_FENCE	CRITICAL	
806-5108	1	CAN,TOPSIDE,COVER,ALT,J41/J43	TBTTOPSIDE_2P_COVER	CRITICAL	
806-3142	1	CAN,TBT,J11/J13	TBTFENCE	CRITICAL	
806-3215	1	CAN,COVER,TBT,J11/J13	TBTCOVER	CRITICAL	
806-3216	1	CAN,MDP,J11/J13	MDPCAN	CRITICAL	
806-3083	1	SHLD,USB,MLB,J11/J13	USBCAN	CRITICAL	

Plated Board Slot



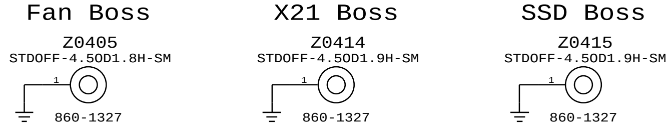
CPU Heat Sink Mounting Bosses



Fan Boss

X21 Boss

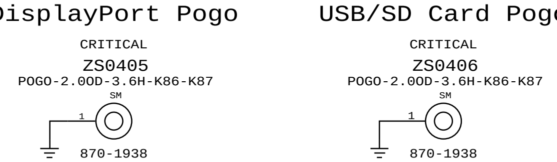
SSD Boss



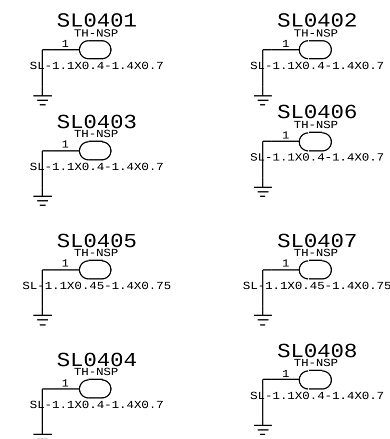
EMI I/O Pogo Pins

DisplayPort Pogo

USB/SD Card Pogo



Can Slots



2x TBT pin diodes

2x MDP Connector

2x TBT chip

2x USB Connector

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PD PARTS

Apple Inc.

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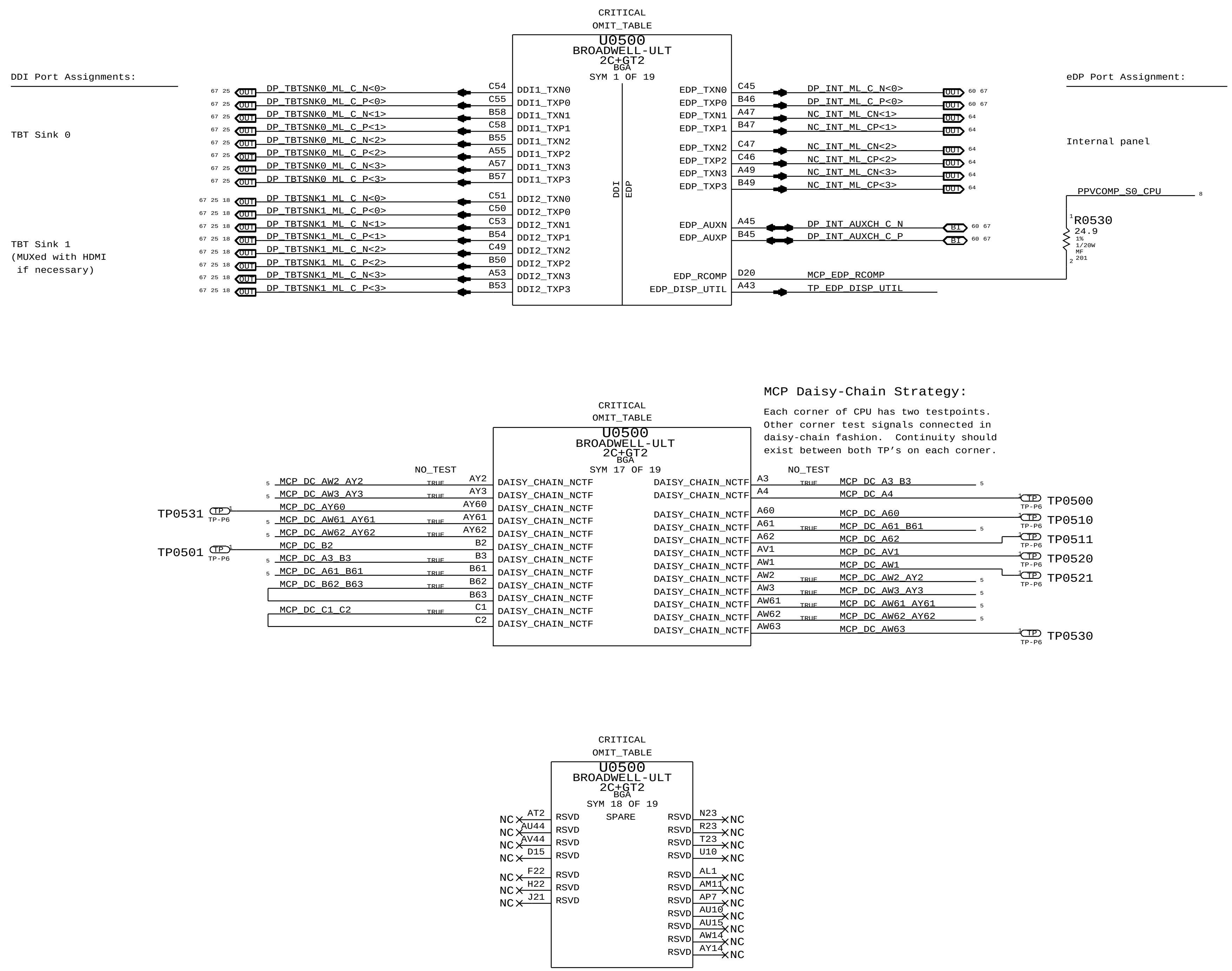

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CPU GFX/NCTF/RSVD

Apple Inc.

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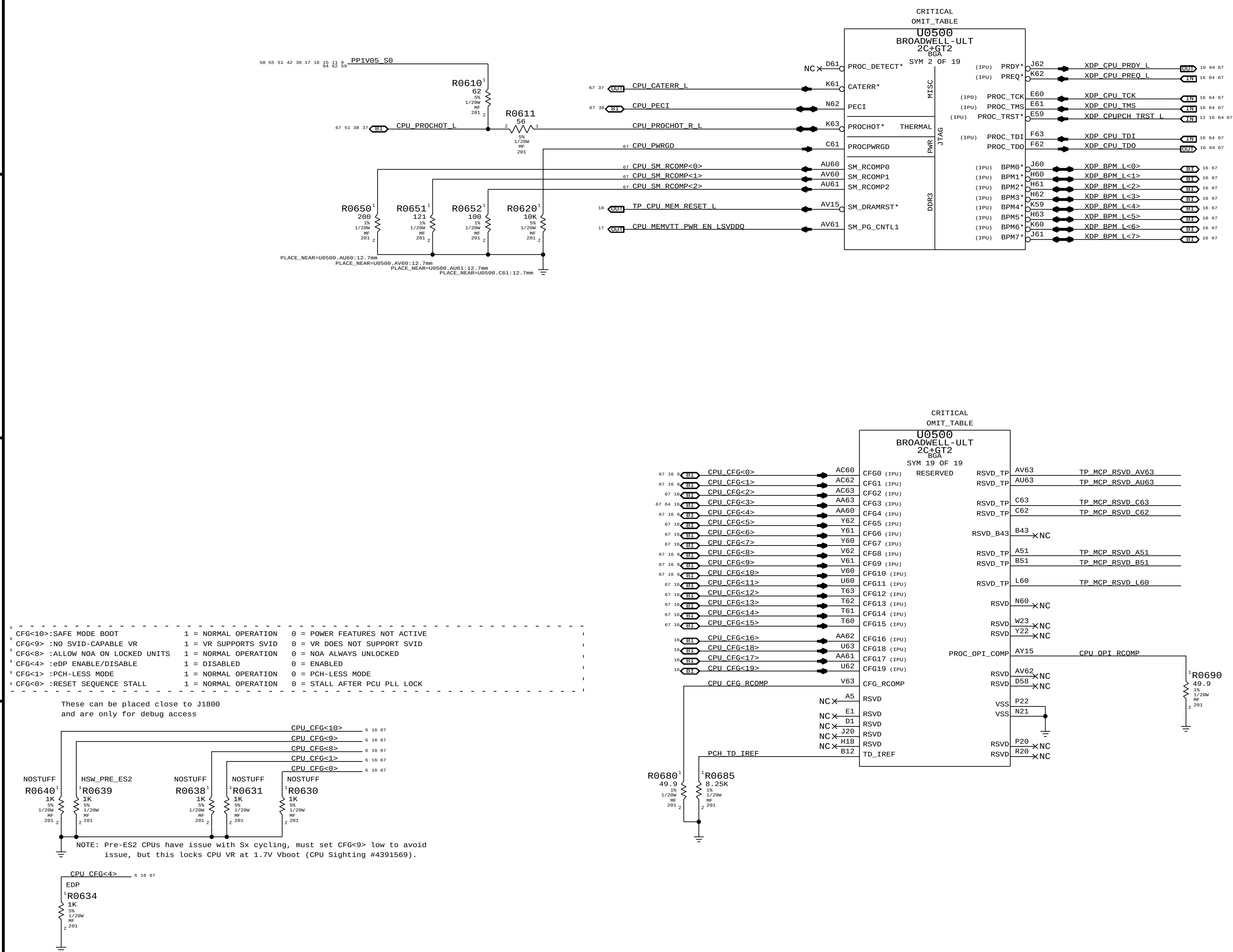

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PAGE TITLE

CPU Misc/JTAG/CFG/RSVD

Apple Inc.

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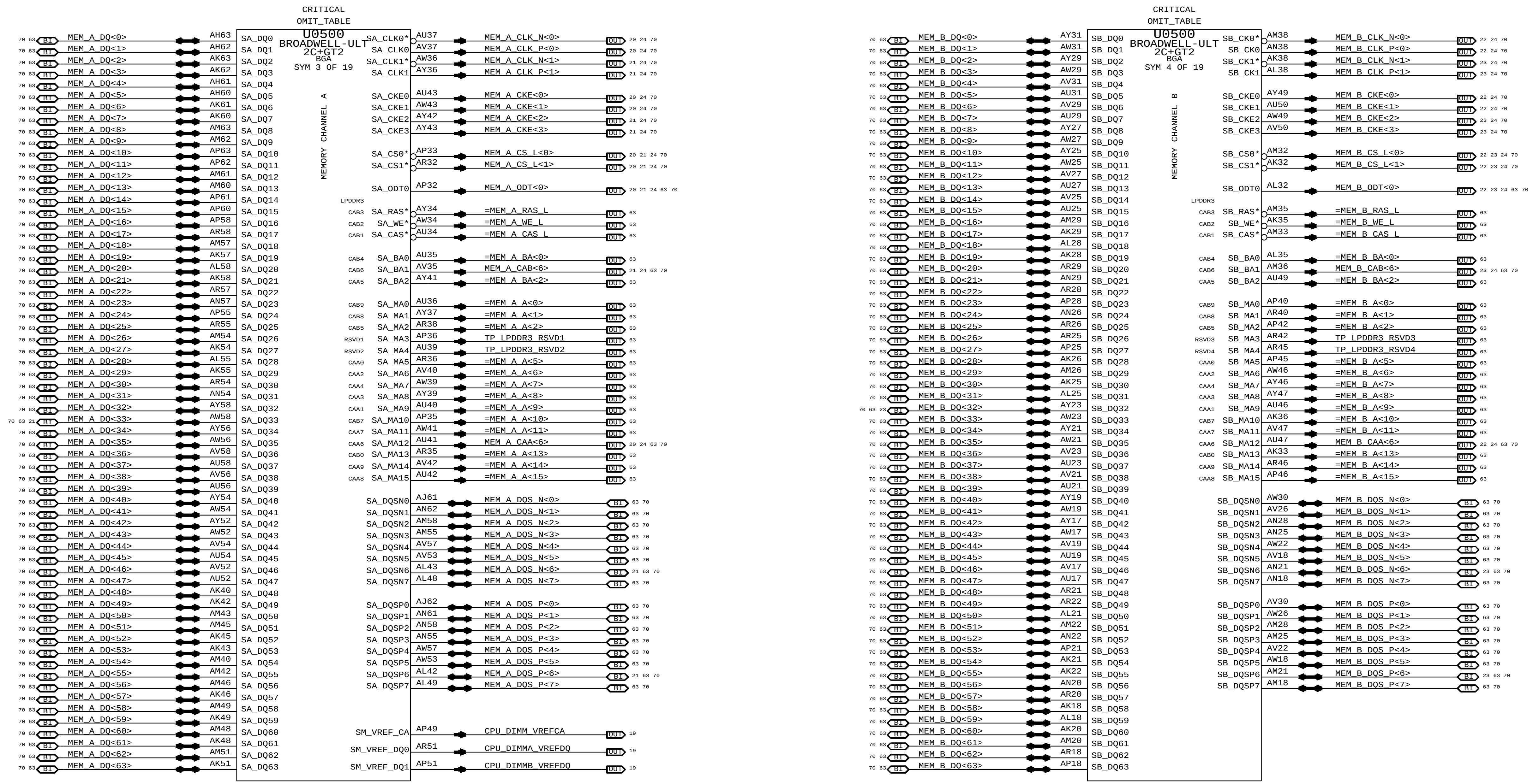

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PAGE TITLE

CPU DDR3/LPDDR3 Interfaces

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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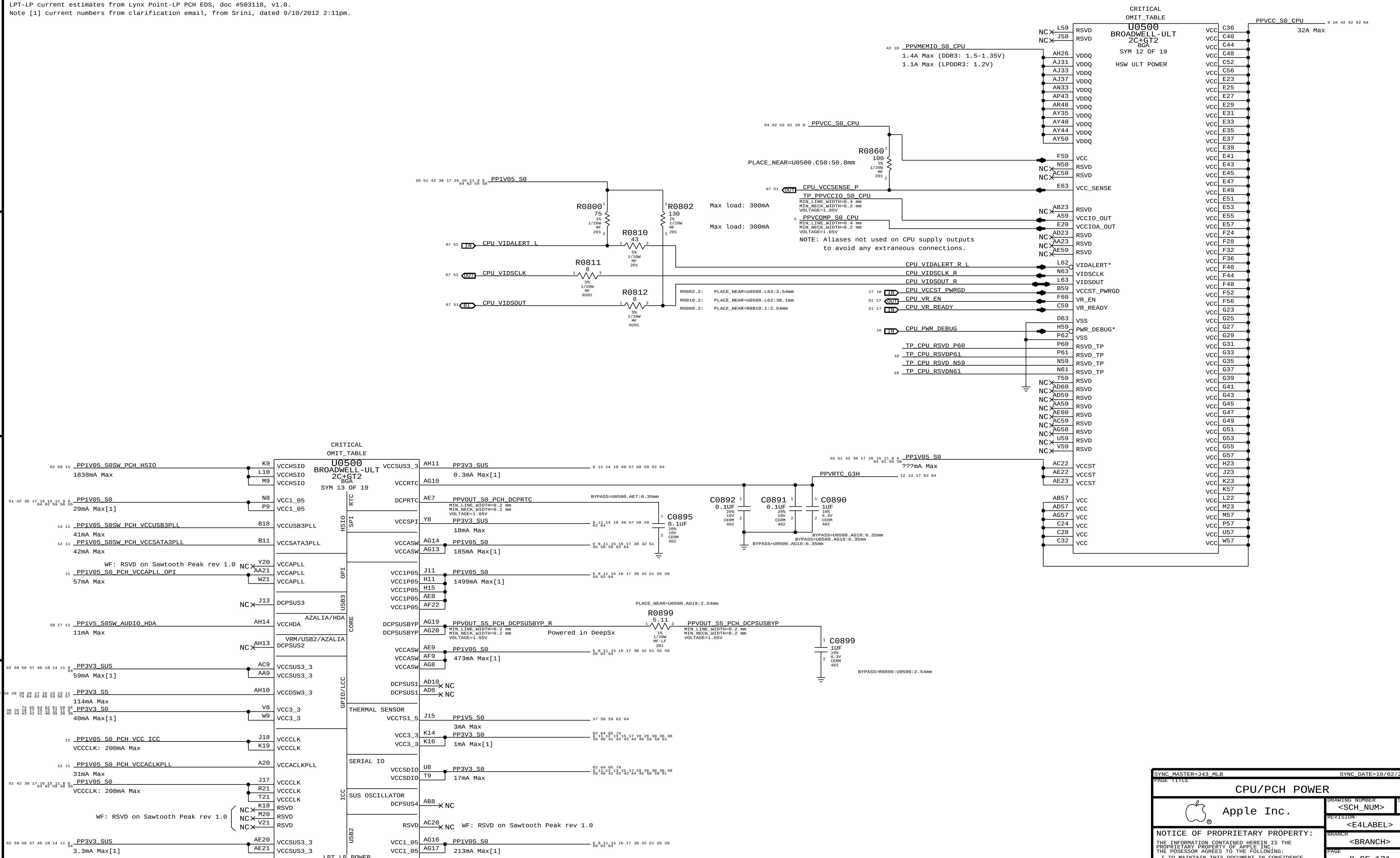
HSW-ULT current estimates from Haswell Mobile ULT Processor EDS vol 1, doc #502406, v0.9.
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.



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CPU/PCH POWER

Apple Inc.

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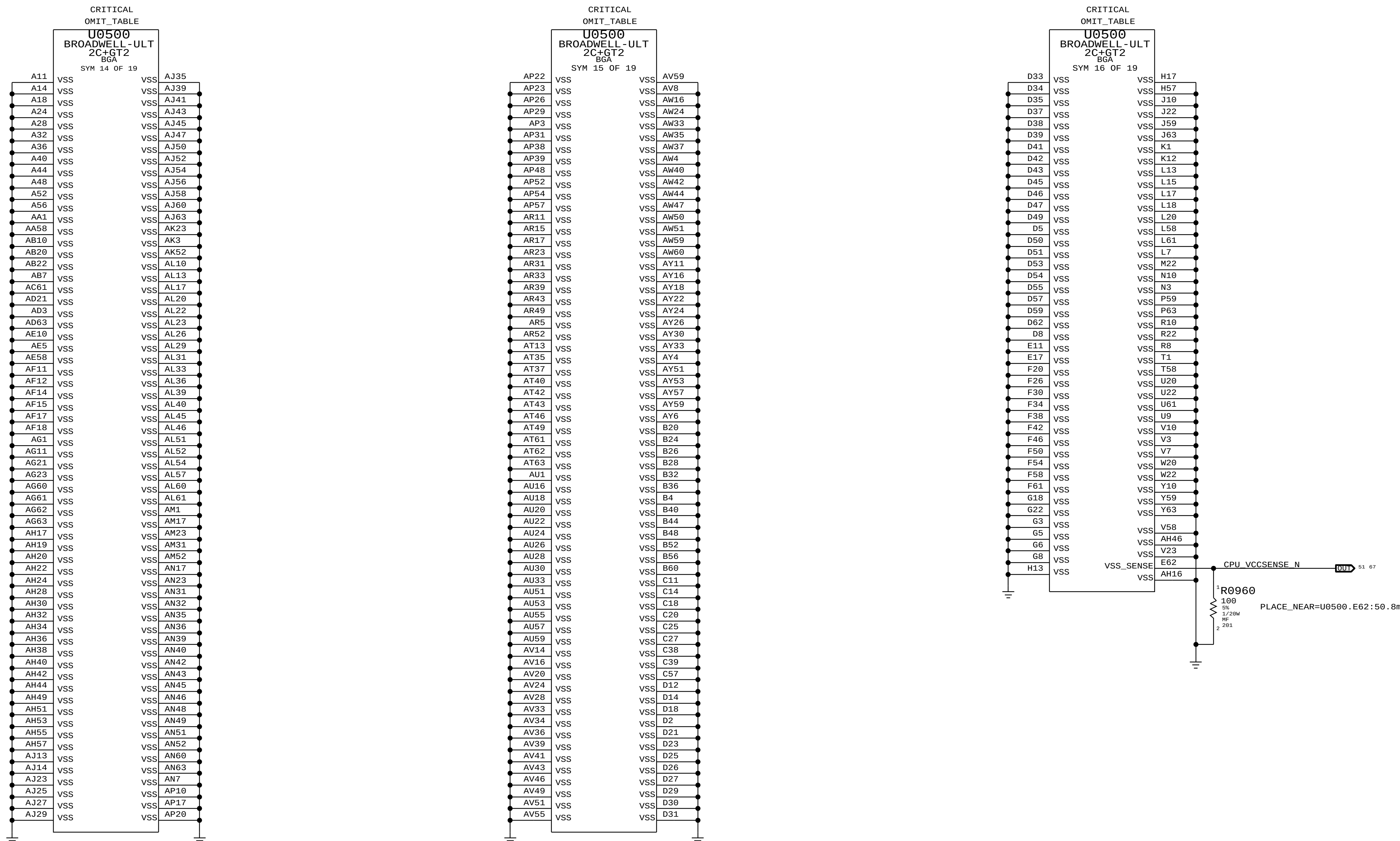

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PAGE TITLE

CPU/PCH GROUNDS

Apple Inc.

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REVISION <E4LABEL>

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SHEET 9 OF 76

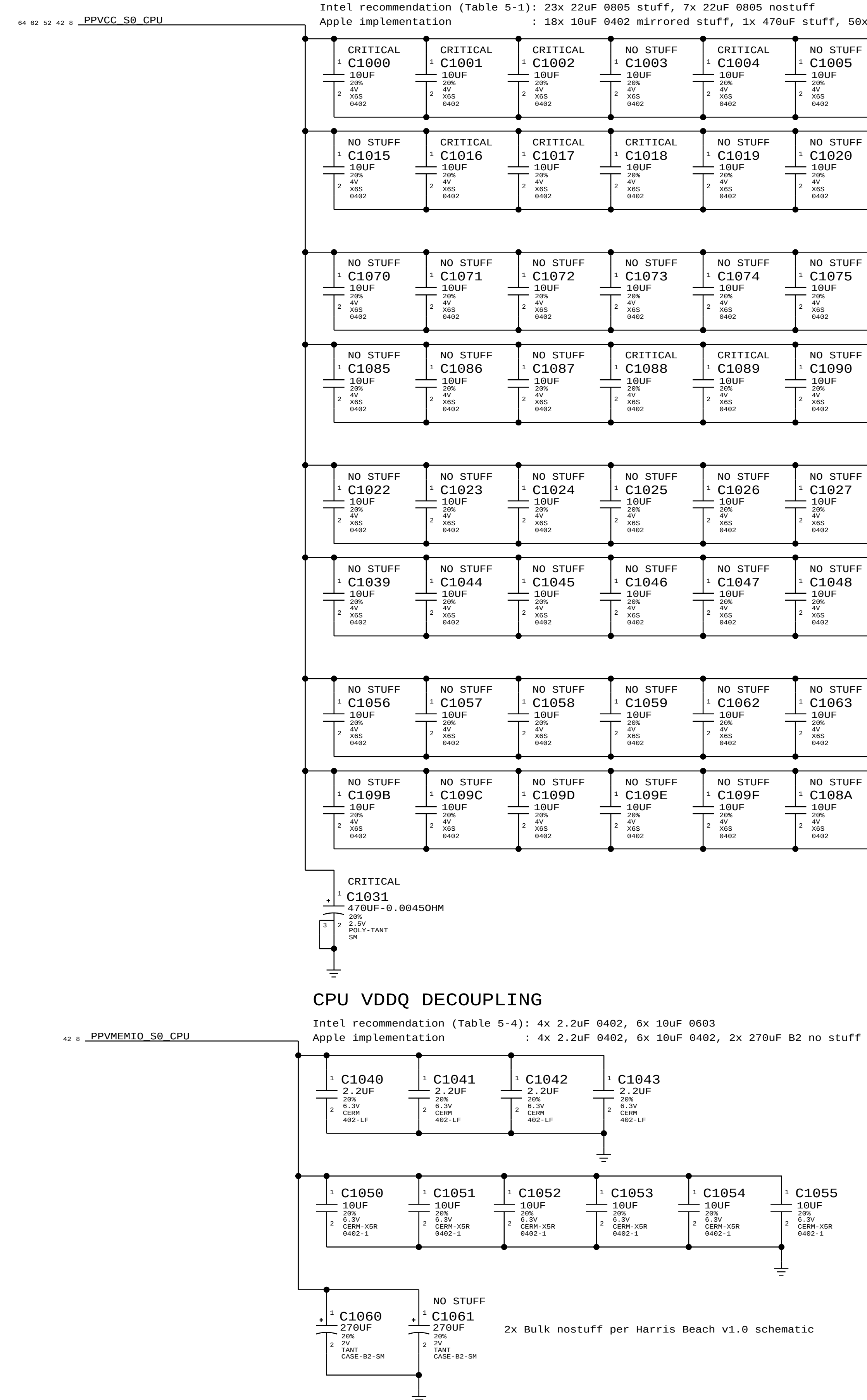
All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 1.0 unless stated otherwise

CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 18x 10uF 0402 mirrored stuff, 1x 470uF stuff, 50x 10uF mirrored no stuff, 50x 10uF single sided no stuff

CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 4x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 4x 2.2uF 0402, 6x 10uF 0402, 2x 270uF B2 no stuff



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CPU Decoupling

Apple Inc.

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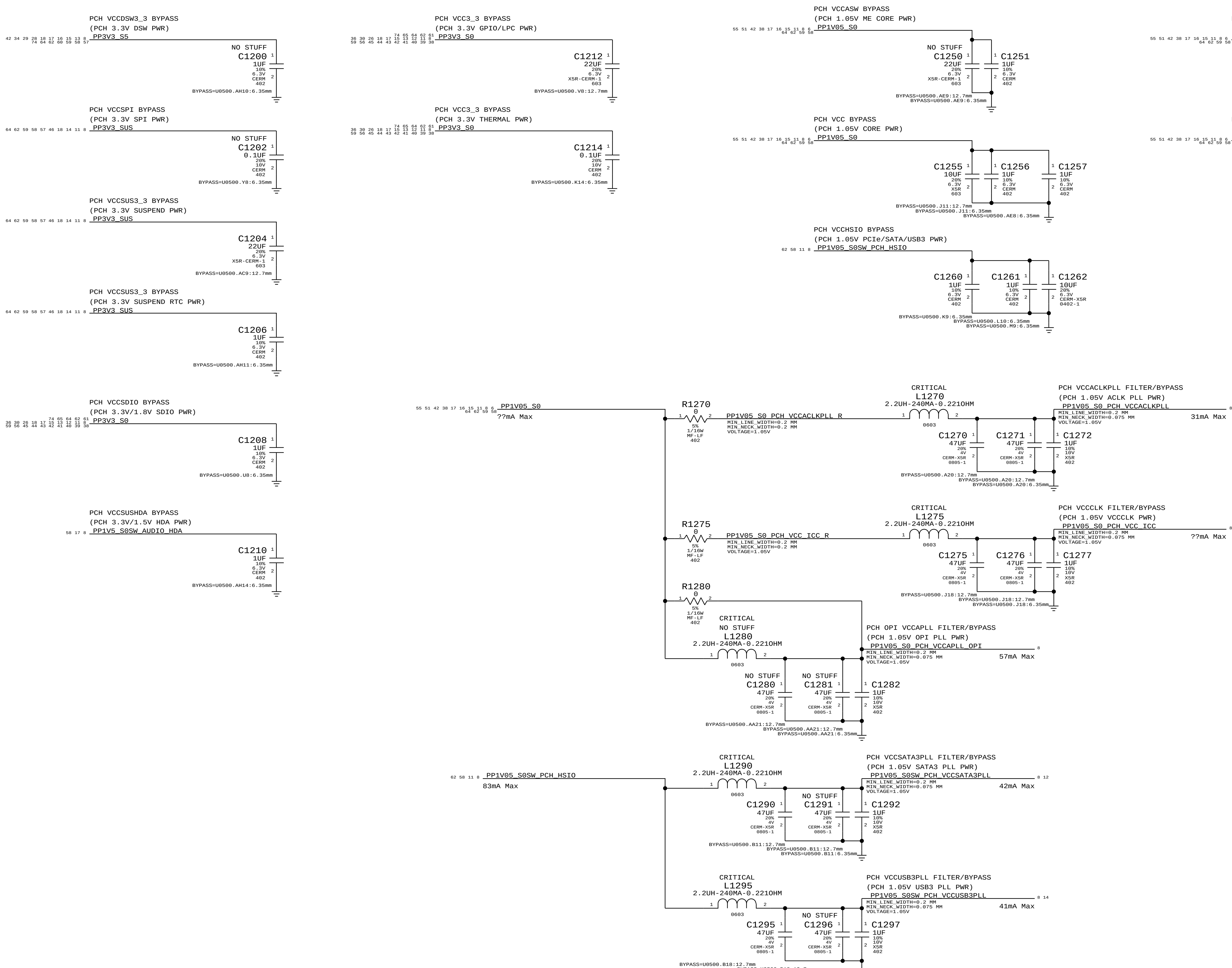

LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

SYNC_MASTER=J41_MLB SYNC_DATE=02/07/2013

PAGE TITLE

PCH Decoupling

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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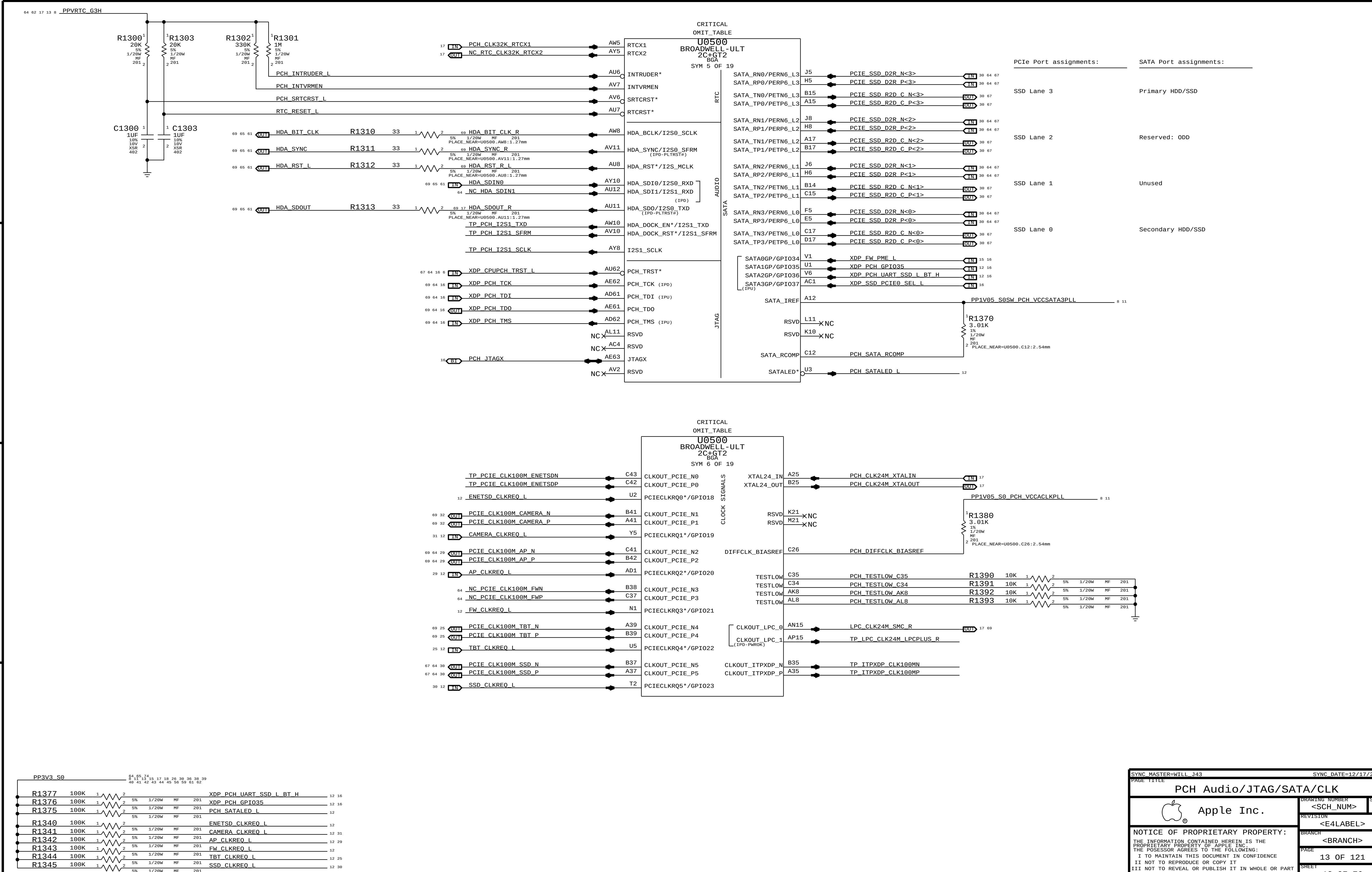

SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE

PCH Audio/JTAG/SATA/CLK

Apple Inc.

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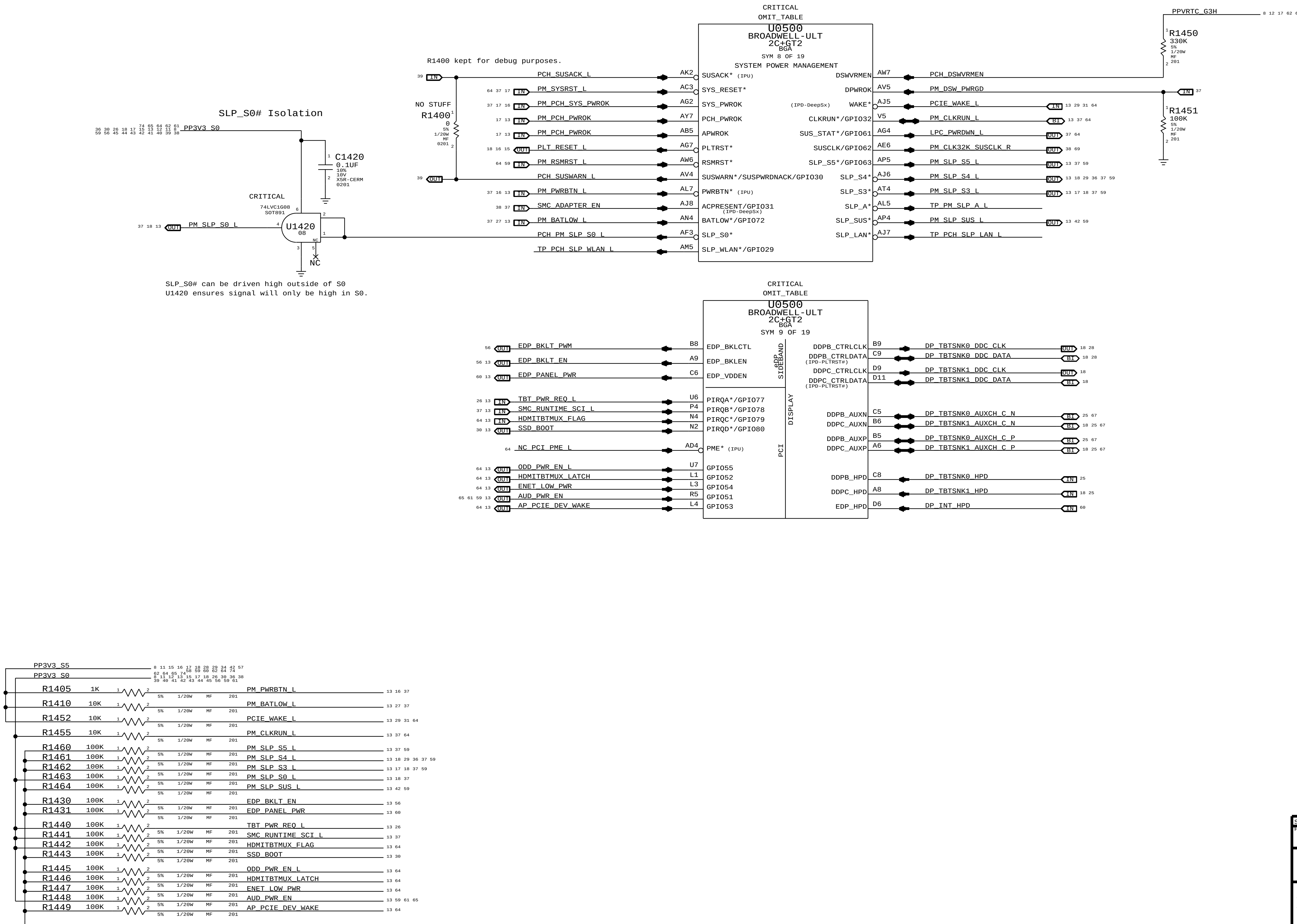

SYNC_MASTER=J43_MLB SYNC_DATE=02/20/2013

PAGE TITLE

PCH PM/PCI/GFX

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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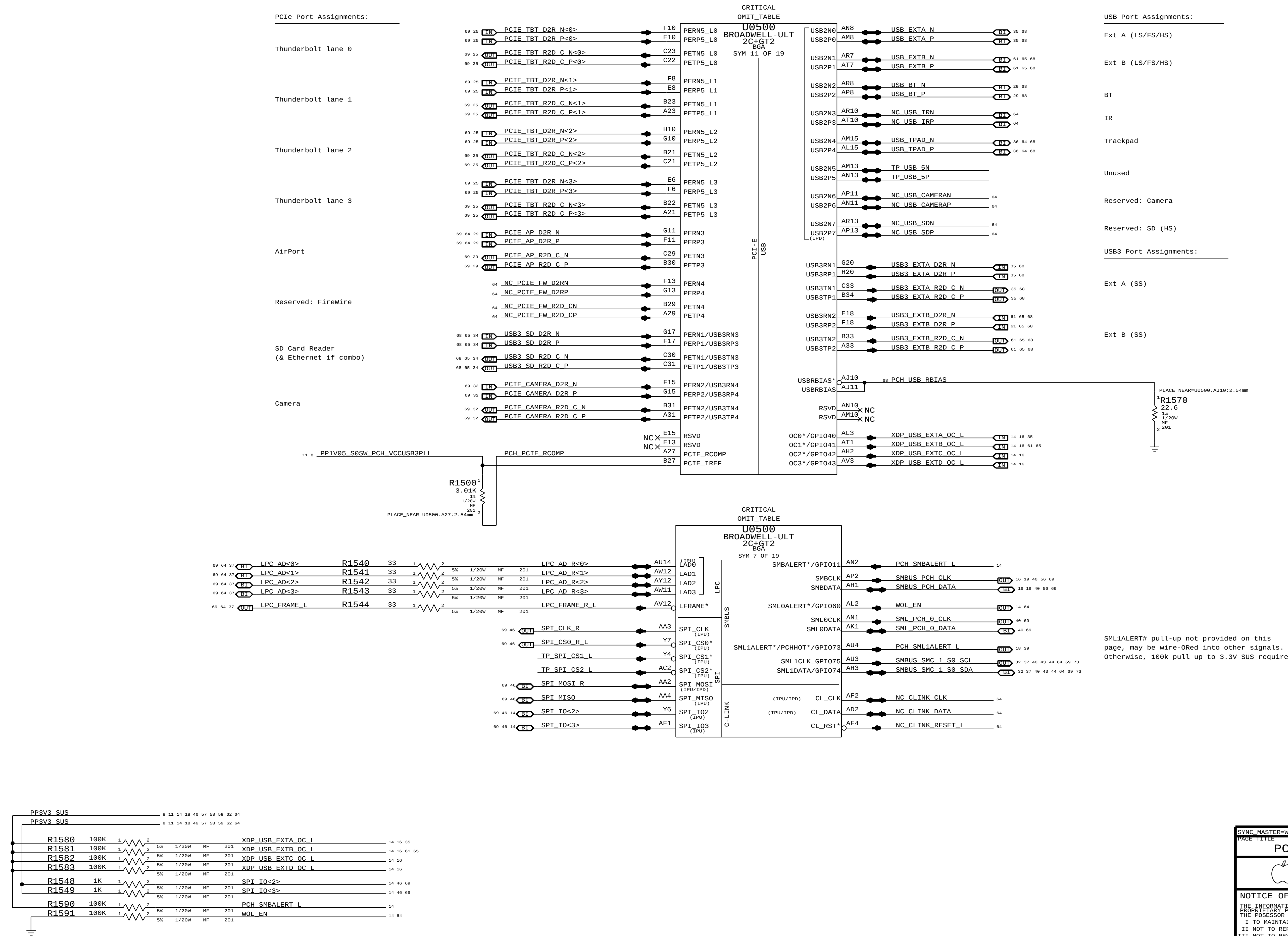

SYNC_MASTER=WILL_J43 SYNC_DATE=09/13/2012

PAGE TITLE

PCH PCIe/USB/LPC/SPI/SMBus

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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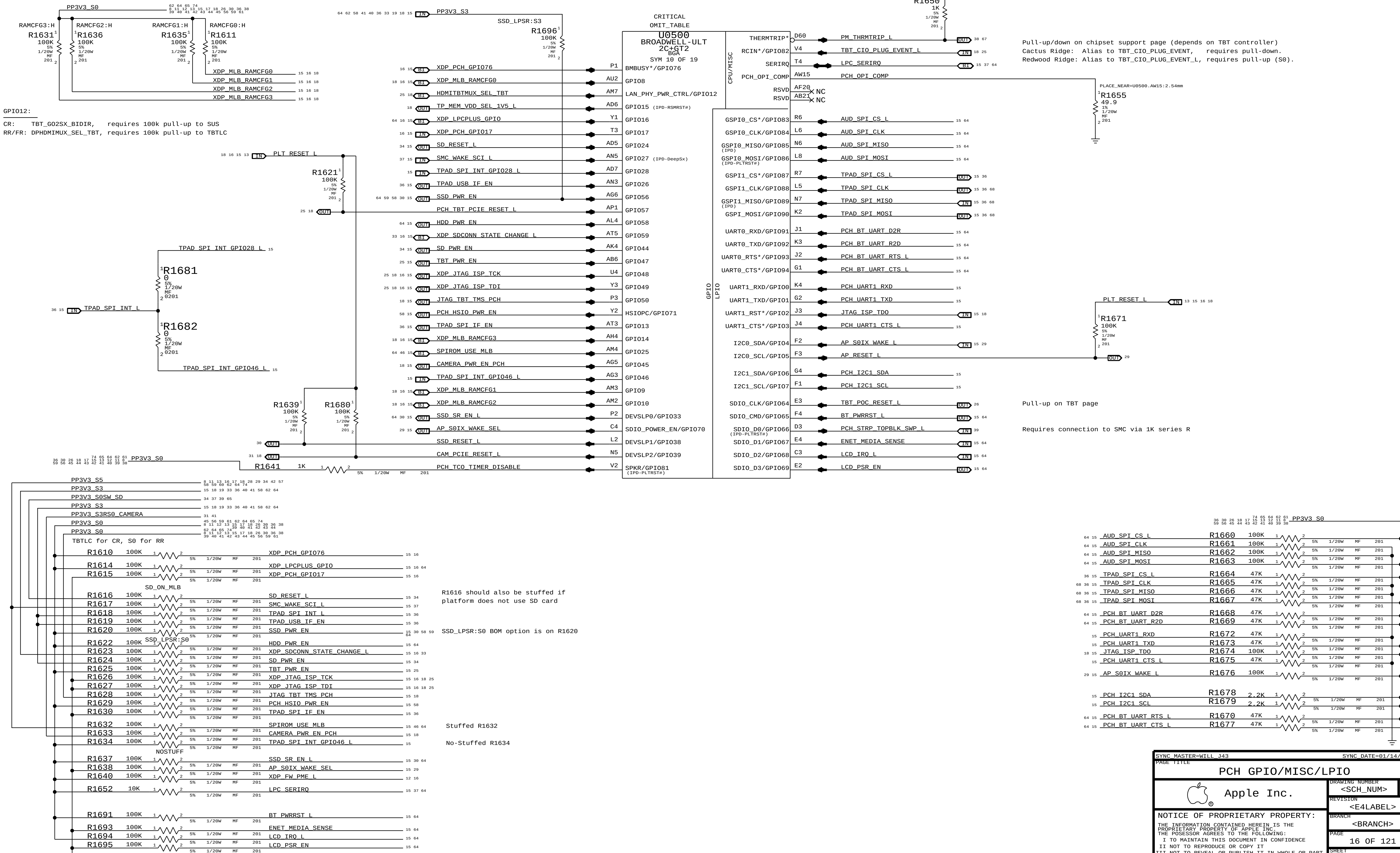
BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H



GPIO12:

CR: TBT_GO2SX_BIDIR, requires 100k pull-up to SUS

RR/FR: DPHDMIMUX_SEL_TBT, requires 100k pull-up to TBTLC

Pull-up/down on chipset support page (depends on TBT controller)

Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.

Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

Pull-up on TBT page

Requires connection to SMC via 1K series R

R1616 should also be stuffed if platform does not use SD card

SSD_LPSR:S0 BOM option is on R1620

Stuffed R1632

No-Stuffed R1634

SYNC_MASTER=WILL_J43 SYNC_DATE=01/14/2013

PAGE TITLE

PCH GPIO/MISC/LPIO

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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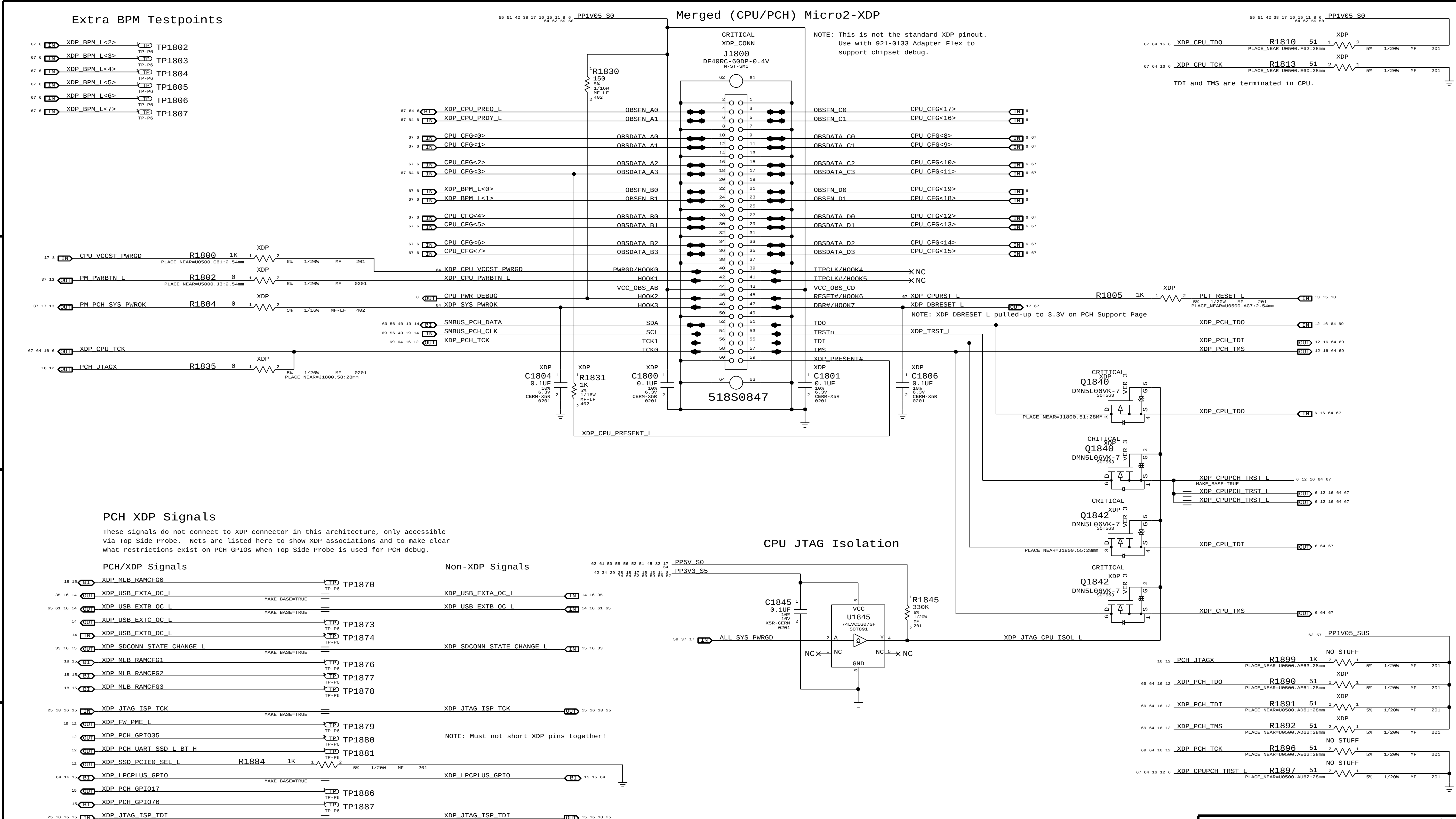

Unused & MLB_RAMCFGx GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD Cards during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

NOTE: Should force PCH GPIO47 high to ensure TBT router powered to avoid leakage/clamping of signals.

SSD_PCIEx_SEL_L straps are connected via 1K to common net.

LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.

SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE

CPU/PCH Merged XDP

Apple Inc.

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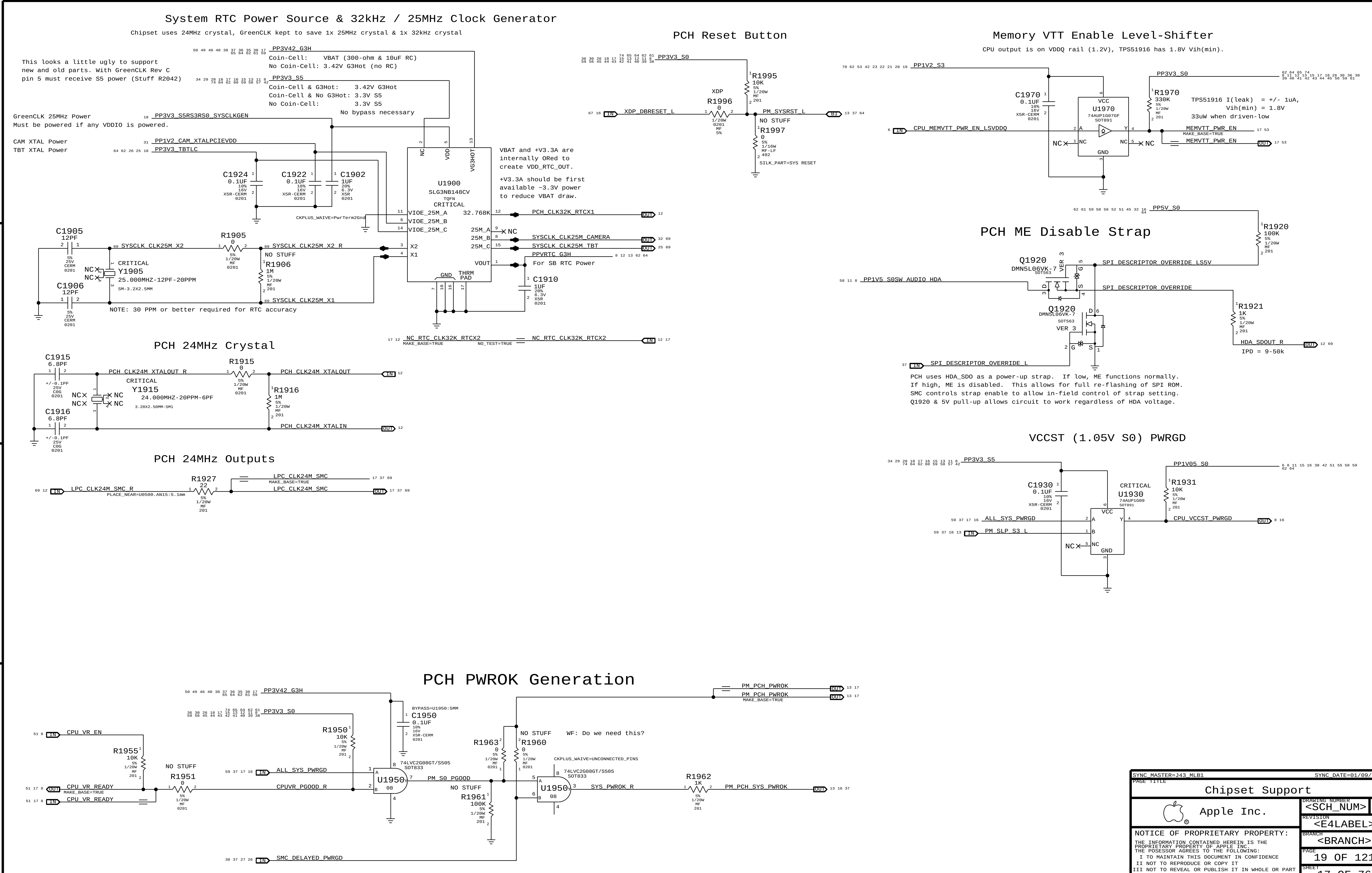

SYNC_MASTER=J43_MLB1 SYNC_DATE=01/09/2013

PAGE TITLE: Chipset Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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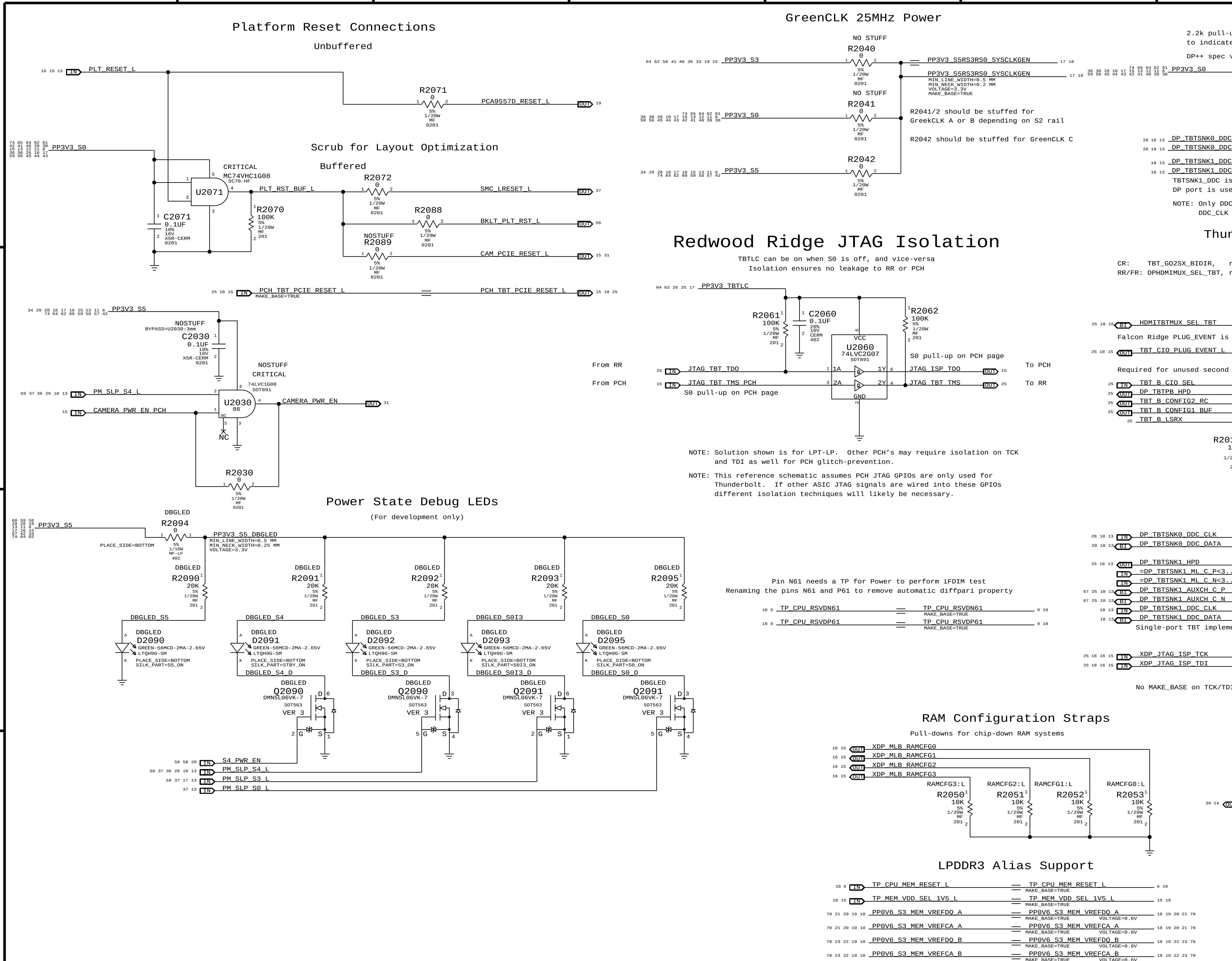

SYNC_MASTER=J43_MLB SYNC_DATE=01/17/2013

PAGE TITLE: Project Chipset Support

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

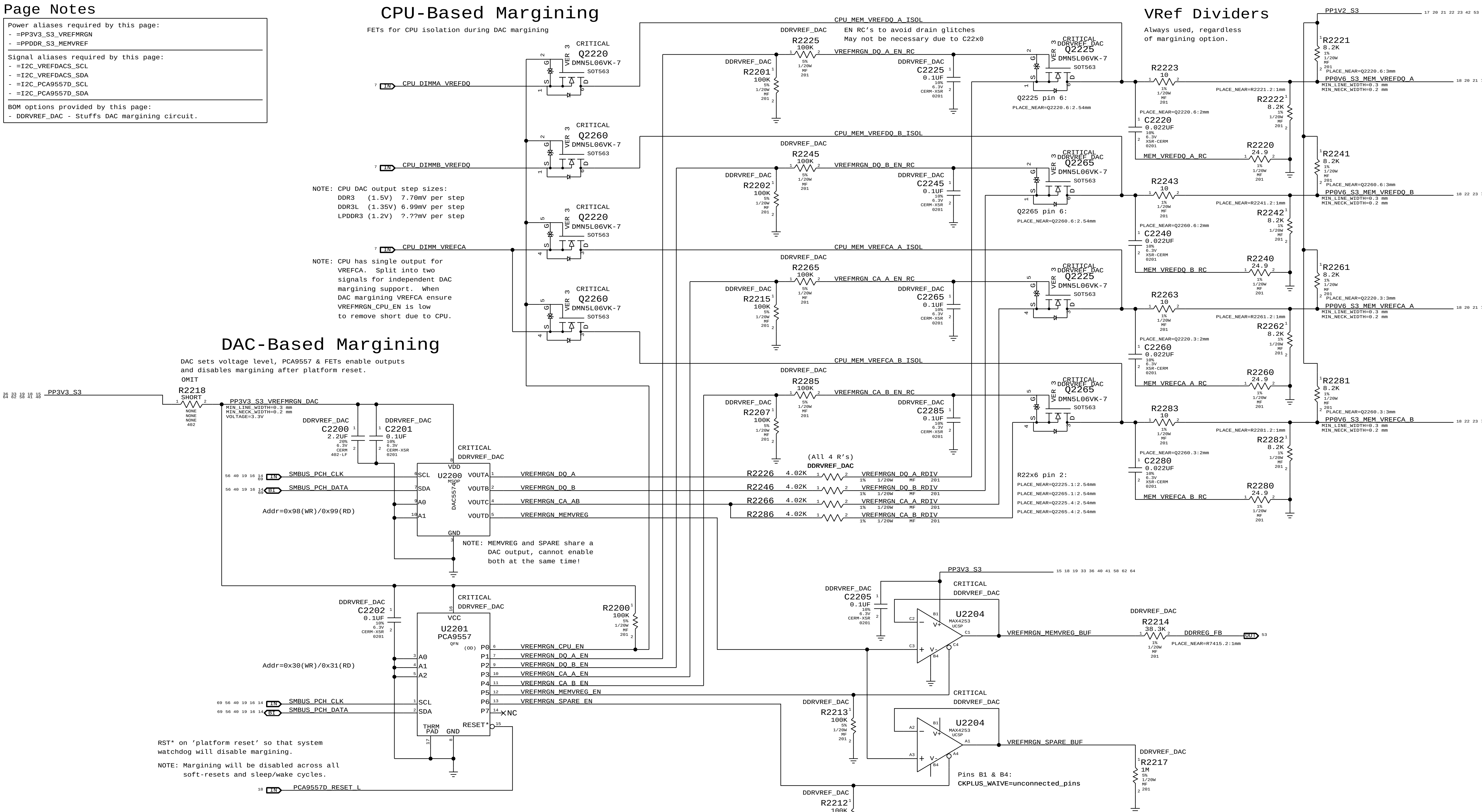
FETs for CPU isolation during DAC margining

VRef Dividers

Always used, regardless of margining option.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	
DAC Channel:	A	B	C	C	D	
PCA9557D Pin:	1	2	3	4	5	
	LPDDR3 (1.2V)		DDR3L (1.35V)		LPDDR3 (1.2V)	DDR3L (1.35V)
Nominal value	0.600V (DAC: 0x2E.5)		0.675V (DAC: 0x34)		1.200V (DAC: 0x5D)	1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)
VRef current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output	3.53mV / step @ output

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=J41_MLB SYNC_DATE=02/12/2013

PAGE TITLE: DDR3 VREF MARGINING

DRAWING NUMBER: <SCH_NUM> SIZE: D

Apple Inc.

REVISION: <E4LABEL>

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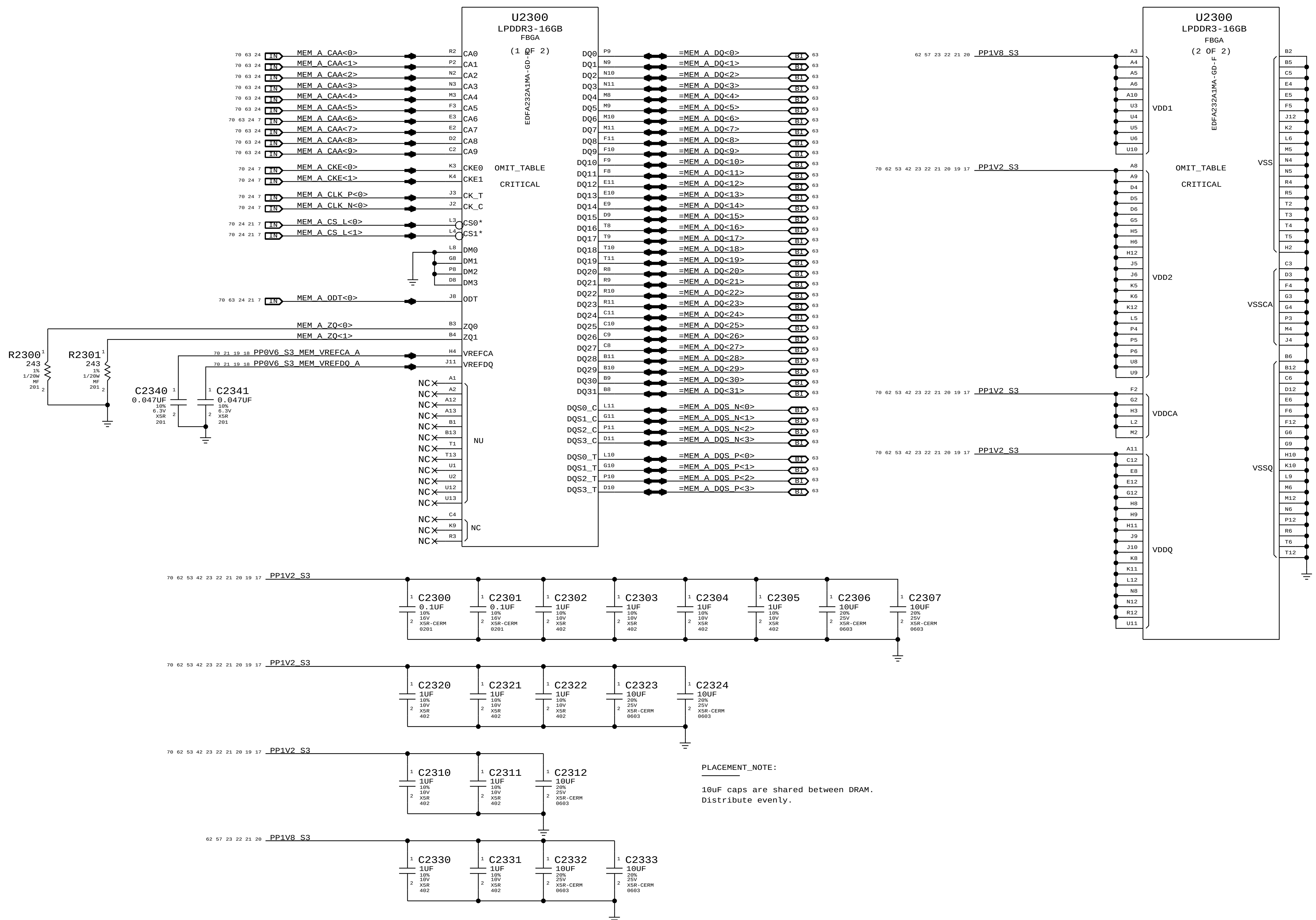
LPDDR3 CHANNEL A (0-31)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

LPDDR3 DRAM Channel A (0-31)

Apple Inc.

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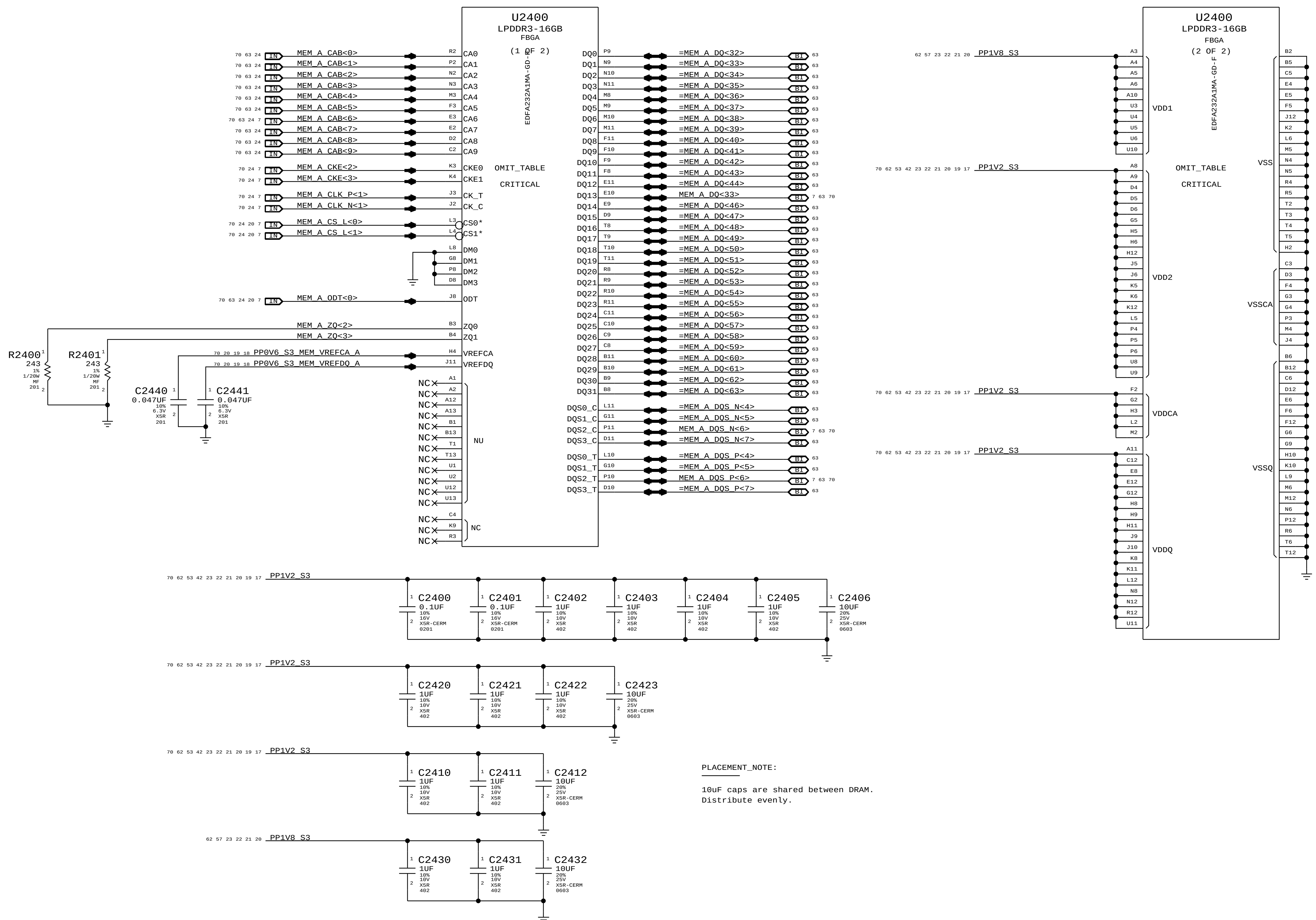
LPDDR3 CHANNEL A (32-63)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

LPDDR3 DRAM Channel A (32-63)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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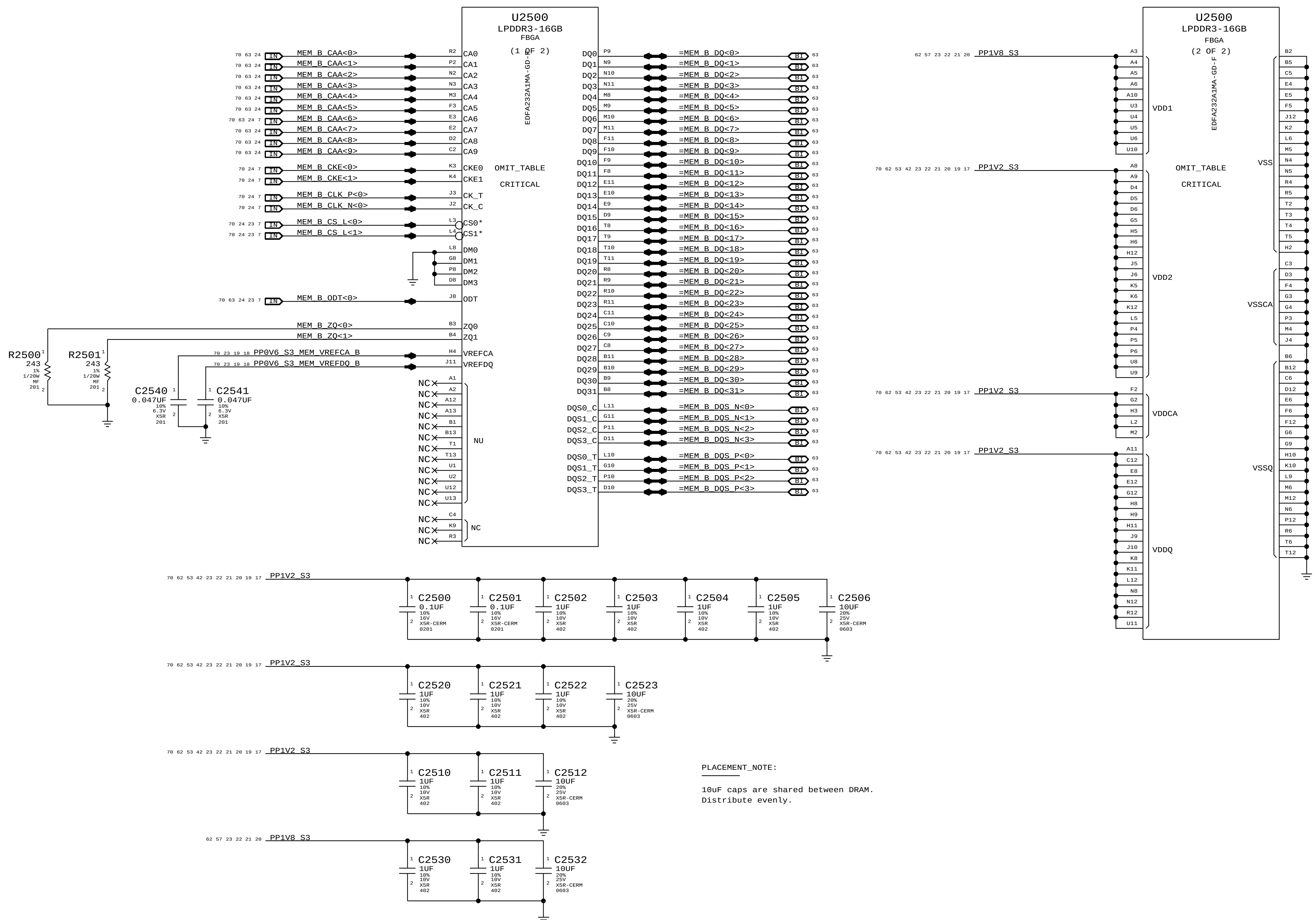
LPDDR3 CHANNEL B (0-31)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

LPDDR3 DRAM Channel B (0-31)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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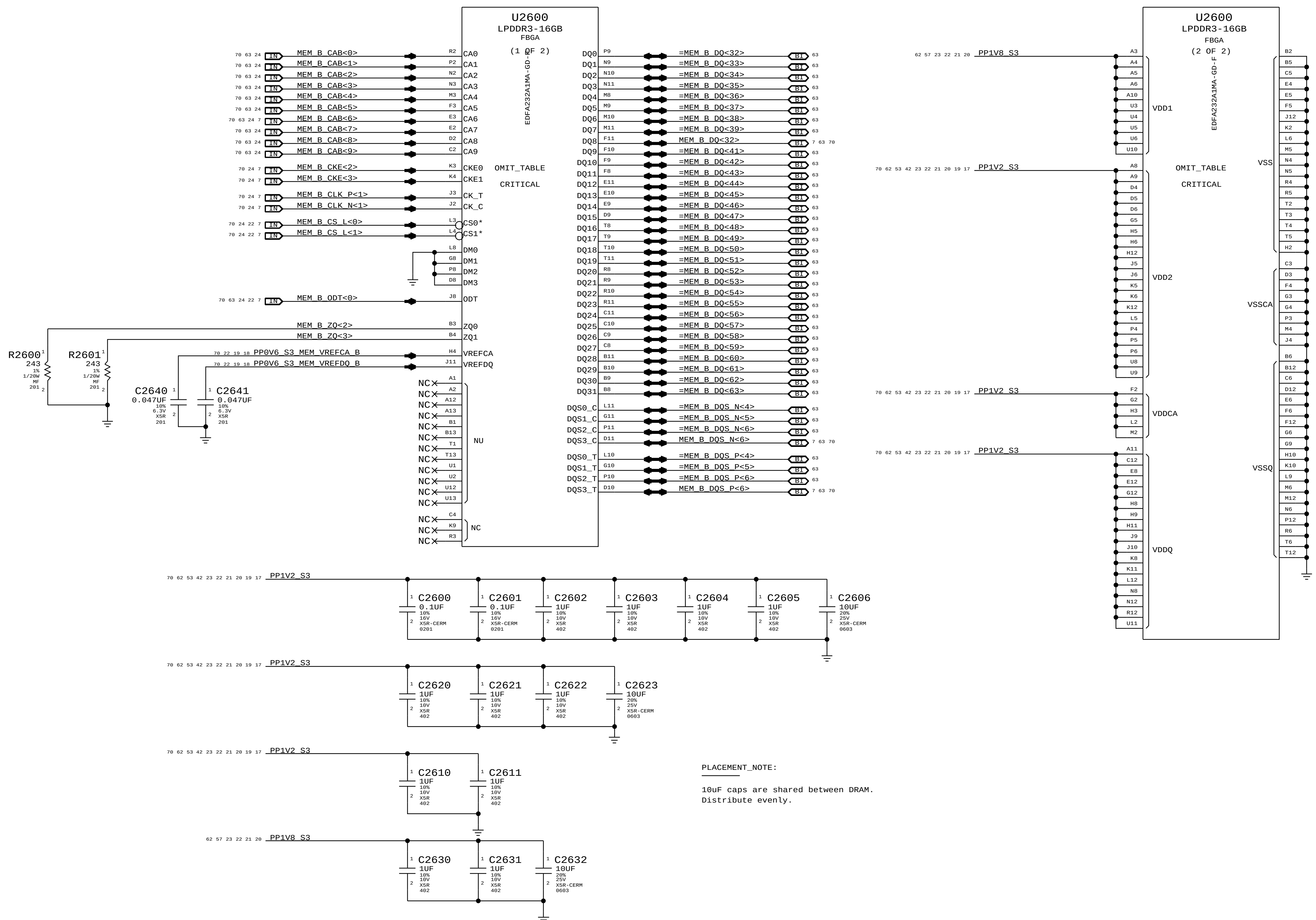
LPDDR3 CHANNEL B (32-63)



PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE
LPDDR3 DRAM Channel B (32-63)

Apple Inc.

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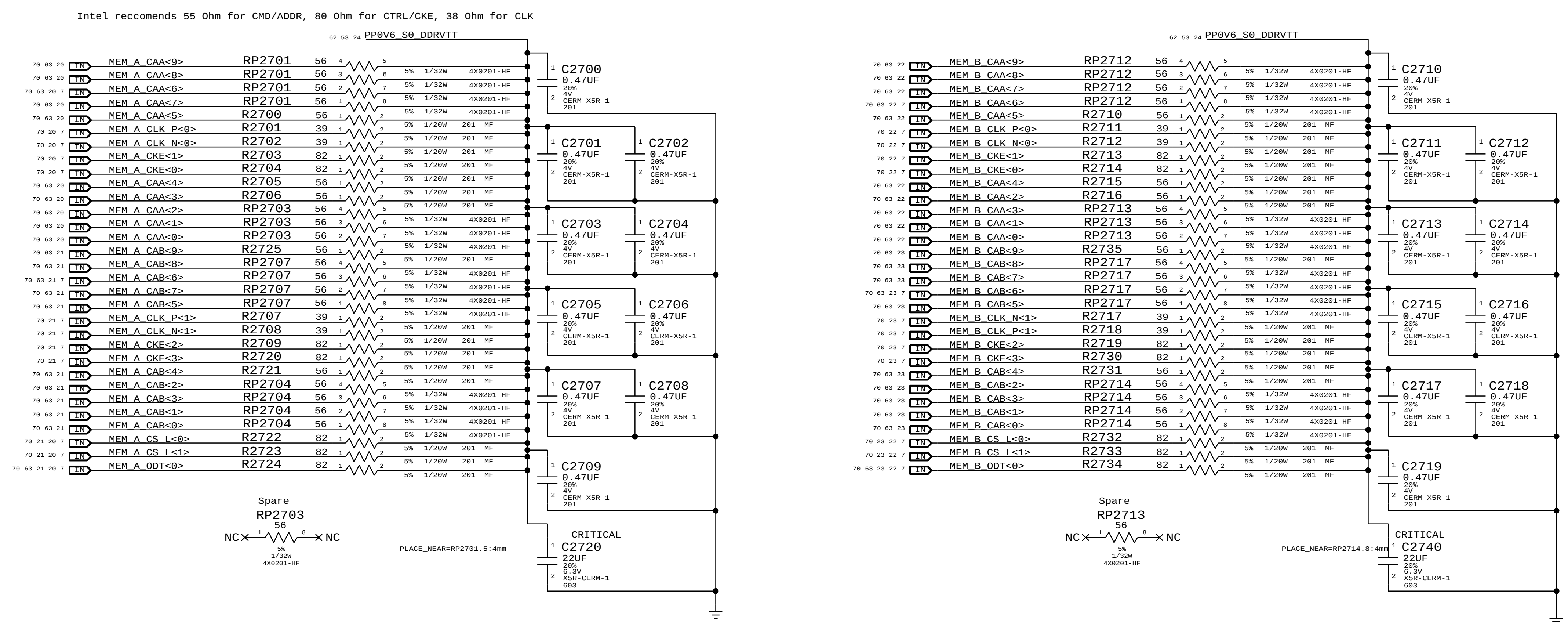

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

LPDDR3 DRAM Termination

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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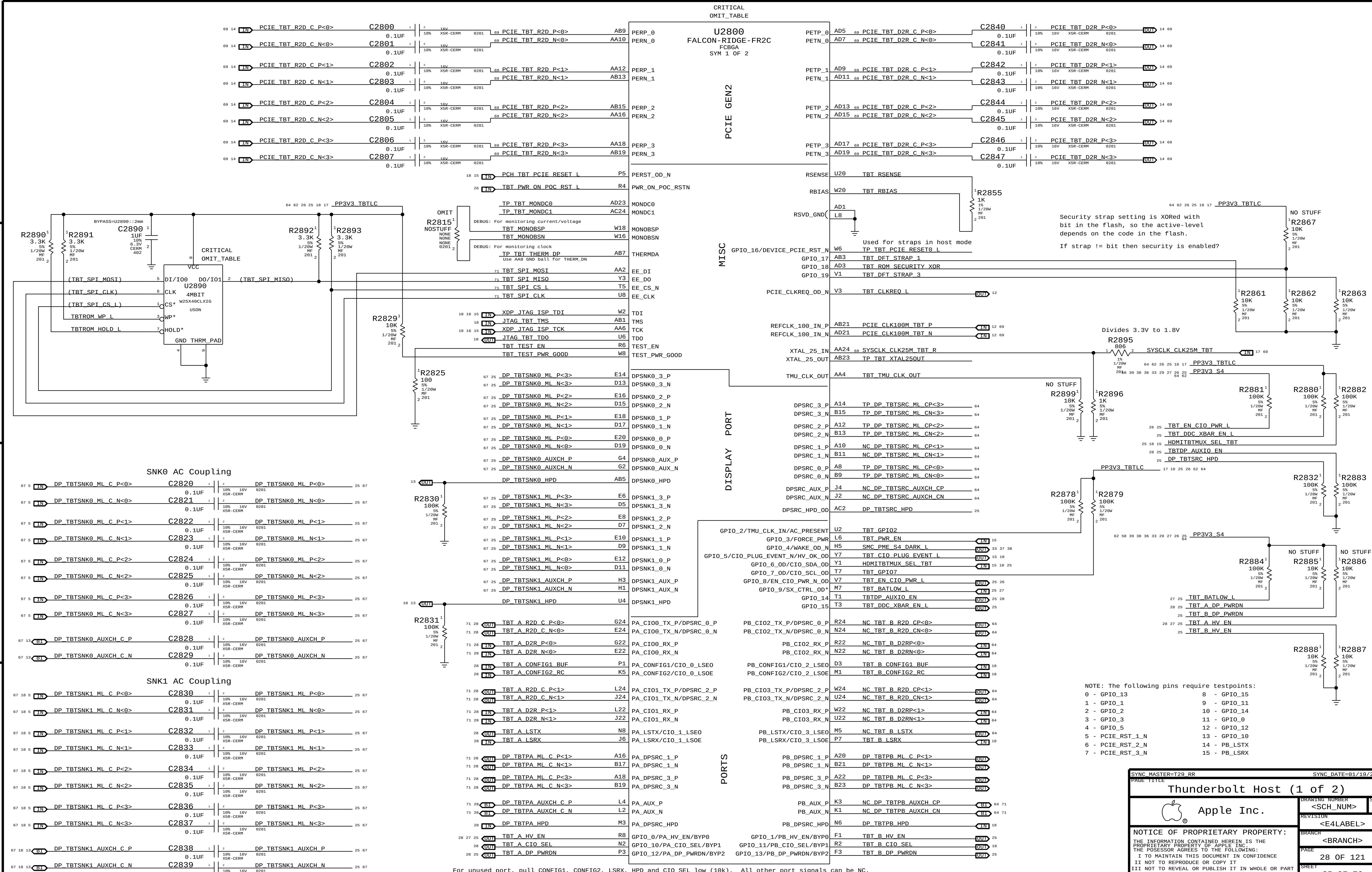

SYNC_MASTER=T29_RR SYNC_DATE=01/19/2013

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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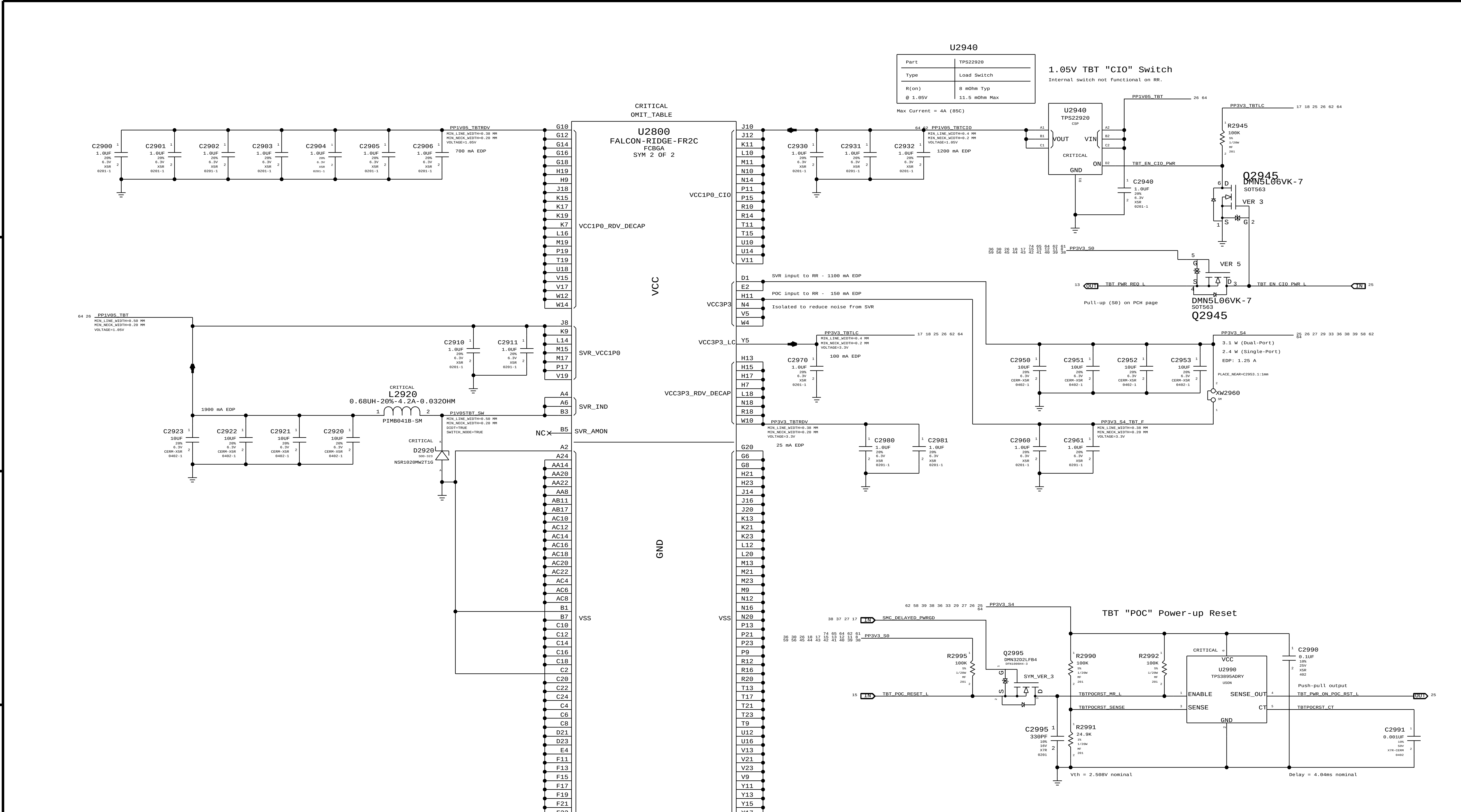

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC_MASTER=T29_RR SYNC_DATE=12/17/2012

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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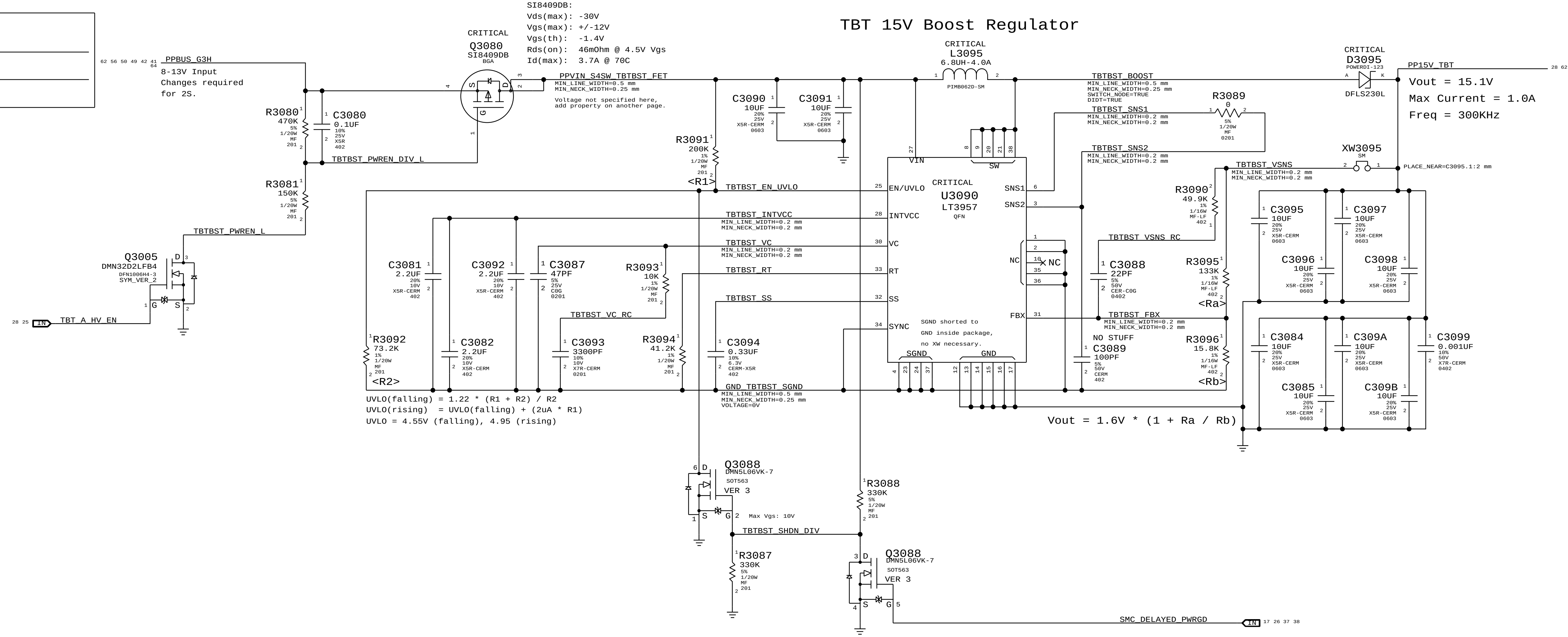
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)

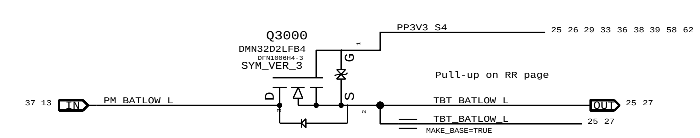
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

TBT 15V Boost Regulator



BATLOW# Isolation



SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE: TBT Power Support

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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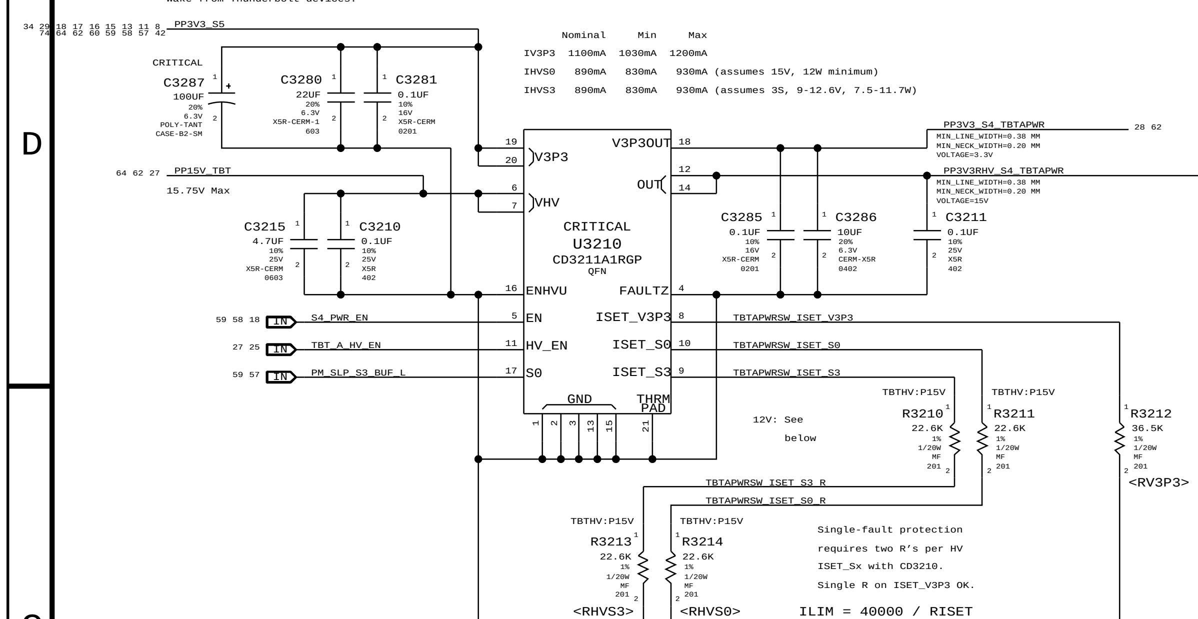
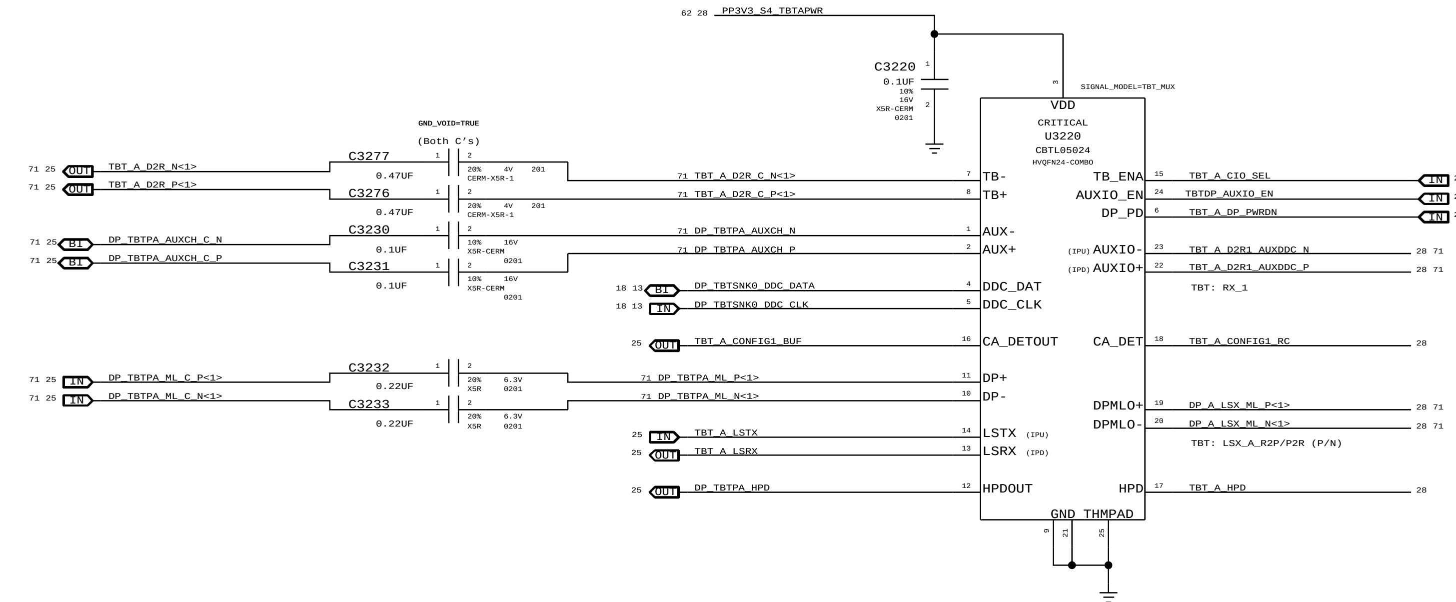
3.3V/HV Power MUX



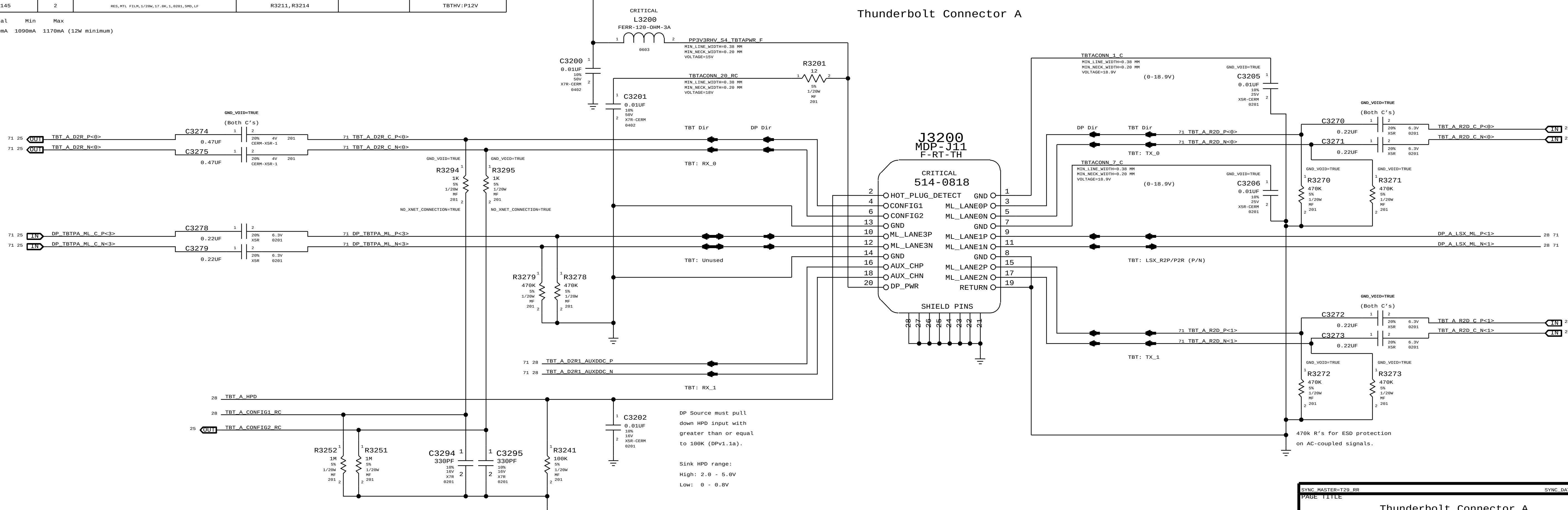
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3210,R3213		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3211,R3214		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)



Thunderbolt Connector A



SYNC_MASTER=T29_RR SYNC_DATE=10/26/2012

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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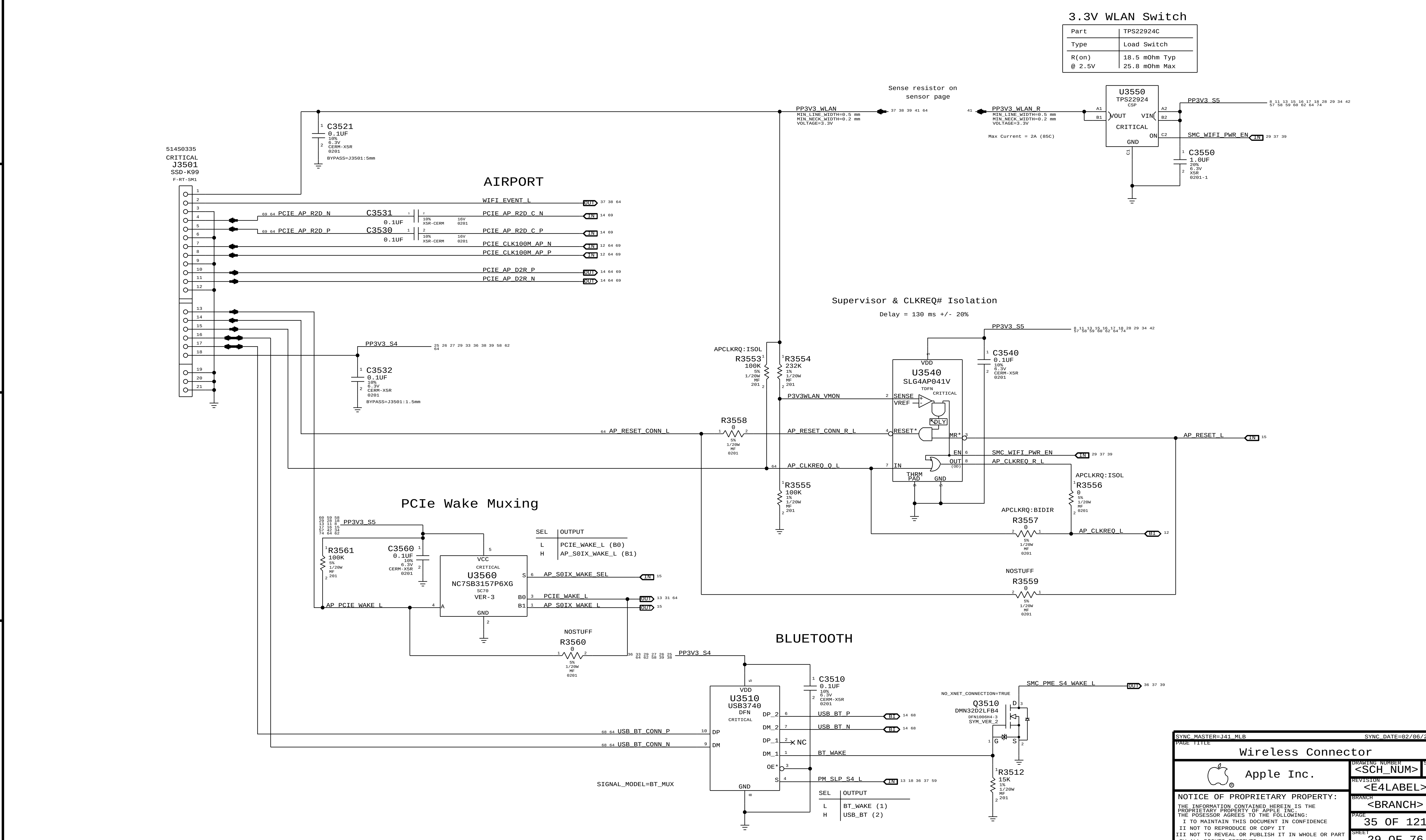

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SEL	OUTPUT
L	PCIE_WAKE_L (B0)
H	AP_S0IX_WAKE_L (B1)

SEL	OUTPUT
L	BT_WAKE (1)
H	USB_BT (2)

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

Wireless Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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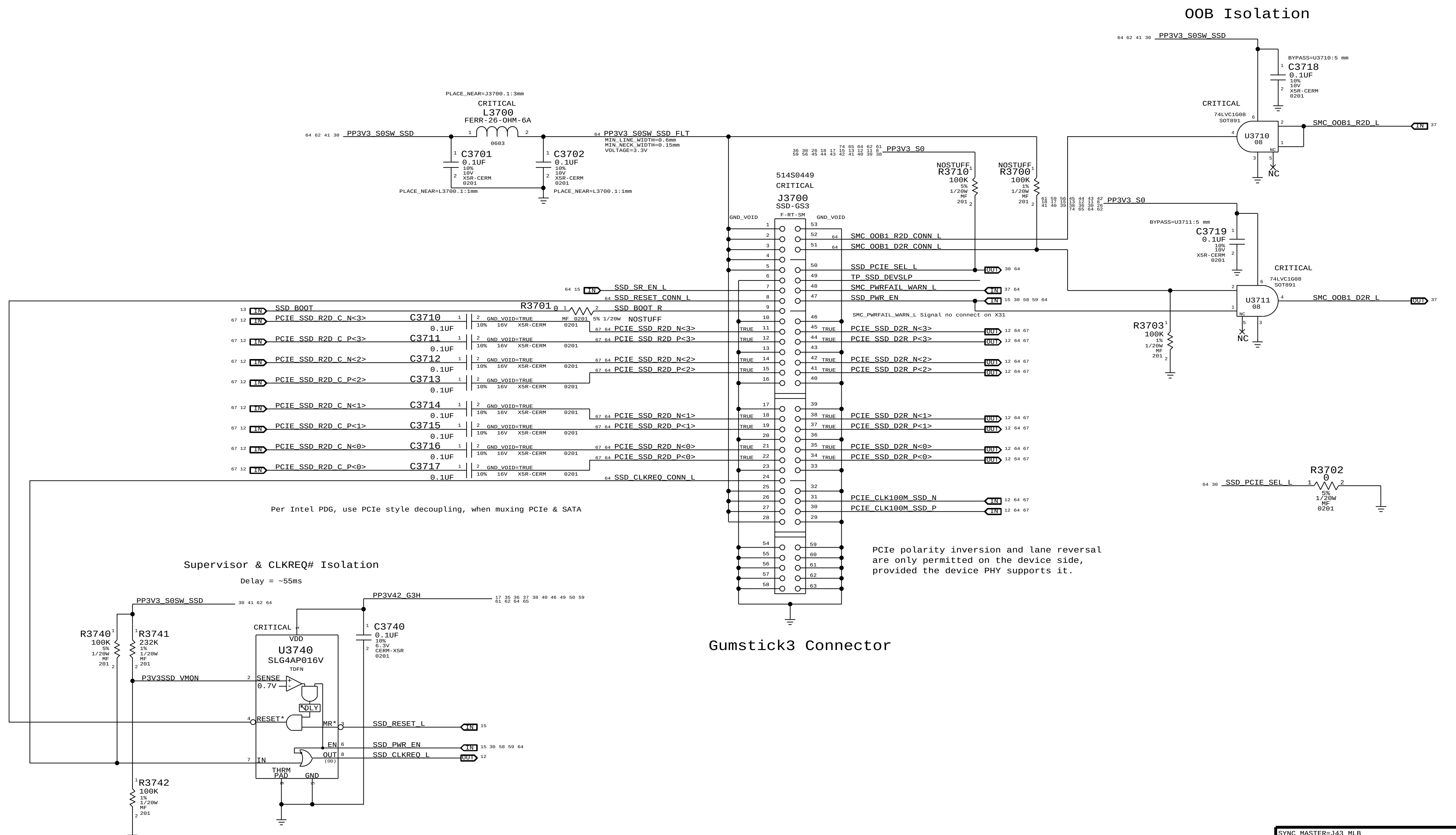

SYNC_MASTER=J43_MLB SYNC_DATE=02/20/2013

PAGE TITLE

SSD Connector

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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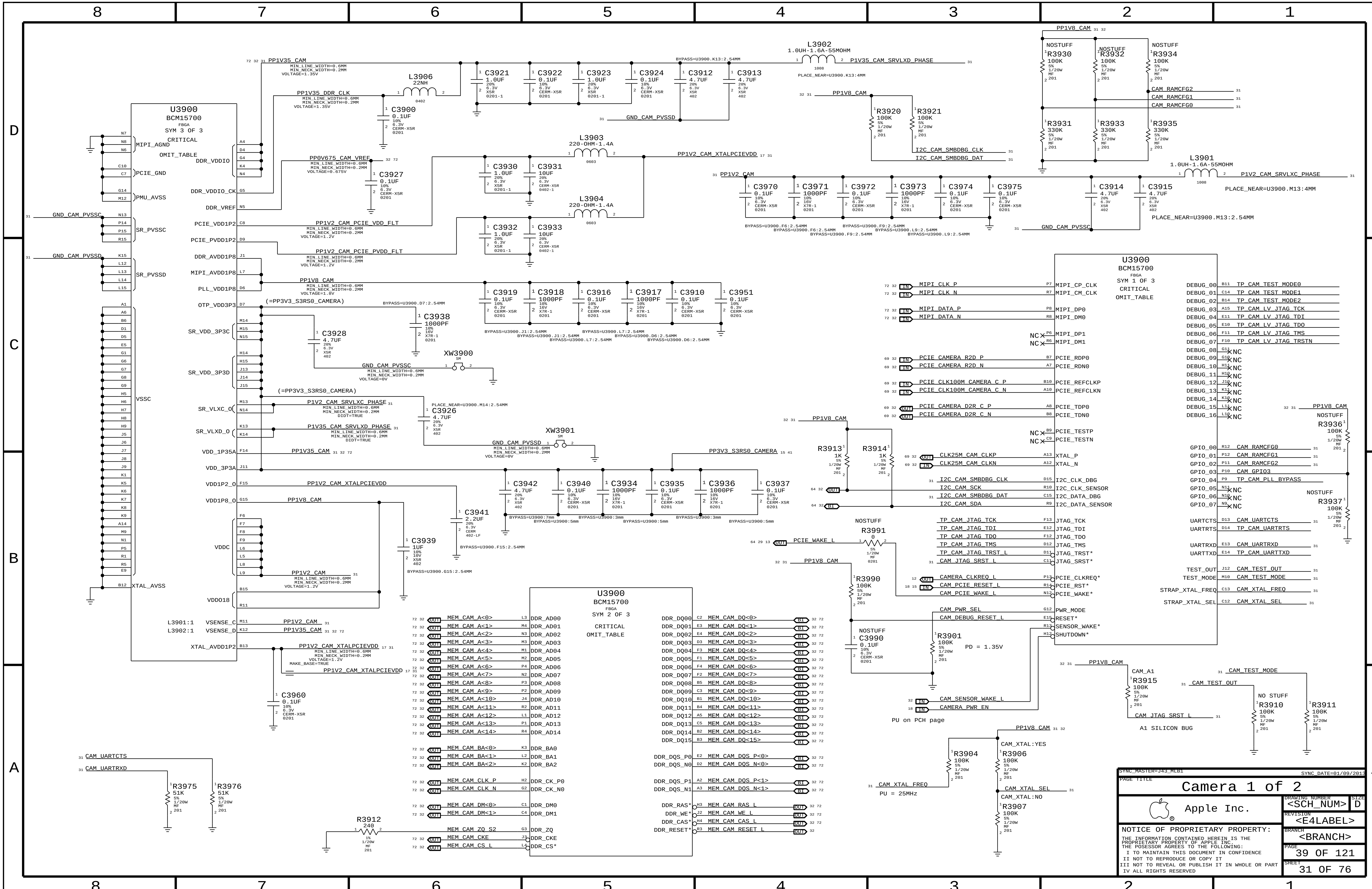

U3900
BCM15700
FBGA
SYM 3 OF 3
CRITICAL
OMIT_TABLE
U3900
BCM15700
FBGA
SYM 2 OF 3
CRITICAL
OMIT_TABLE
U3900
BCM15700
FBGA
SYM 1 OF 3
CRITICAL
OMIT_TABLE
PP1V35_CAM
PP1V8_CAM
PP1V2_CAM
GND_CAM_PVSSC
GND_CAM_PVSSD
PP1V35_CAM_SRVLXD_PHASE
P1V2_CAM_SRVLXC_PHASE
PP1V2_CAM_XTALPCIEVDD
PP3V3_S3RS0_CAMERA
PU on PCH page
PU = 25MHz
PD = 1.35V
A1 SILICON BUG
SYNC_MASTER=J43_MLB1
SYNC_DATE=01/09/2013
PAGE TITLE
Camera 1 of 2
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
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REVISION
<E4LABEL>
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<BRANCH>
PAGE
39 OF 121
SHEET
31 OF 76
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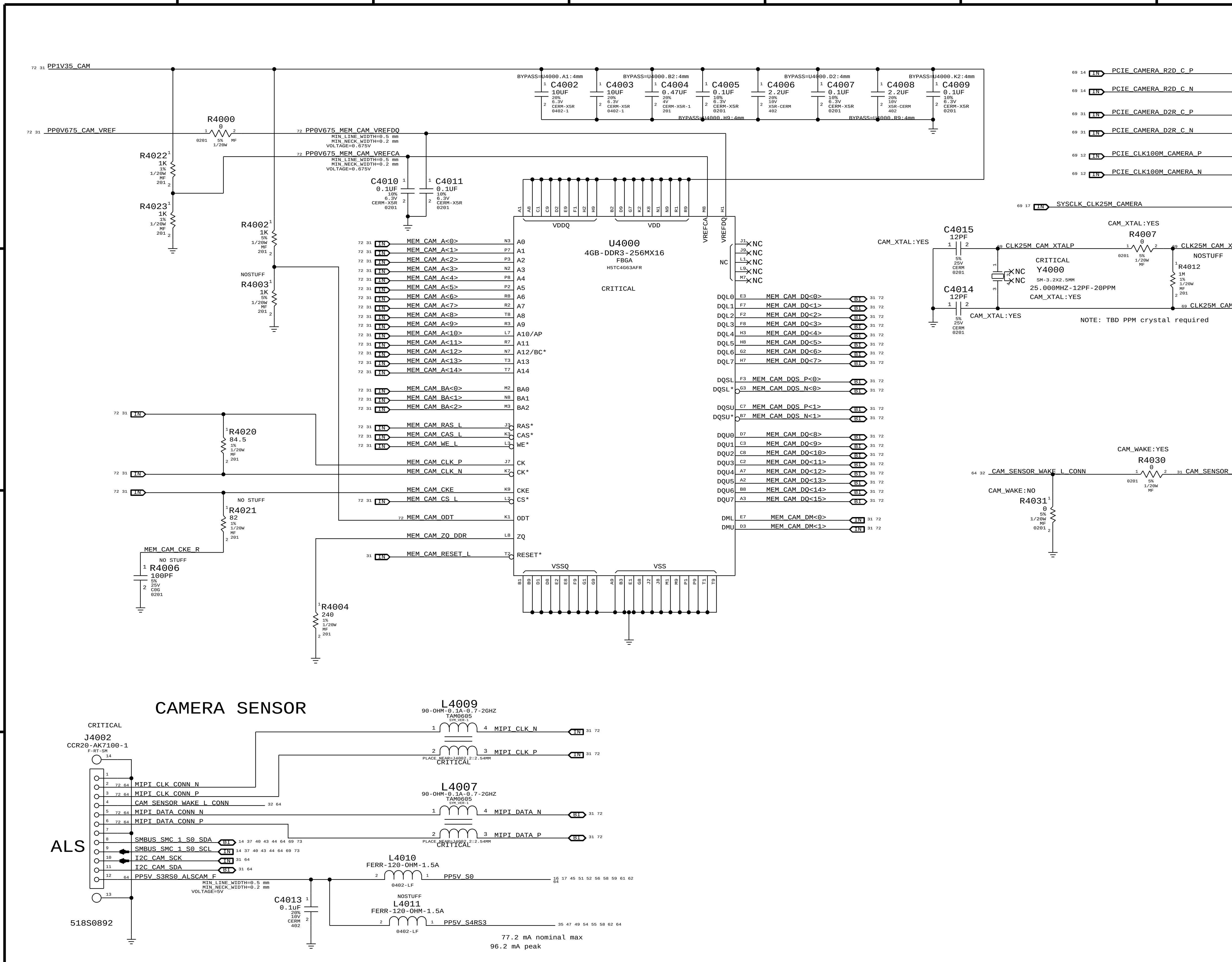

SYNC_MASTER=J43_MLB SYNC_DATE=09/14/2012

PAGE TITLE

Camera 2 of 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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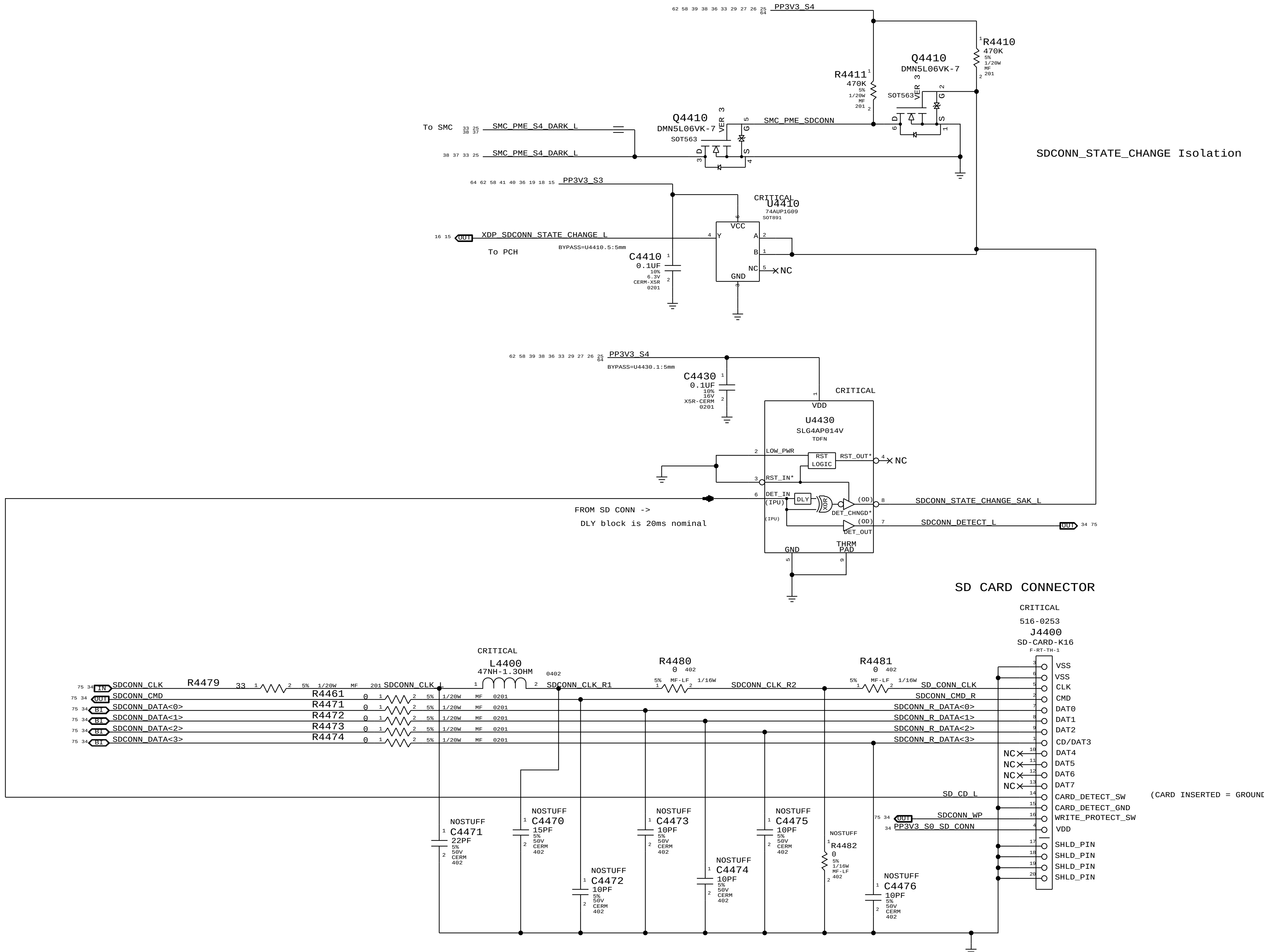

SD CARD

SYNC_MASTER=MASTER SYNC_DATE=07/01/2011

PAGE TITLE

SD READER CONNECTOR

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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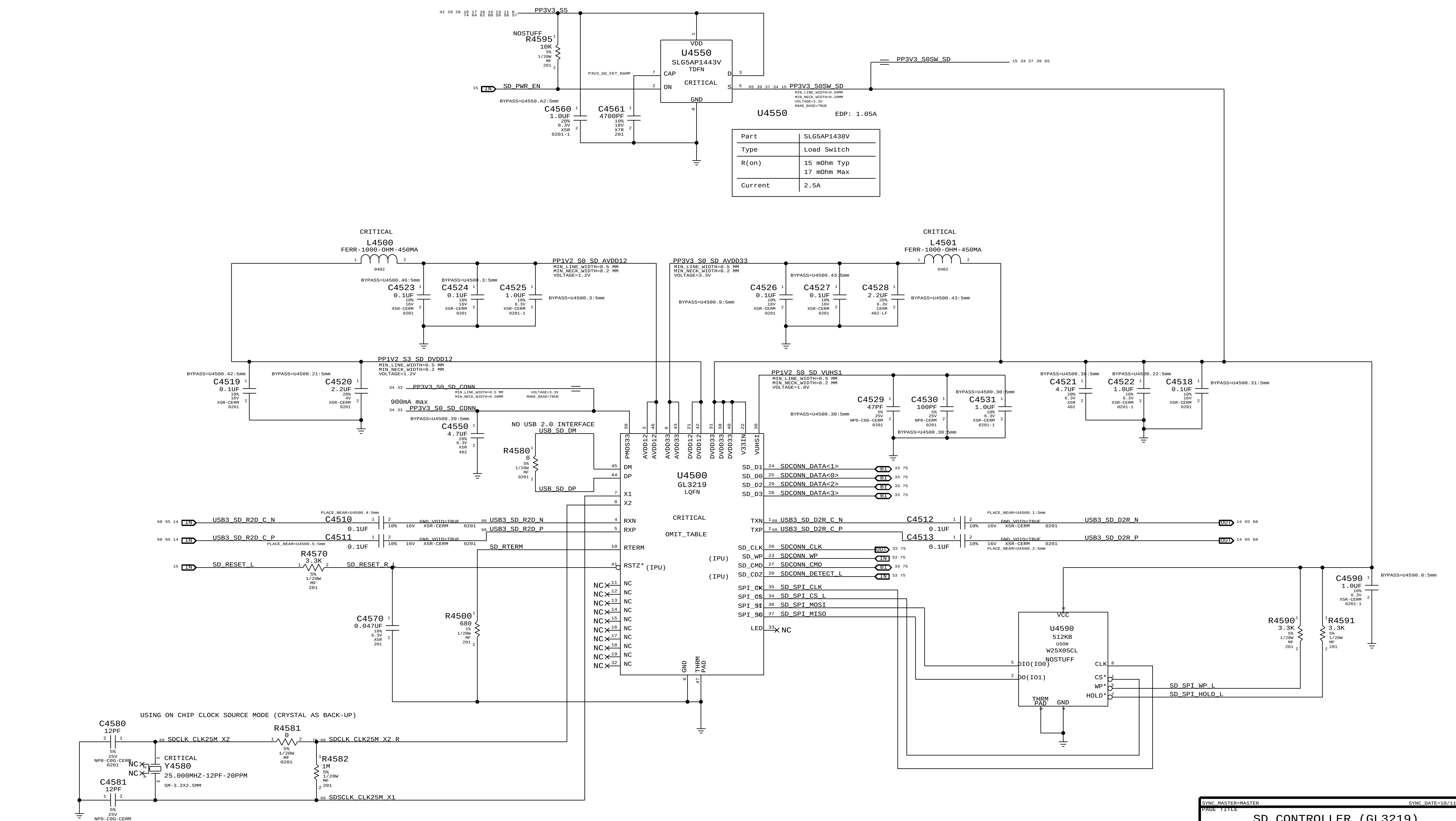
3.3V S3 SD Card Switch



U4550

Part	SLG5AP1438V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

SYNC_MASTER=MASTER SYNC_DATE=10/11/2010

PAGE TITLE

SD CONTROLLER (GL3219)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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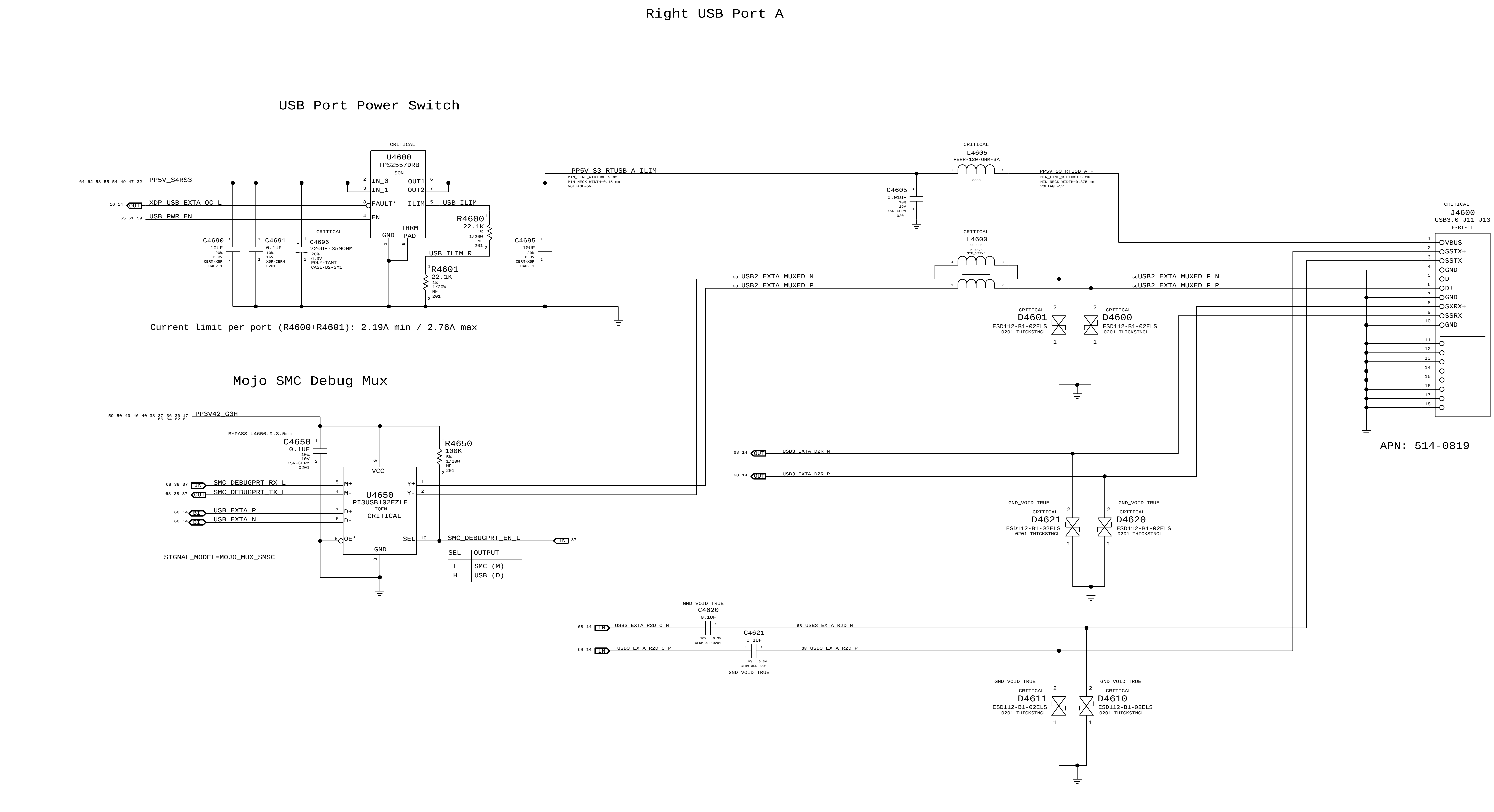
Right USB Port A



SYNC_MASTER=J43_MLB SYNC_DATE=02/20/2013

PAGE TITLE: External A USB3 Connector

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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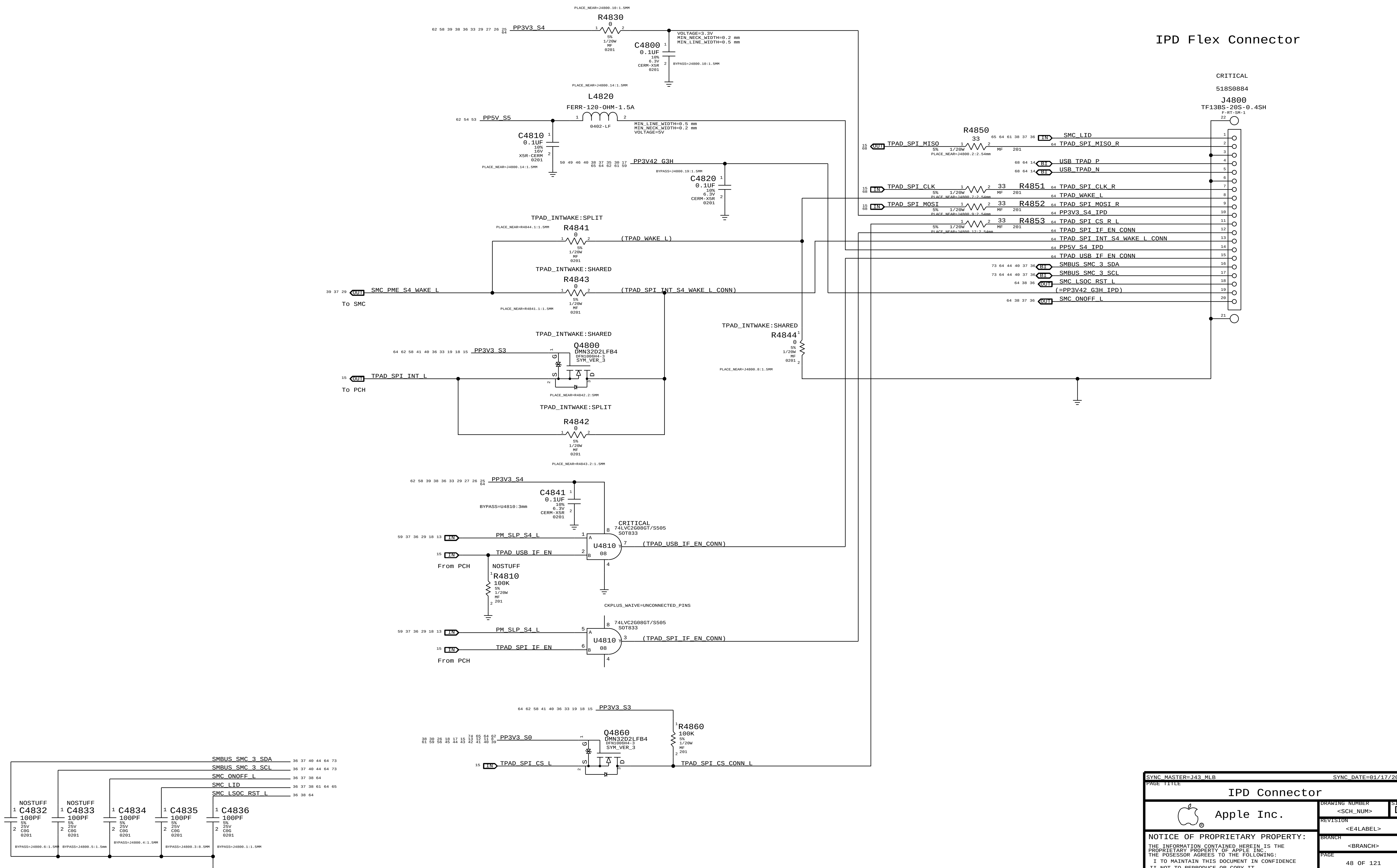
IPD Flex Connector



SYNC_MASTER=J43_MLB SYNC_DATE=01/17/2013

PAGE TITLE

IPD Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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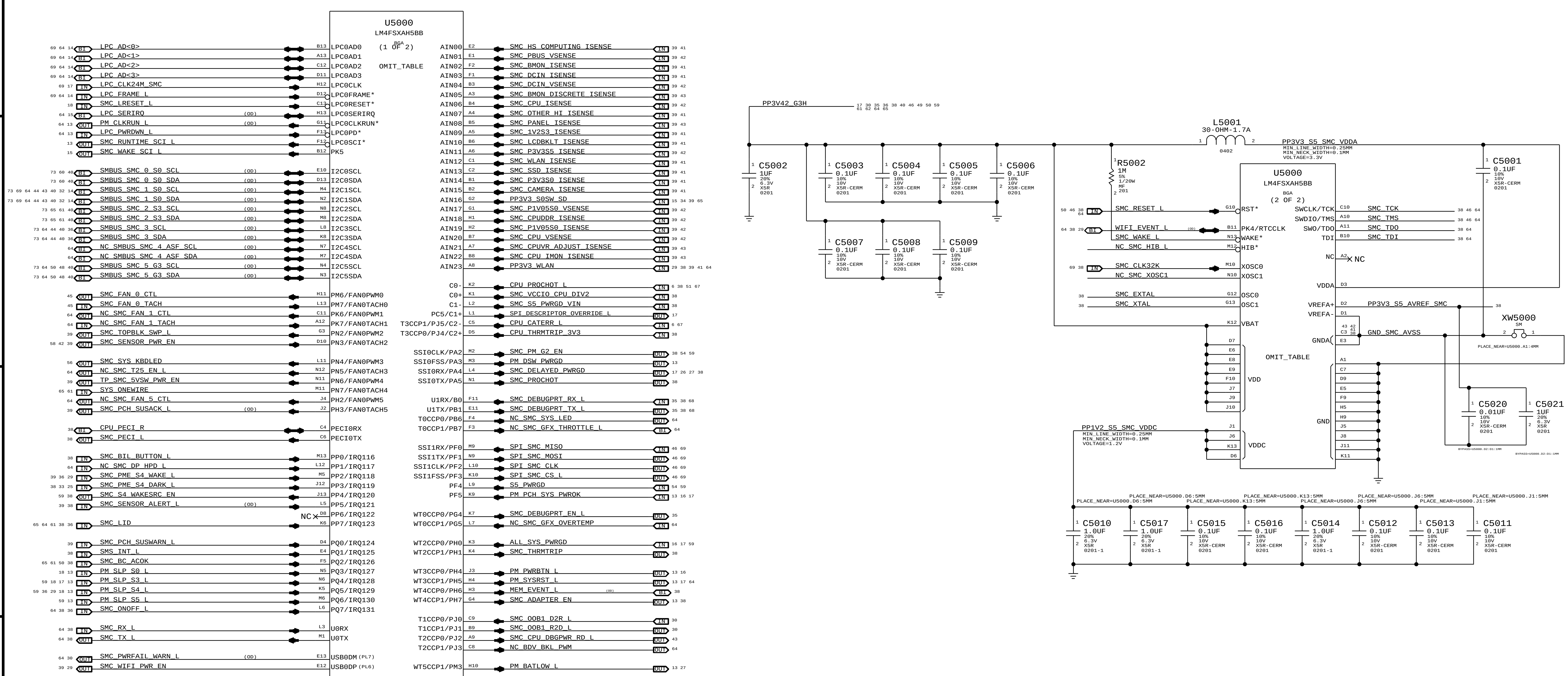

NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

SMC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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BRANCH <BRANCH>

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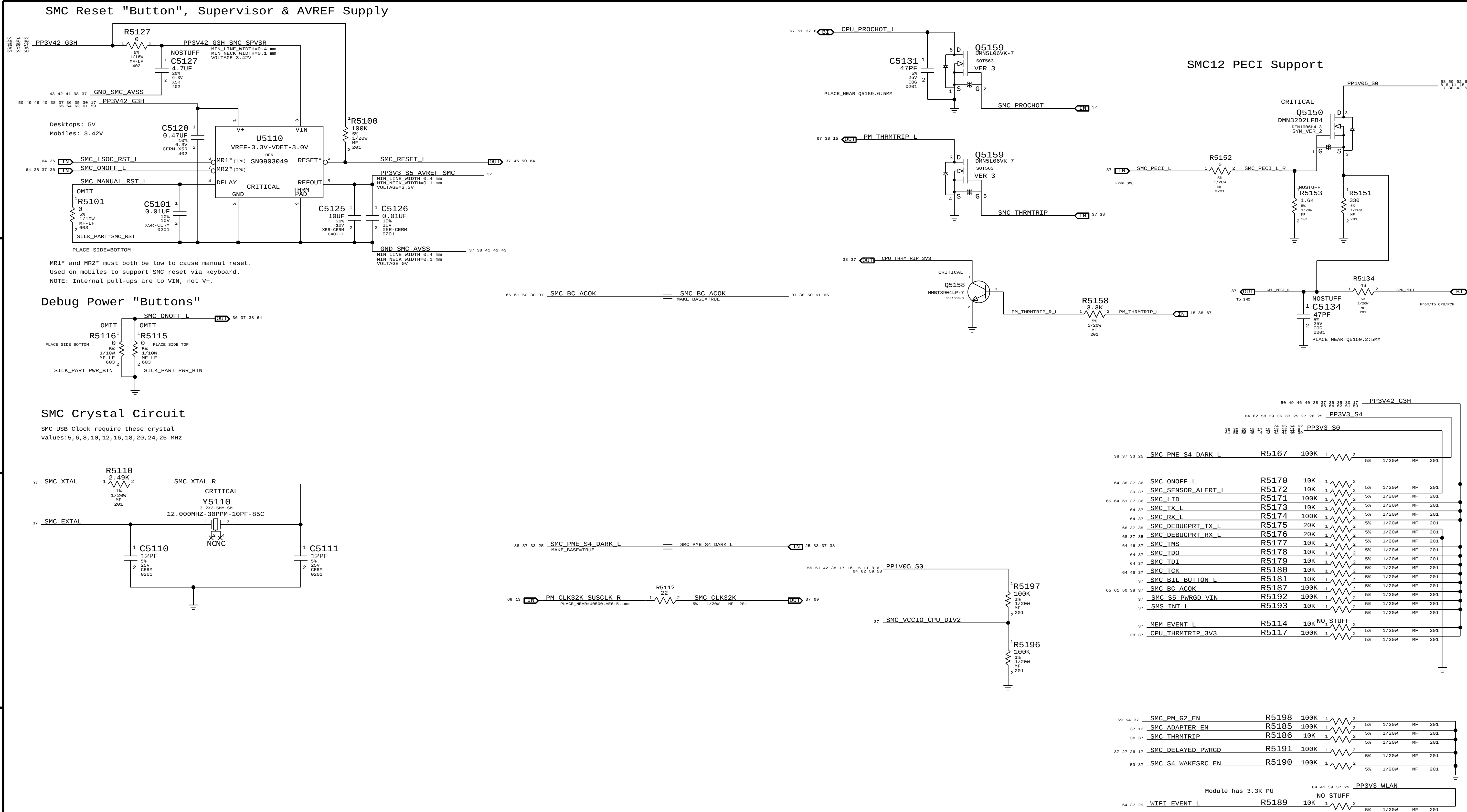

SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE

SMC Shared Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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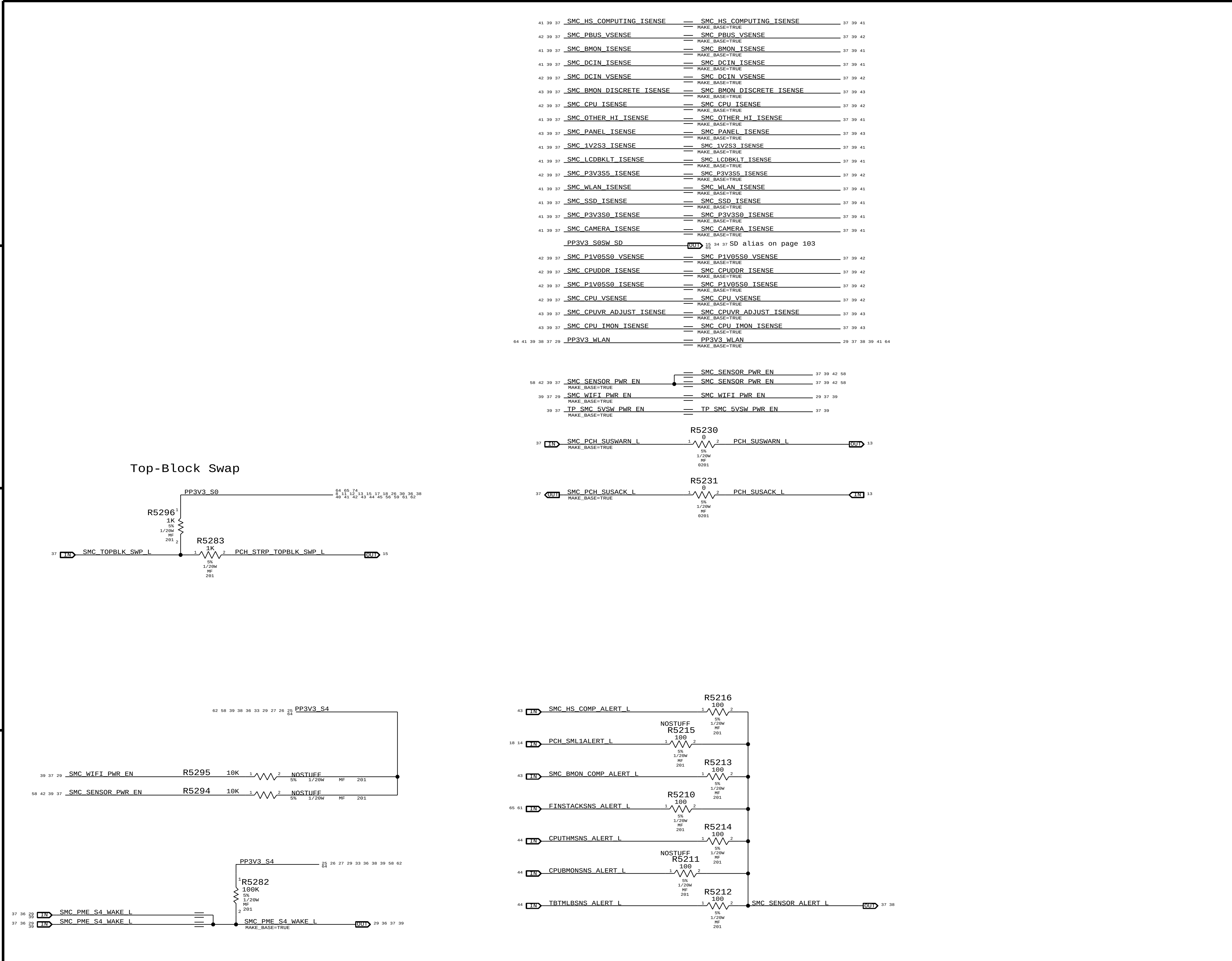

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

SMC Project Support

Apple Inc.

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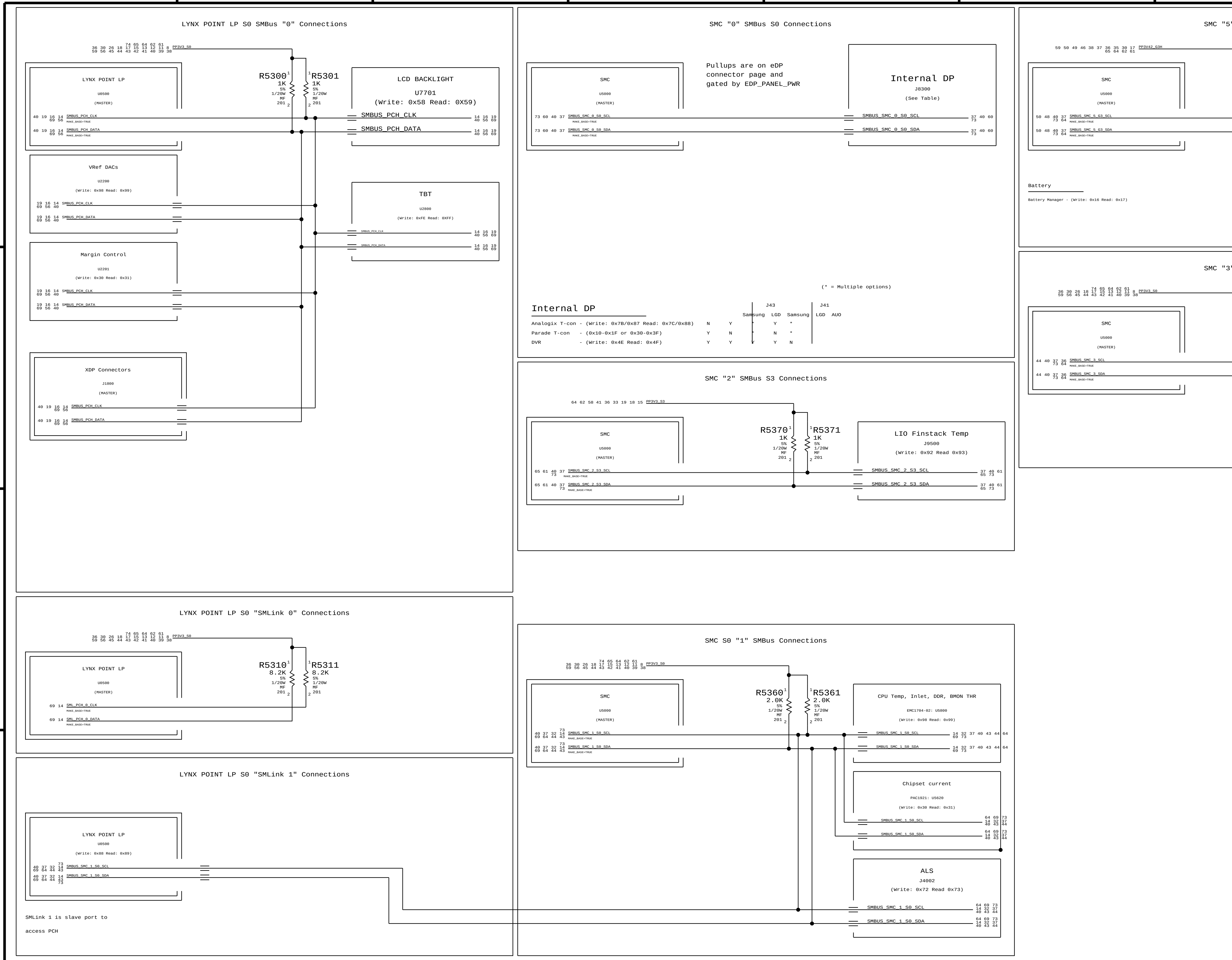

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

SMBus Connections

Apple Inc.

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REVISION <E4LABEL>

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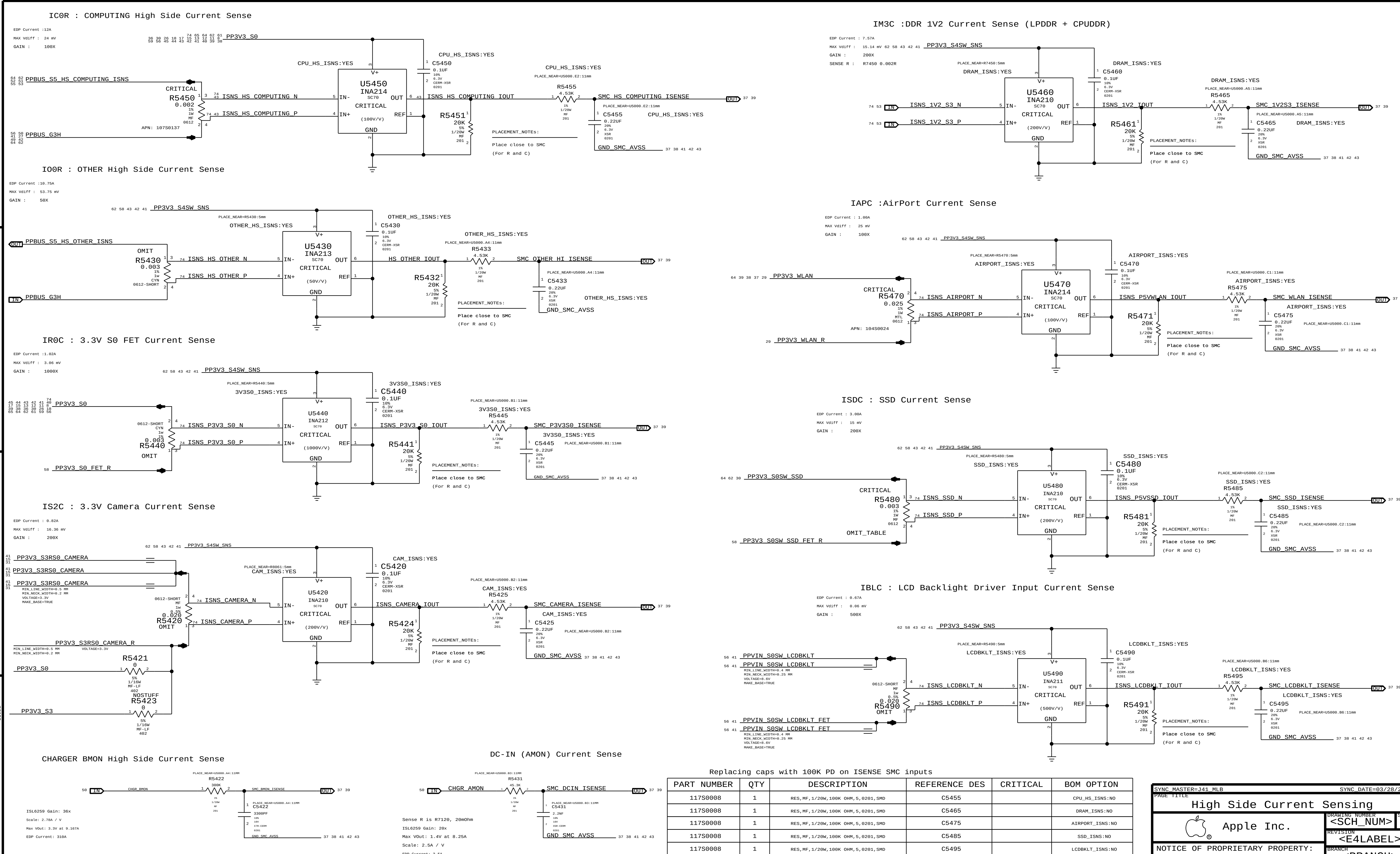
SHEET 40 OF 76



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
107S0248	1	RES,SENSE,0.003OHM,1W,4-TERM,1%,0612,TFT	R5480	CRITICAL	

Replacing caps with 100K PD on ISENSE SMC inputs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5455		CPU_HS_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5465		DRAM_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5475		AIRPORT_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5485		SSD_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5495		LCDBKLT_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5433		OTHER_HS_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5425		CAM_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5445		3V3S0_ISNS:NO

SYNC_MASTER=J41_MLB SYNC_DATE=03/28/2013

PAGE TITLE

High Side Current Sensing

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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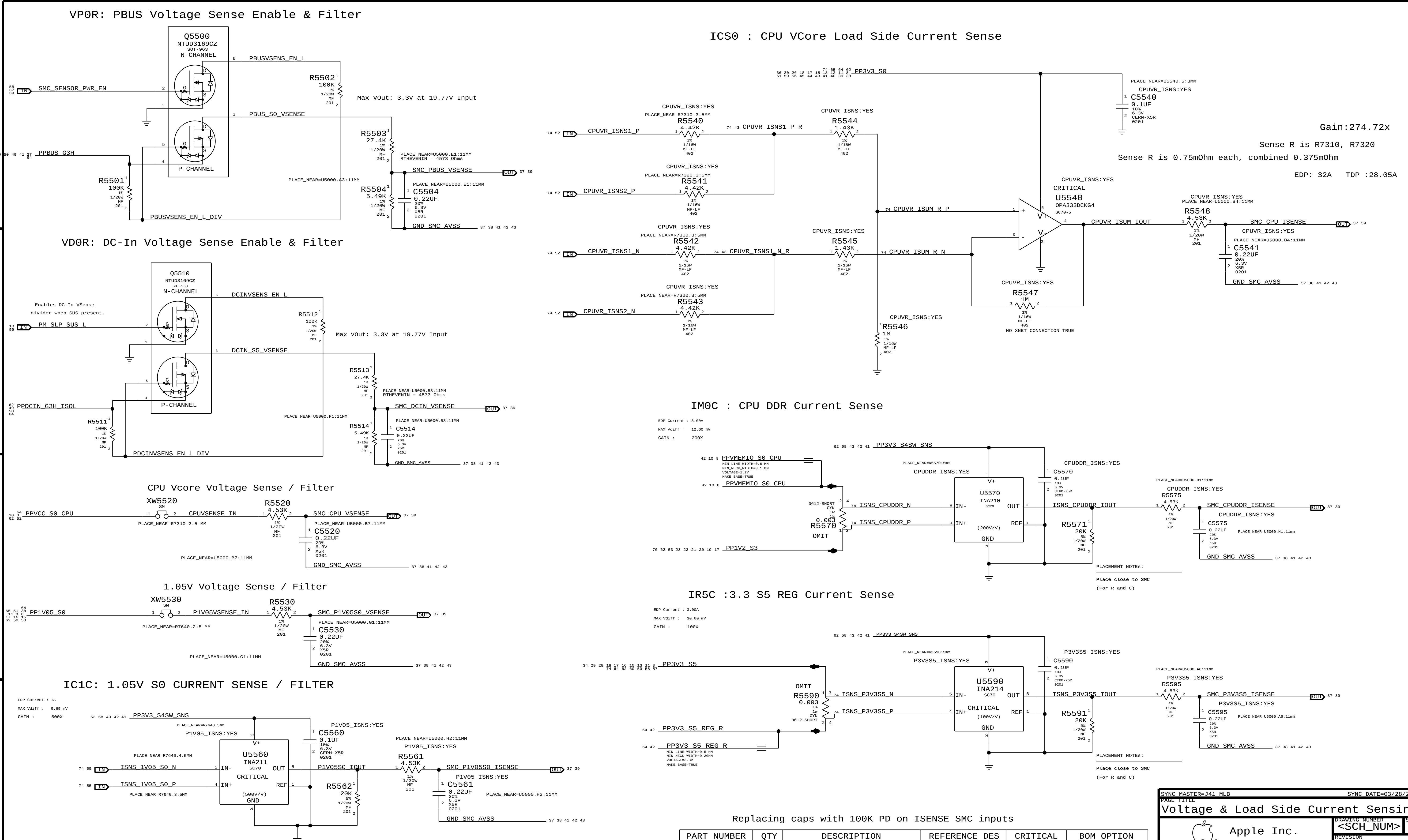

Replacing caps with 100K PD on ISENSE SMC inputs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5541		CPUVR_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5561		P1V05_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5595		P3V3S5_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5575		CPUDDR_ISNS:NO

SYNC_MASTER=J41_MLB SYNC_DATE=03/28/2013

PAGE TITLE

Voltage & Load Side Current Sensing

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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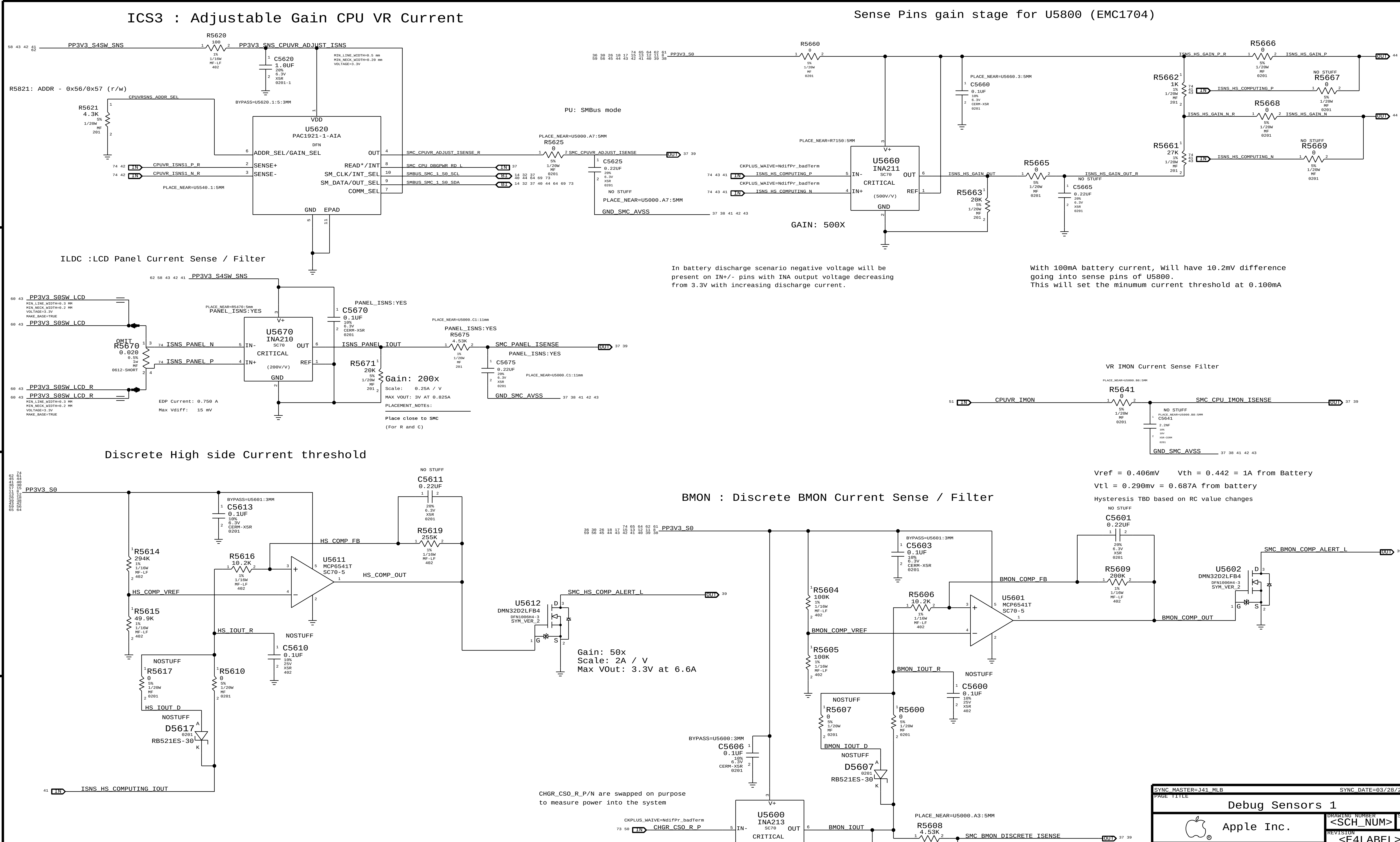

Replacing caps with 100K PD on ISENSE SMC inputs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5675		PANEL_ISNS:NO

SYNC_MASTER=J41_MLB SYNC_DATE=03/28/2013

PAGE TITLE

Debug Sensors 1

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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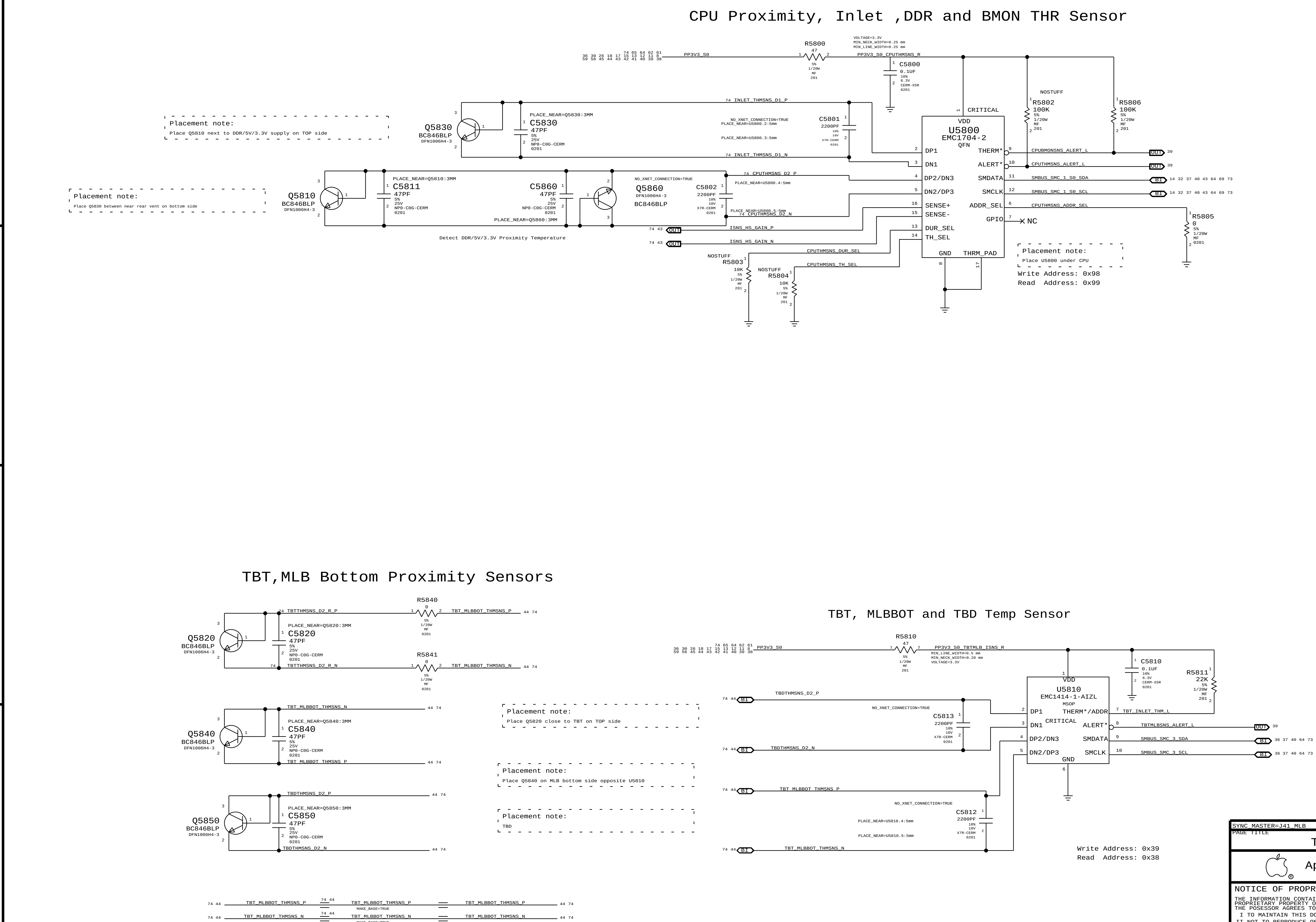
SHEET 43 OF 76



SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE

Thermal Sensors

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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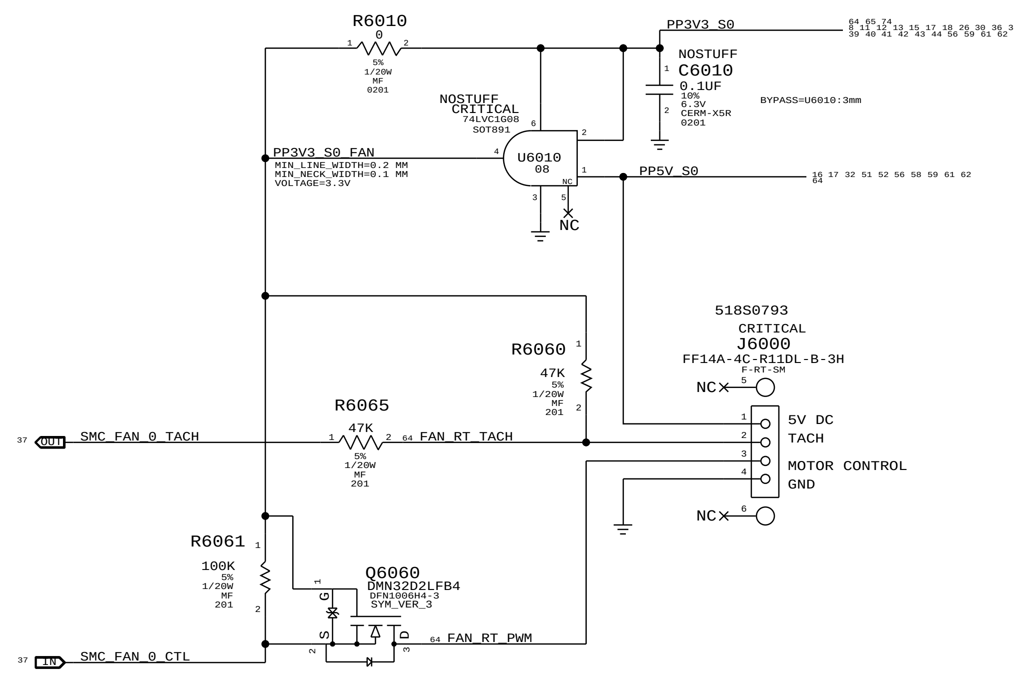
FAN CONNECTOR



SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE
Fan

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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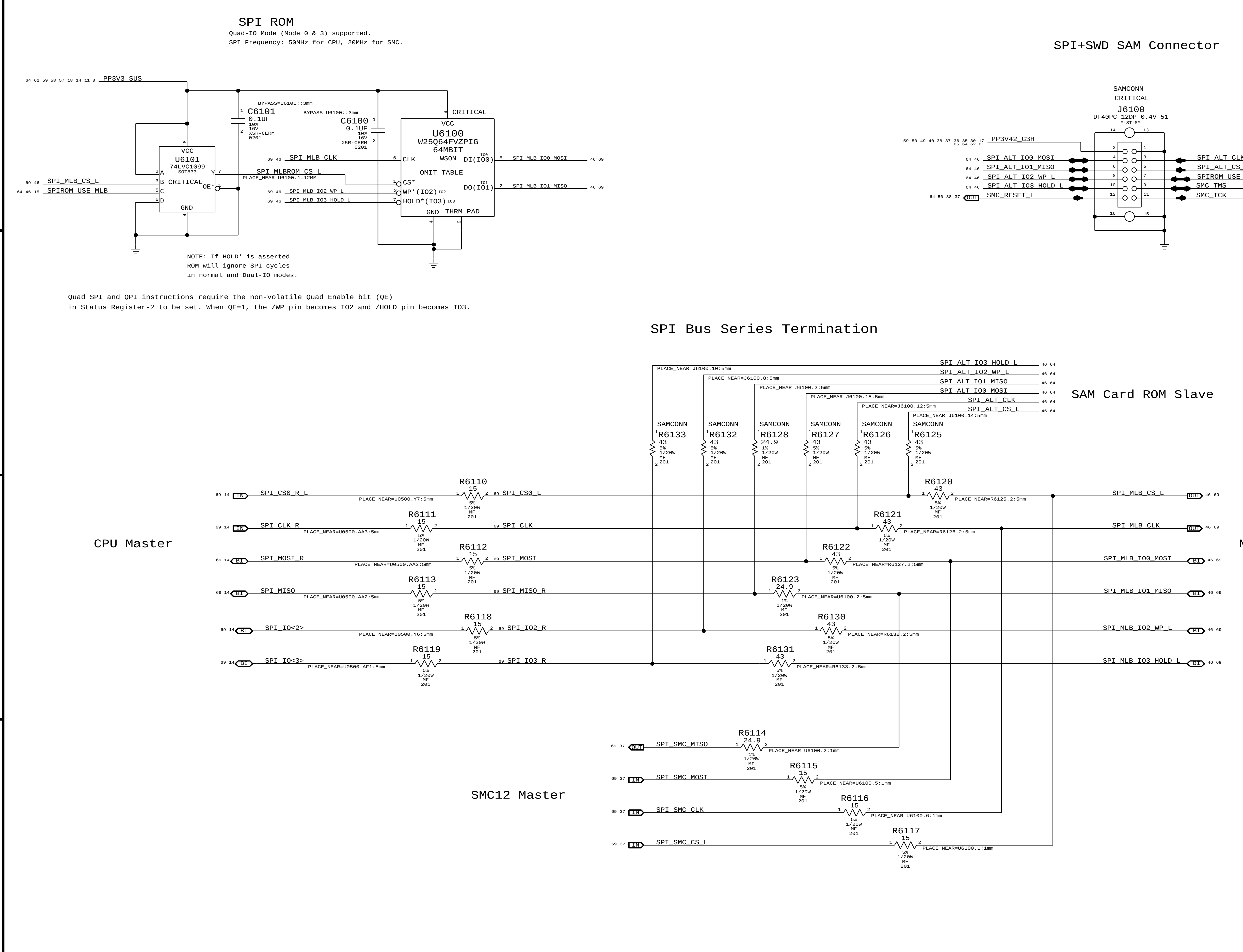


SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/09/2013

PAGE TITLE: SPI Debug Connector

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

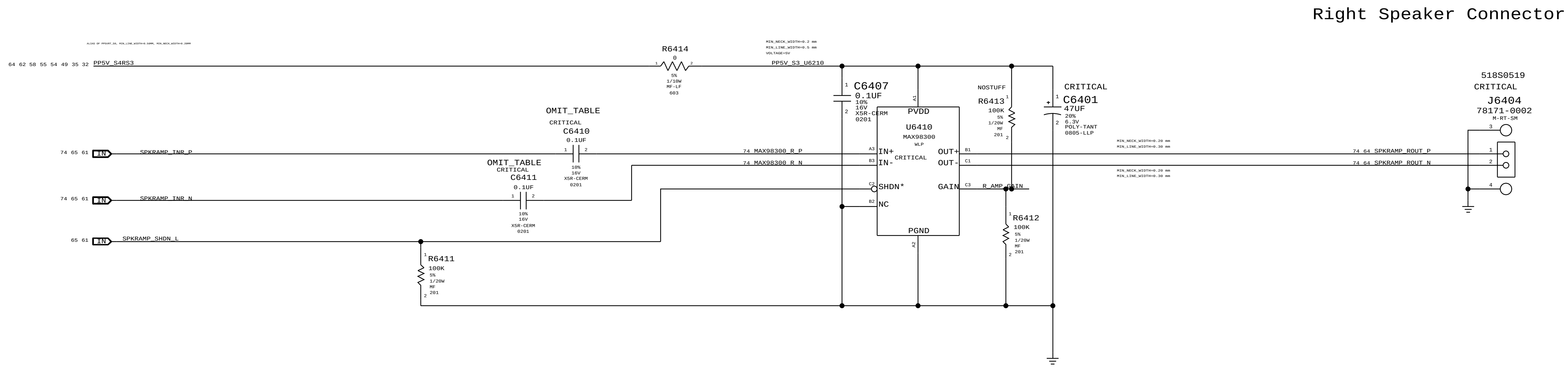
REVISION: <E4LABEL>

BRANCH: <BRANCH>

SPEAKER AMPLIFIERS

APN:353S2888

SPEAKER LOWPASS	80 HZ < FC < 132 HZ
GAIN	6DB



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
132S0460	2	CAP,CER,X5R,0.1UF,10%,16V,0201,MURATA	C6410,C6411	CRITICAL	

SYNC_MASTER=J41_MLB SYNC_DATE=04/26/2013

PAGE TITLE

Audio: Speaker Amp

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

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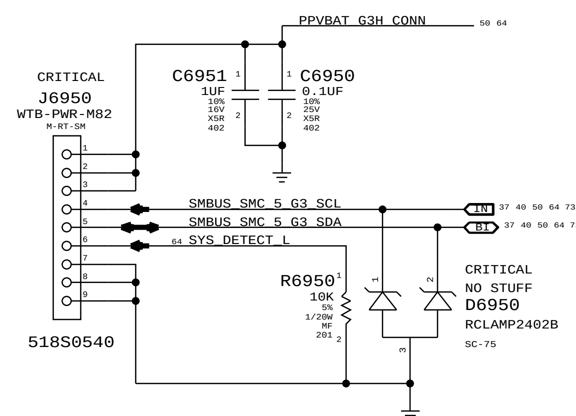
13" SPECIFIC

Battery Connector



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PAGE TITLE: Battery Connector

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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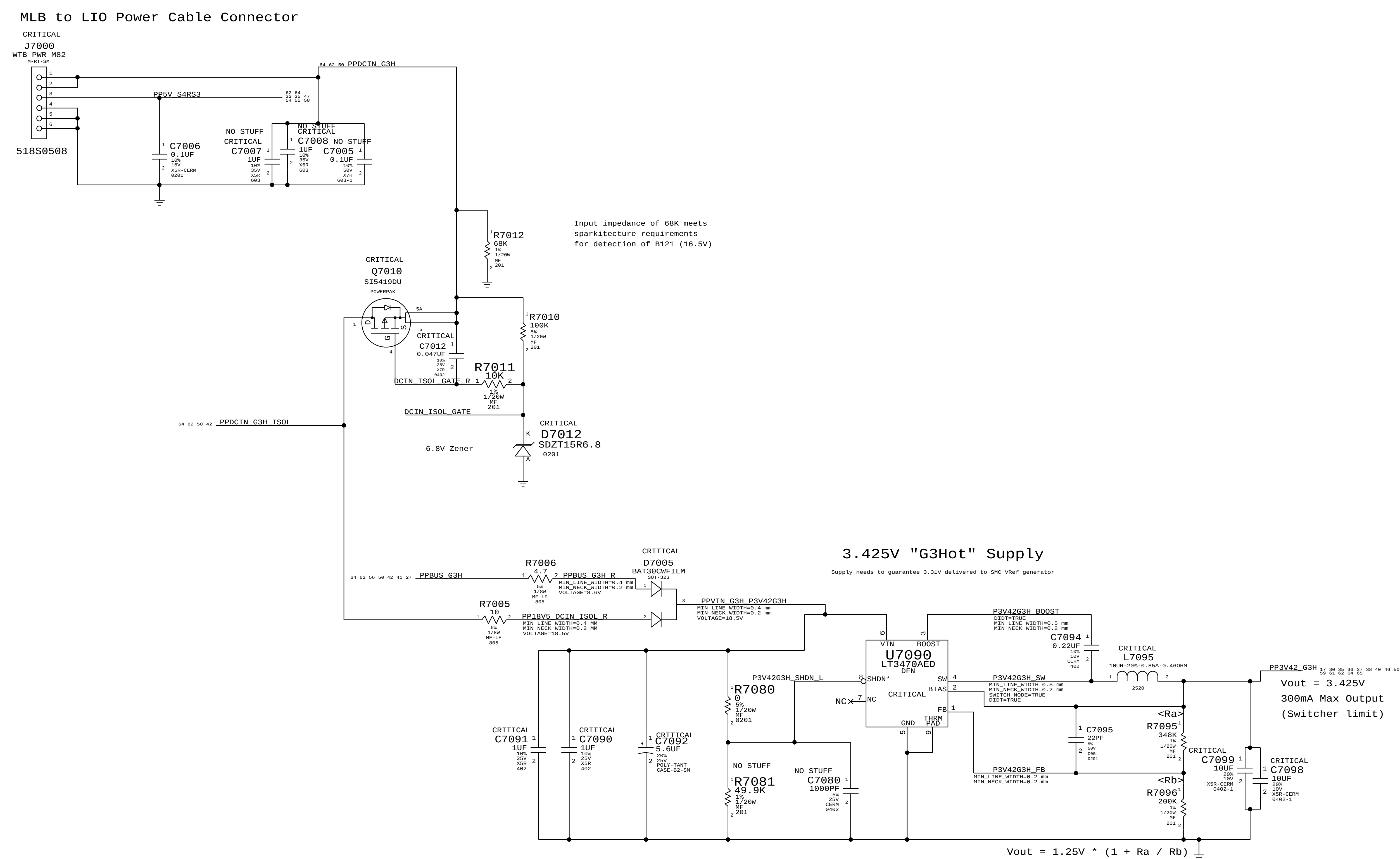
SHEET: 48 OF 76



SYNC_MASTER=J43_MLB SYNC_DATE=09/13/2012

PAGE TITLE

DC-In & G3H Supply

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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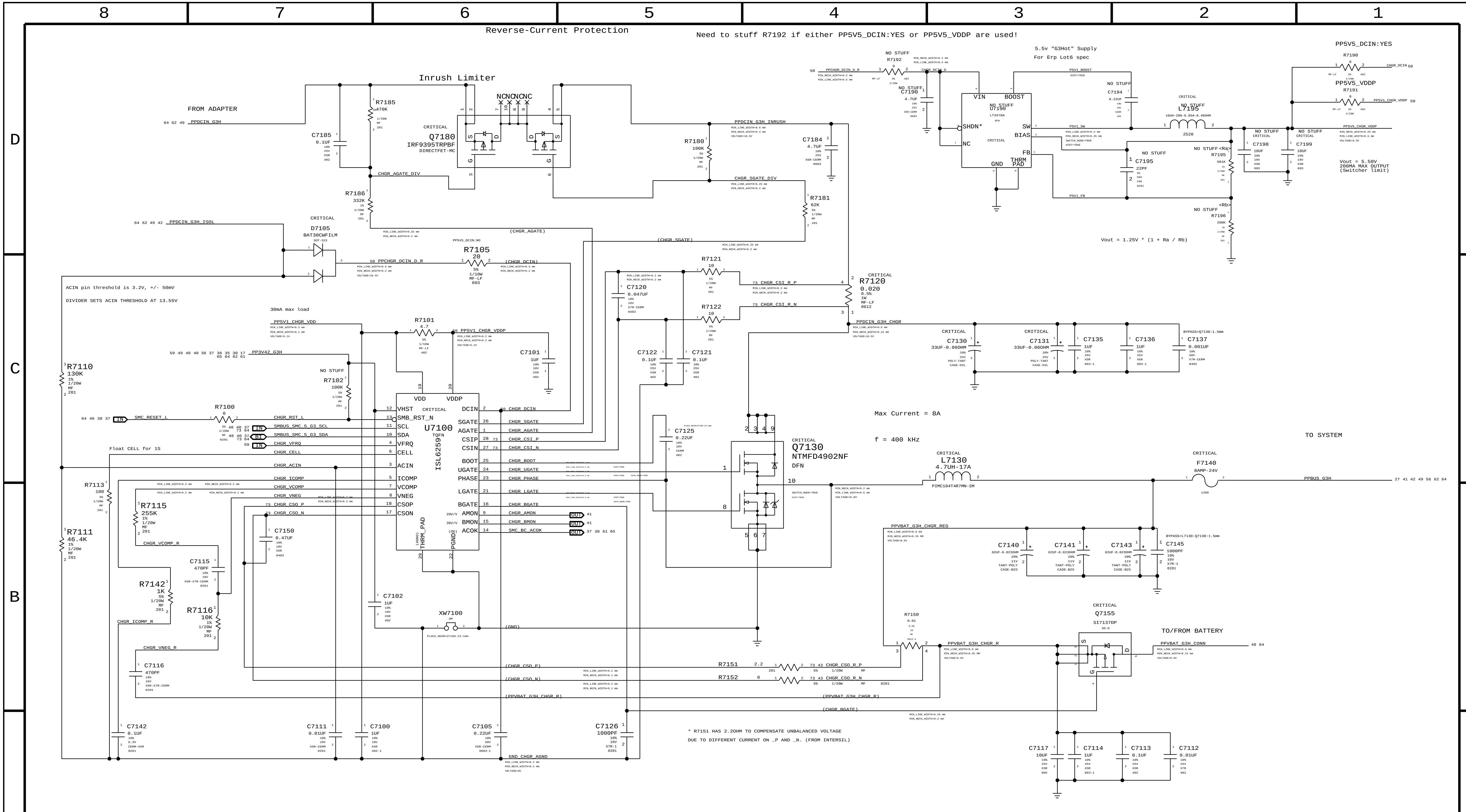

SYNC_MASTER=J43_MLB SYNC_DATE=09/14/2012

PAGE TITLE

PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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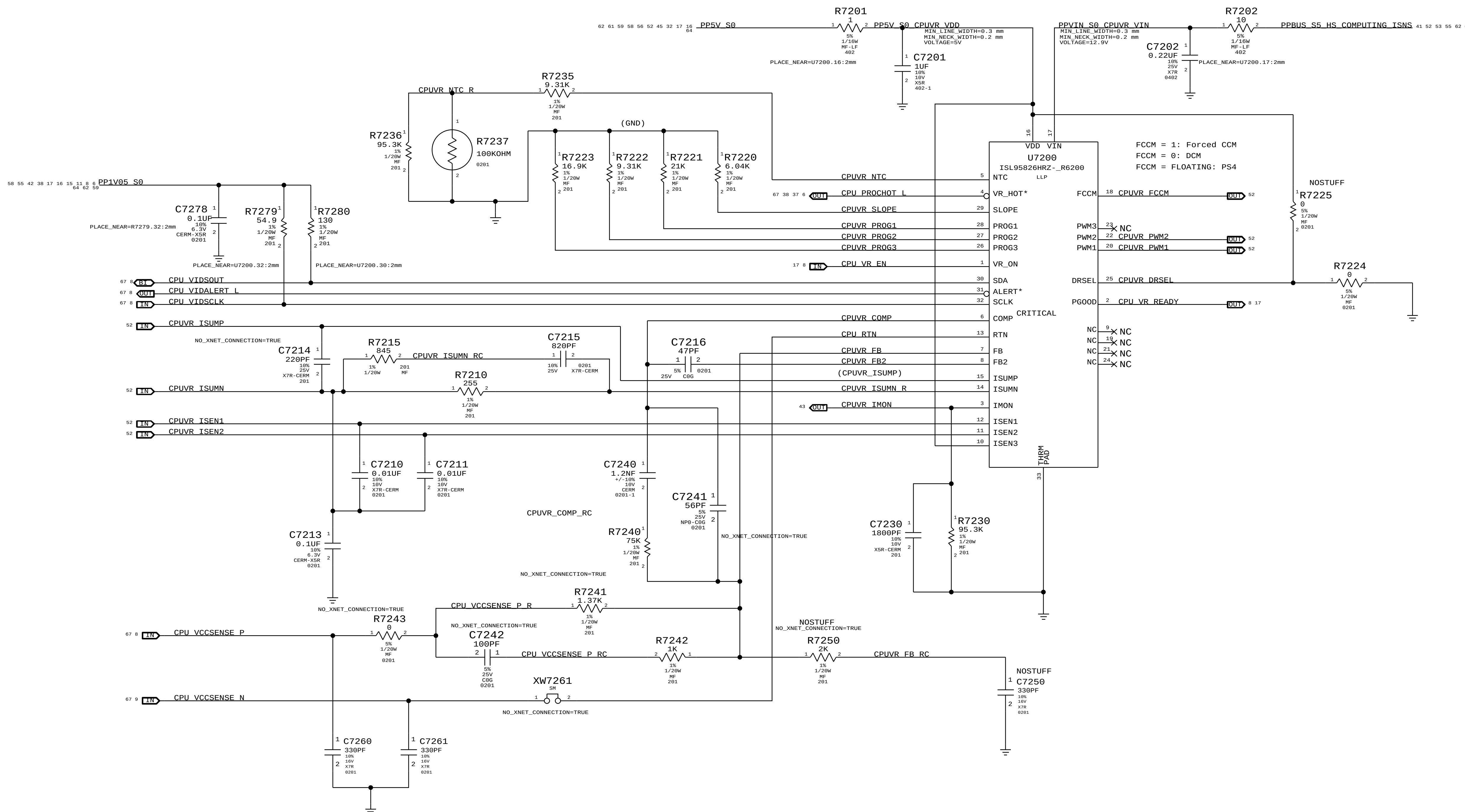

SYNC_MASTER=J43_MLB SYNC_DATE=10/09/2012

PAGE TITLE

CPU VR12.6 VCC Regulator IC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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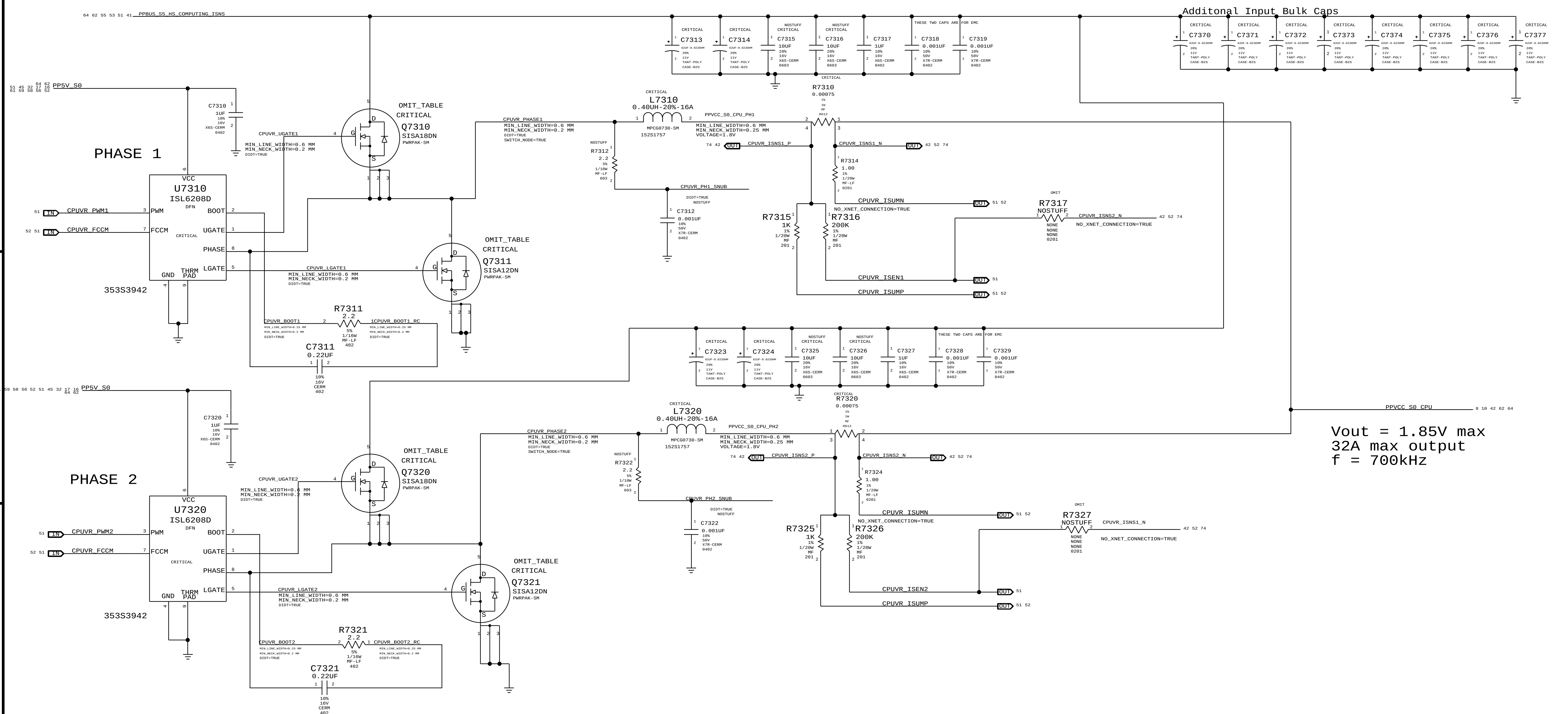
D

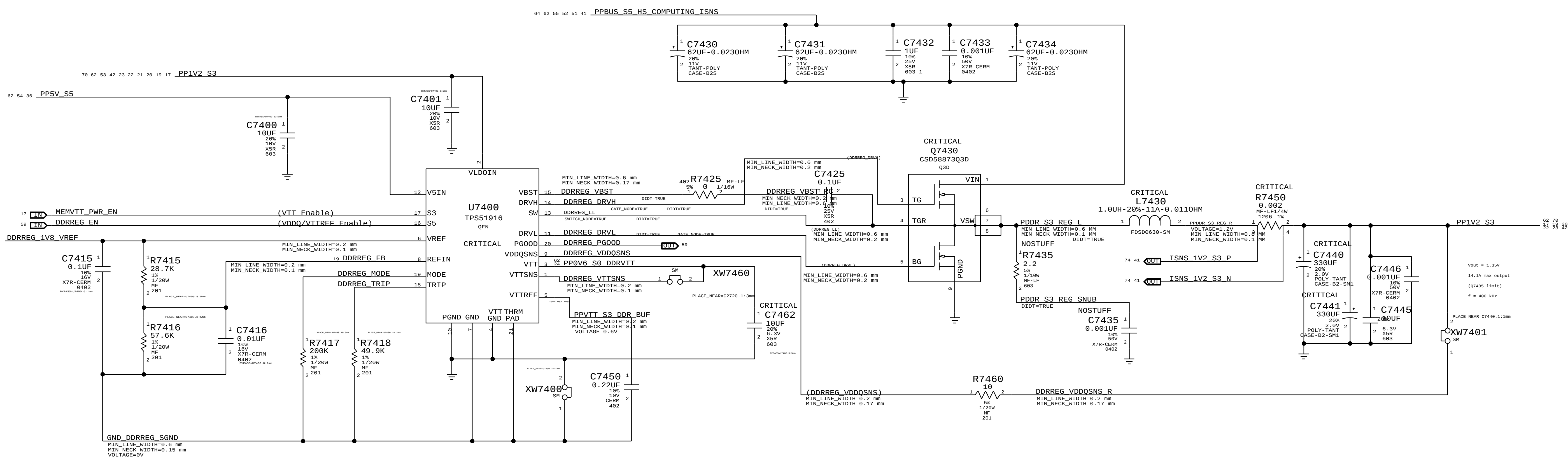
C

B

A



SYNC_MASTER=J41_MLB	SYNC_DATE=05/21/2013
PAGE TITLE	CPU VR12.5 VCC Power Stage
Apple Inc.	DRAWING NUMBER <SCH_NUM> SIZE D
	REVISION <E4LABEL>
	BRANCH <BRANCH>
	PAGE 73 OF 121
	SHEET 52 OF 76



SYNC_MASTER=J41_MLB SYNC_DATE=05/21/2013

PAGE TITLE

LPDDR3 Supply

Apple Inc.

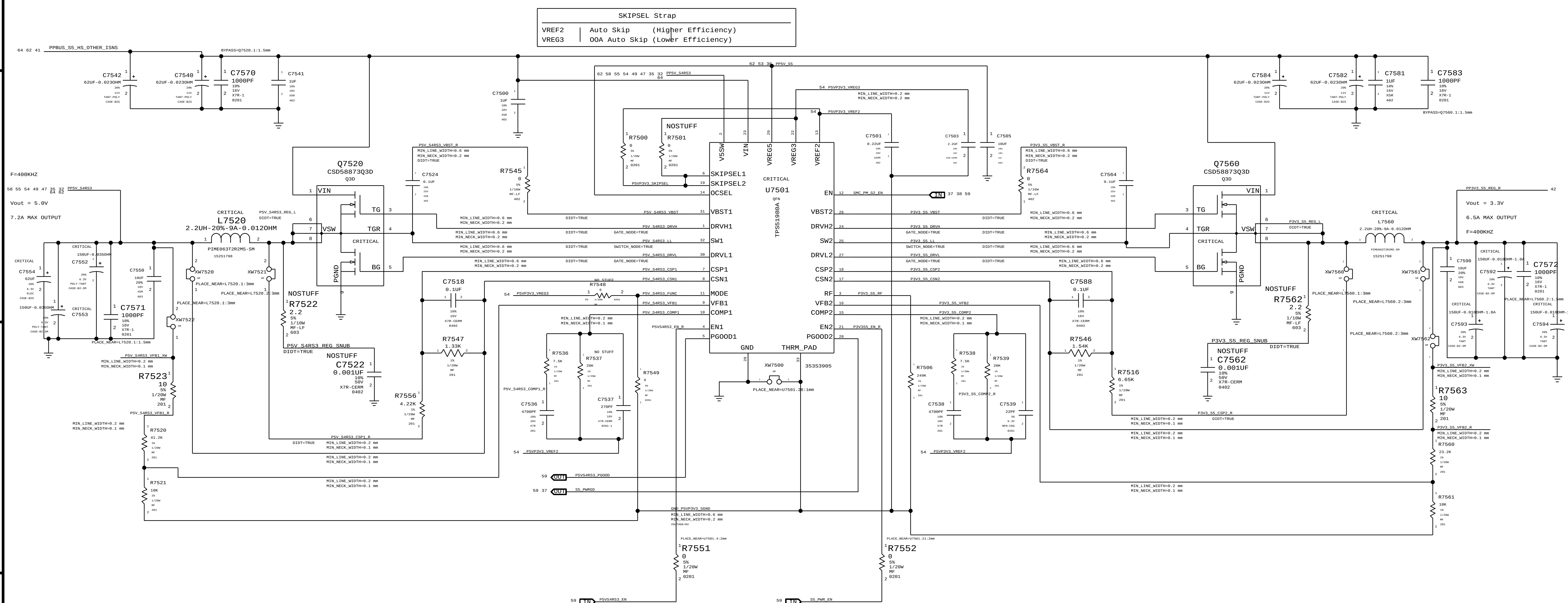
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 74 OF 121

SHEET 53 OF 76



SYNC_MASTER=J41_MLB SYNC_DATE=09/17/2012

PAGE TITLE

5V S4RS3 / 3.3V S5 Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM>

SIZE D

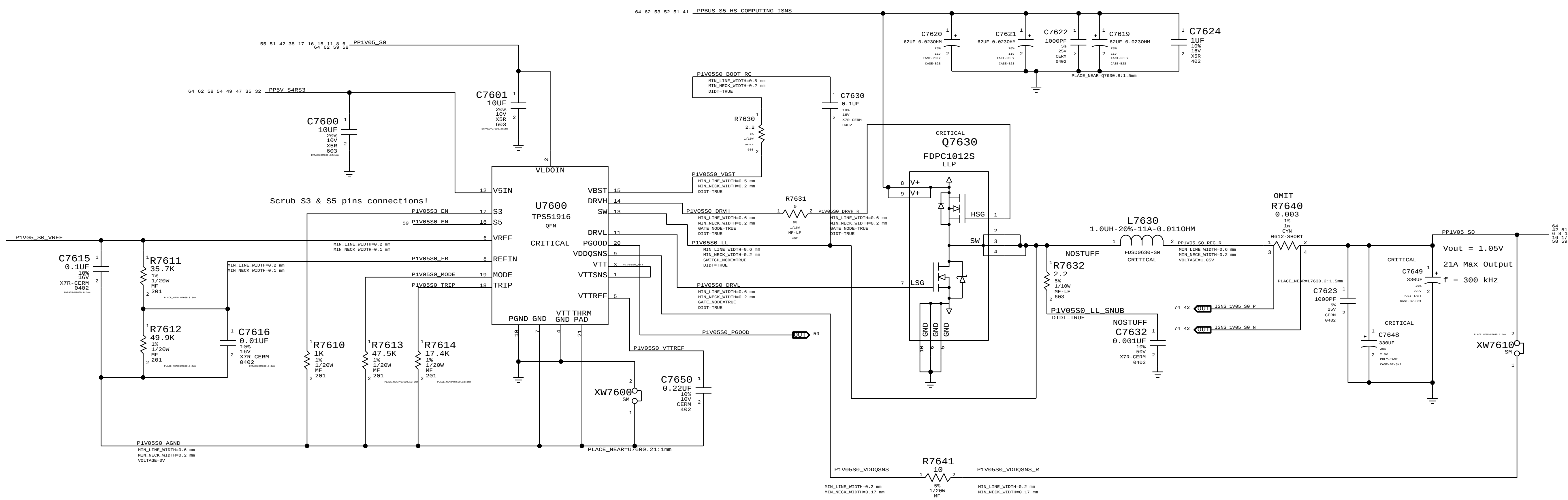
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 75 OF 121

SHEET 54 OF 76

1.05V S0 Regulator



SYNC_MASTER=J41_MLB SYNC_DATE=05/21/2013

PAGE TITLE

1.05V S0 Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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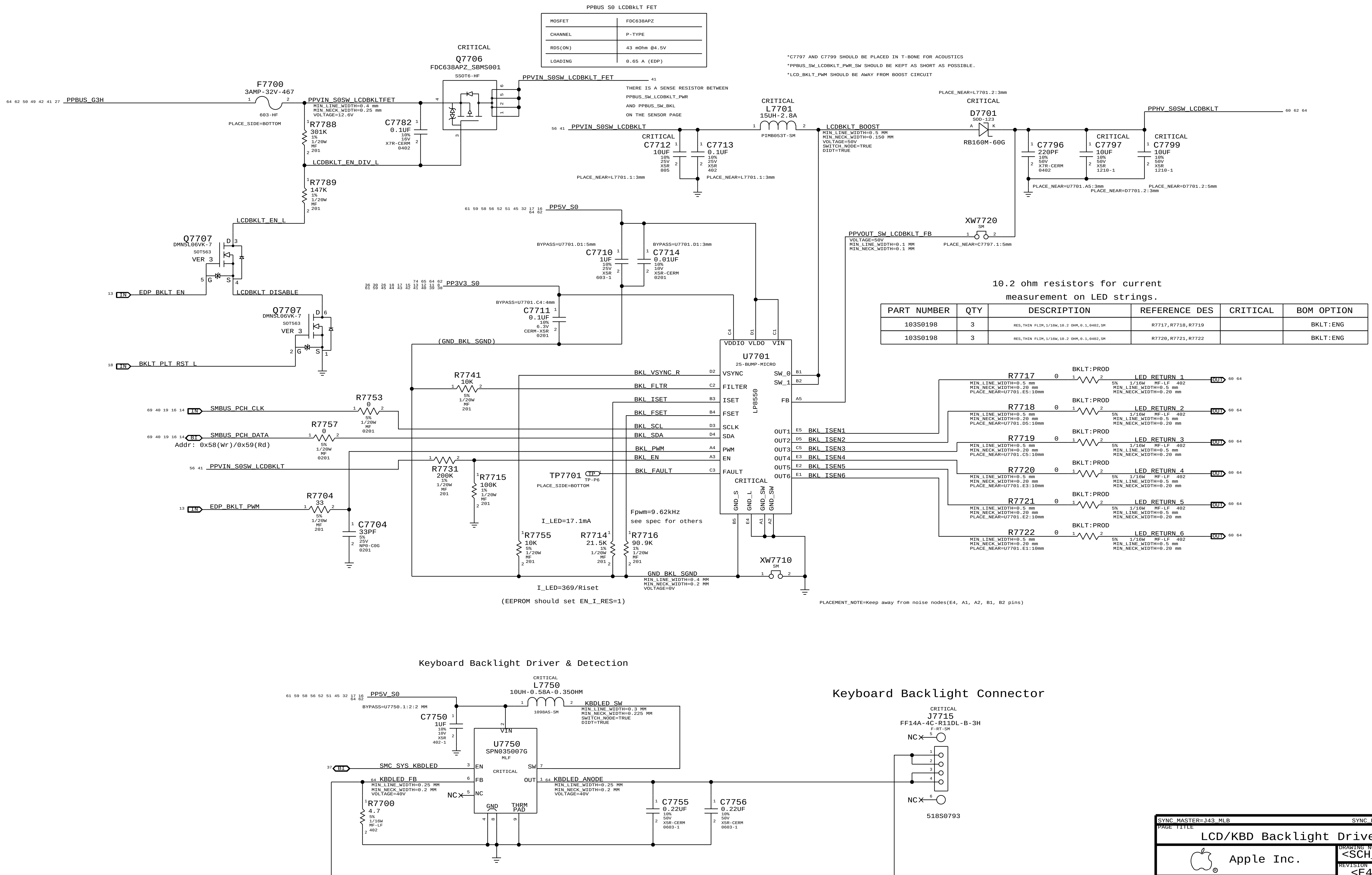
SHEET 55 OF 76



MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.65 A (EDP)

10.2 ohm resistors for current measurement on LED strings.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R7717,R7718,R7719		BKLT:ENG
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R7720,R7721,R7722		BKLT:ENG

SYNC_MASTER=J43_MLB SYNC_DATE=09/13/2012

PAGE TITLE

LCD/KBD Backlight Driver

Apple Inc.

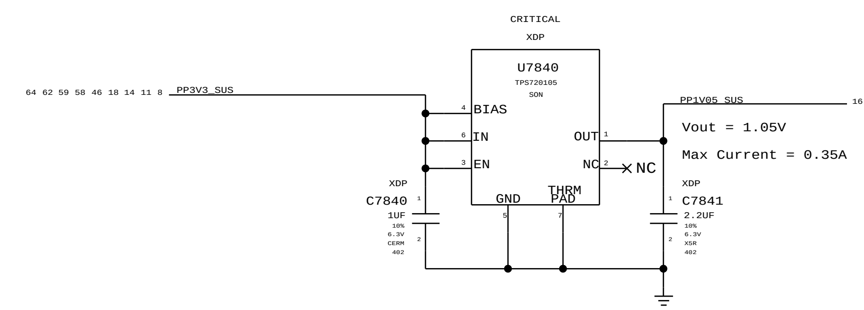
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

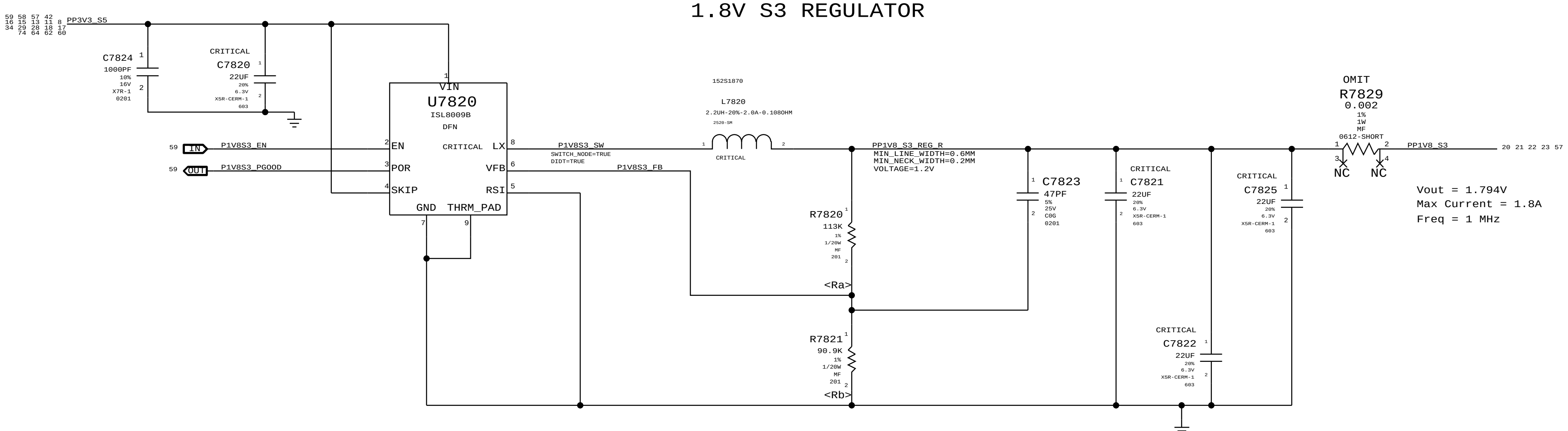
BRANCH <BRANCH>

1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.

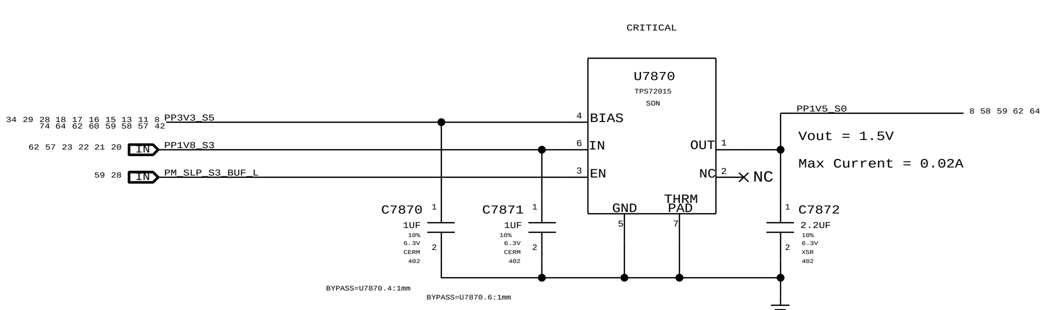


1.8V S3 REGULATOR



Vout = 0.8V * (1 + Ra / Rb)

1.5V S0 LDO



SYNC_MASTER=J43_MLB SYNC_DATE=10/04/2012

PAGE TITLE: Misc Power Supplies

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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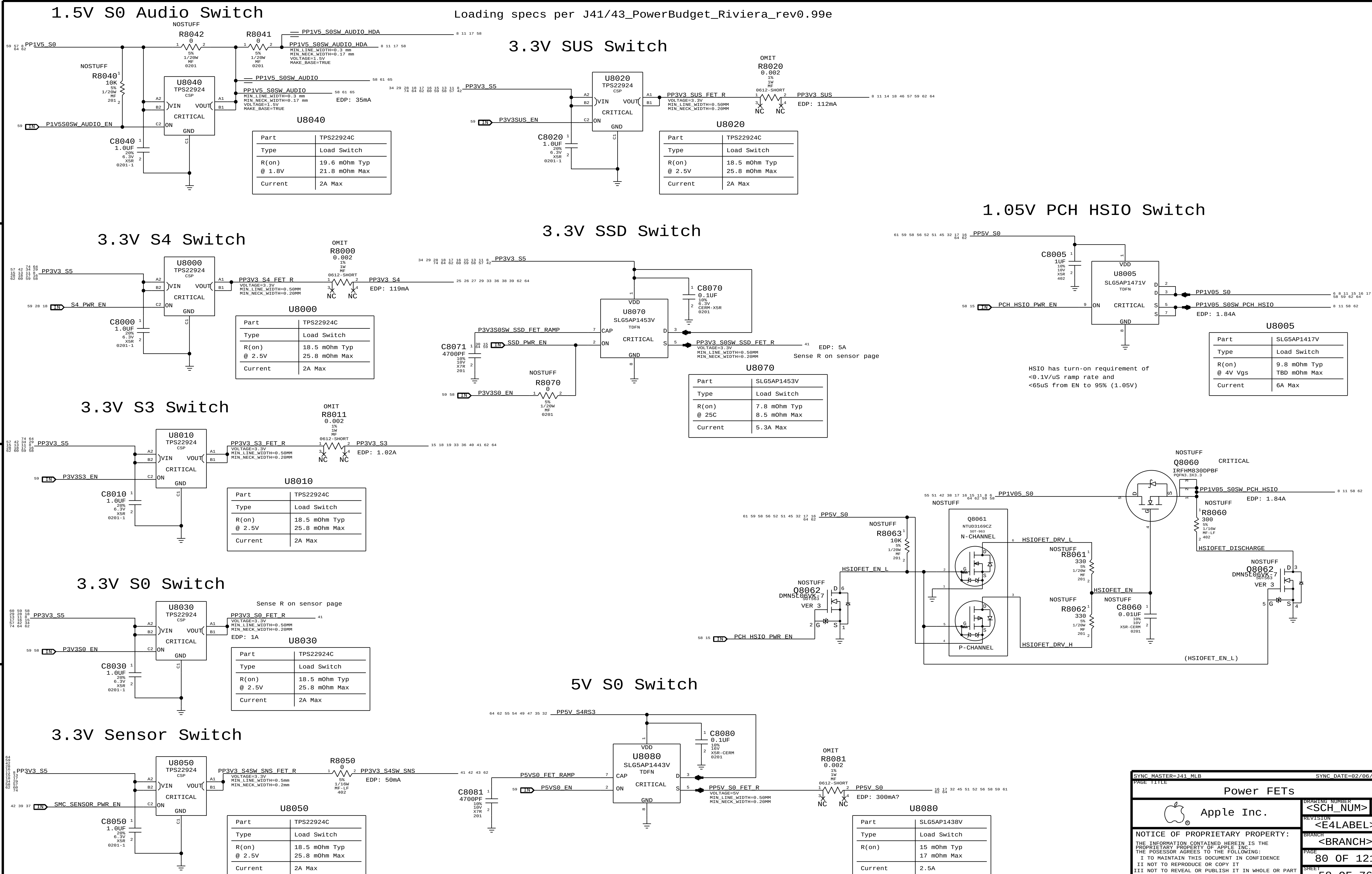

U8040

Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

U8020

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

U8000

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

U8070

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

U8005

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

U8010

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

U8030

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

U8050

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

U8080

Part	SLG5AP1438V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

SYNC_MASTER=J41_MLB SYNC_DATE=02/06/2013

PAGE TITLE: Power FETs

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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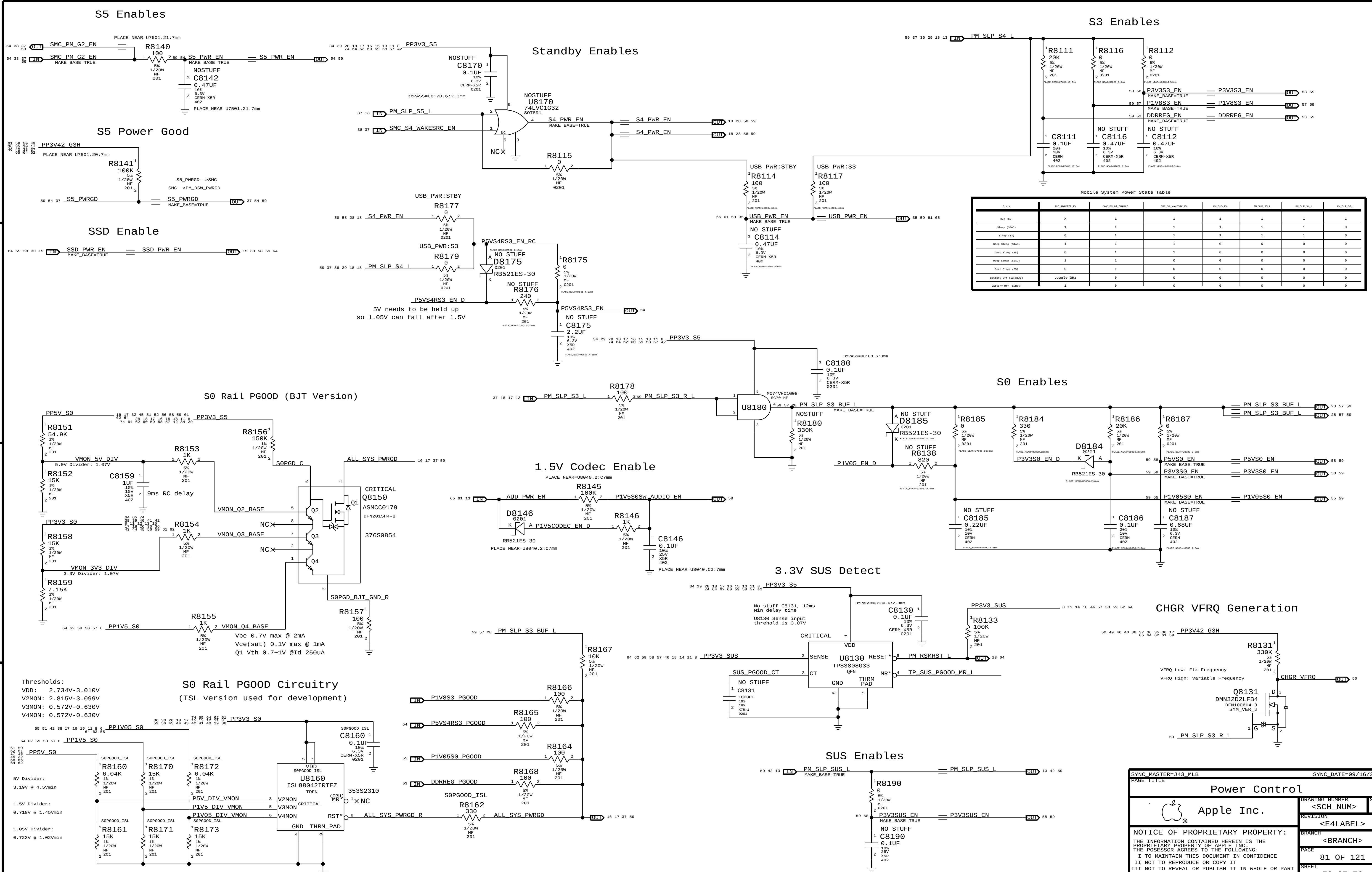

Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S5AC)	1	1	0	0	0	0	0
Deep Sleep (S5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC_MASTER=J43_MLB SYNC_DATE=09/16/2012

PAGE TITLE: Power Control

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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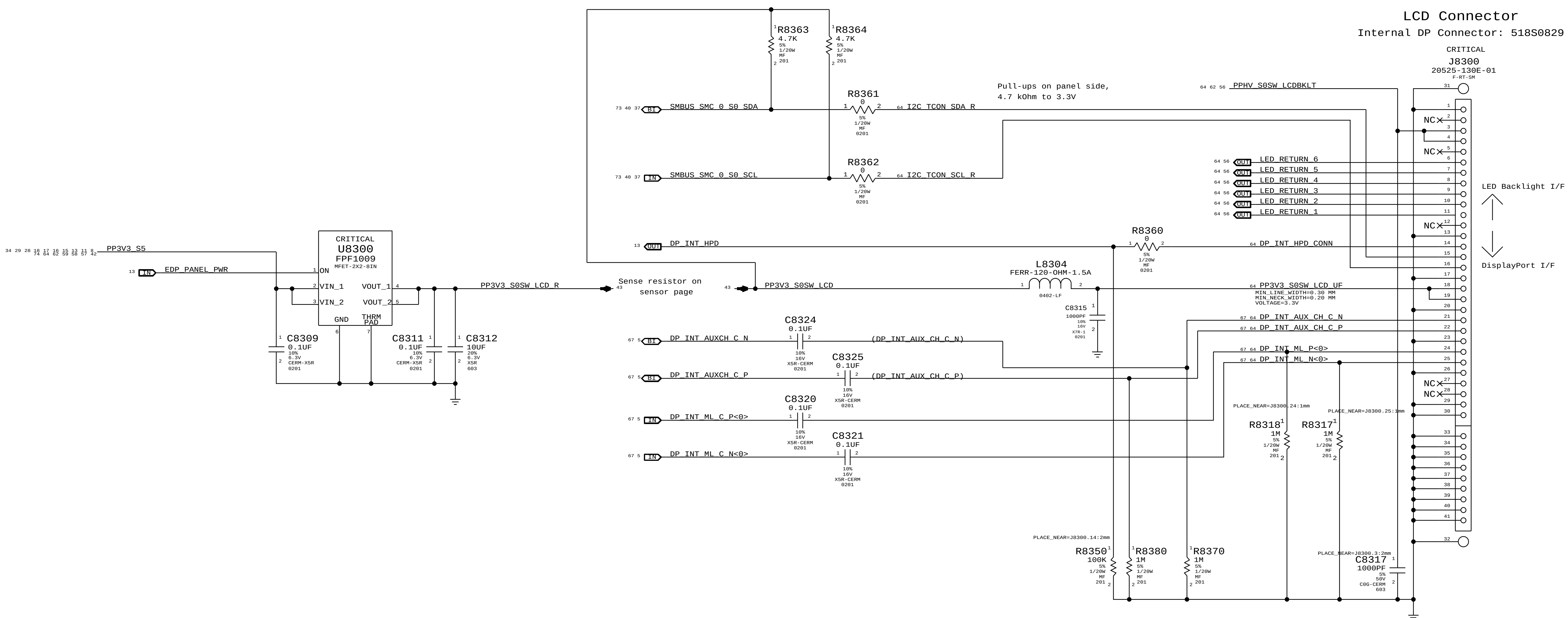
SHEET: 59 OF 76



SYNC_MASTER=J43_MLB SYNC_DATE=09/11/2012

PAGE TITLE

Internal DisplayPort Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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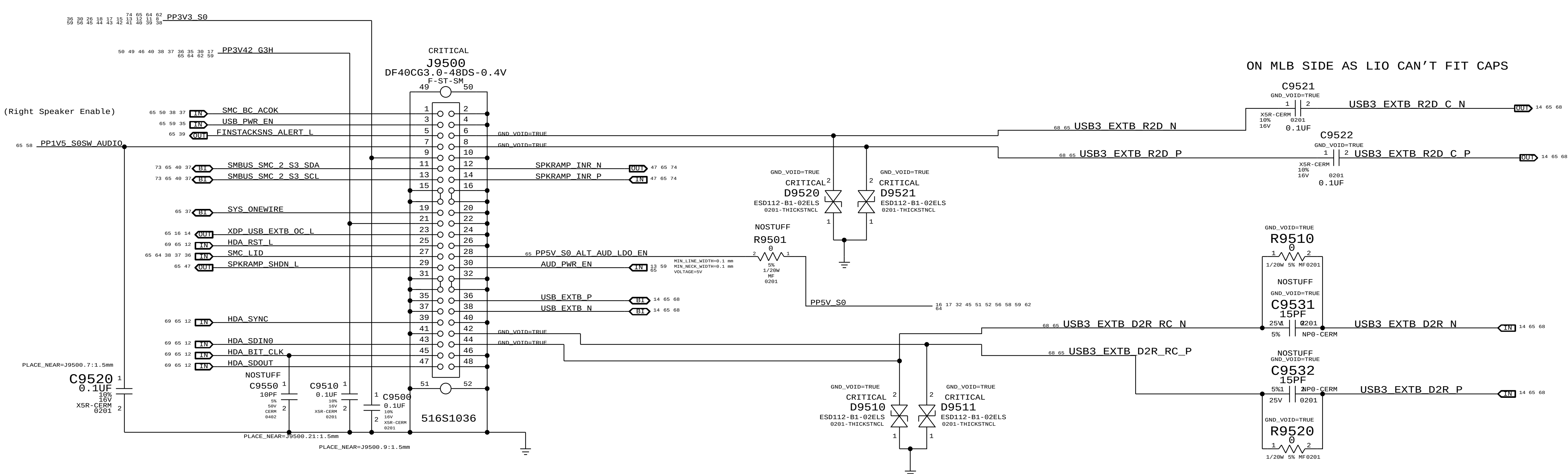
SHEET 60 OF 76



IO Ports

SYNC_MASTER=CLEAN_J43 SYNC_DATE=11/13/2012

PAGE TITLE

Left I/O (LIO) Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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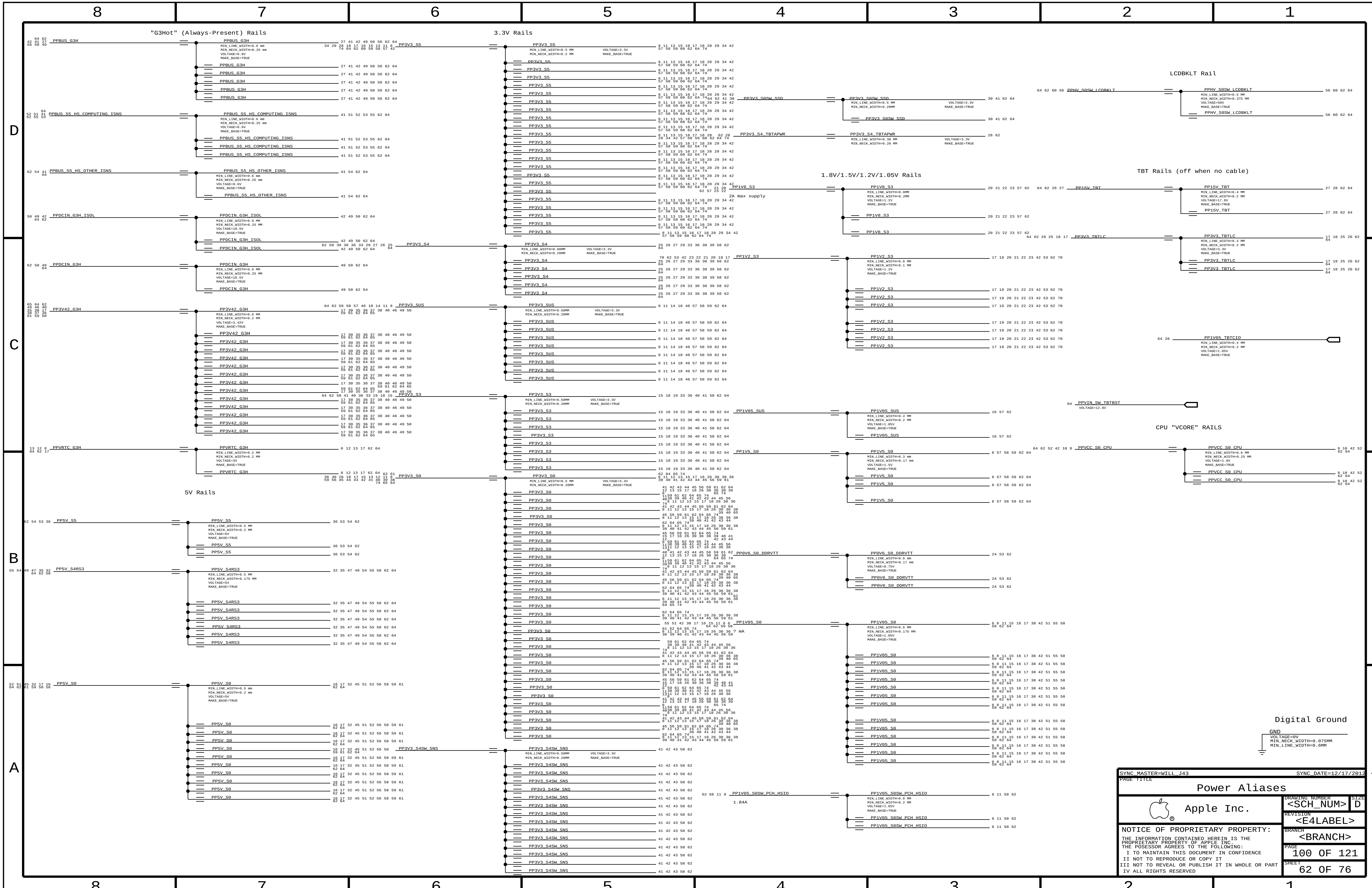

SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE

Power Aliases

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

8 7 6 5 4 3 2 1

D C B A

Memory Bit/Byte Swizzle

LPDDR3 Command/Address

Pages	Net	MAKE_BASE	Alias	Pages
7	=MEM_A_A<5>	TRUE	MEM_A_CAA<0>	20 24 70
7	=MEM_A_A<9>	TRUE	MEM_A_CAA<1>	20 24 70
7	=MEM_A_A<6>	TRUE	MEM_A_CAA<2>	20 24 70
7	=MEM_A_A<8>	TRUE	MEM_A_CAA<3>	20 24 70
7	=MEM_A_A<7>	TRUE	MEM_A_CAA<4>	20 24 70
7	=MEM_A_BA<2>	TRUE	MEM_A_CAA<5>	20 24 70
70 63 24 20 7	MEM_A_CAA<6>	TRUE	MEM_A_CAA<6>	7 20 24 63 70
7	=MEM_A_A<11>	TRUE	MEM_A_CAA<7>	20 24 70
7	=MEM_A_A<15>	TRUE	MEM_A_CAA<8>	20 24 70
7	=MEM_A_A<14>	TRUE	MEM_A_CAA<9>	20 24 70
7	=MEM_A_A<13>	TRUE	MEM_A_CAB<0>	21 24 70
7	=MEM_A_CAS_L	TRUE	MEM_A_CAB<1>	21 24 70
7	=MEM_A_WE_L	TRUE	MEM_A_CAB<2>	21 24 70
7	=MEM_A_RAS_L	TRUE	MEM_A_CAB<3>	21 24 70
7	=MEM_A_BA<0>	TRUE	MEM_A_CAB<4>	21 24 70
7	=MEM_A_A<2>	TRUE	MEM_A_CAB<5>	21 24 70
70 63 24 21 7	MEM_A_CAB<6>	TRUE	MEM_A_CAB<6>	7 21 24 63 70
7	=MEM_A_A<10>	TRUE	MEM_A_CAB<7>	21 24 70
7	=MEM_A_A<1>	TRUE	MEM_A_CAB<8>	21 24 70
7	=MEM_A_A<0>	TRUE	MEM_A_CAB<9>	21 24 70
70 63 24 21 20 7	MEM_A_ODT<0>	TRUE	MEM_A_ODT<0>	7 20 21 24 63 70
63 7	TP_LPDDR3_RSVD1	TRUE	TP_LPDDR3_RSVD1	7 63
63 7	TP_LPDDR3_RSVD2	TRUE	TP_LPDDR3_RSVD2	7 63
7	=MEM_B_A<5>	TRUE	MEM_B_CAA<0>	22 24 70
7	=MEM_B_A<9>	TRUE	MEM_B_CAA<1>	22 24 70
7	=MEM_B_A<6>	TRUE	MEM_B_CAA<2>	22 24 70
7	=MEM_B_A<8>	TRUE	MEM_B_CAA<3>	22 24 70
7	=MEM_B_A<7>	TRUE	MEM_B_CAA<4>	22 24 70
7	=MEM_B_BA<2>	TRUE	MEM_B_CAA<5>	22 24 70
70 63 24 22 7	MEM_B_CAA<6>	TRUE	MEM_B_CAA<6>	7 22 24 63 70
7	=MEM_B_A<11>	TRUE	MEM_B_CAA<7>	22 24 70
7	=MEM_B_A<15>	TRUE	MEM_B_CAA<8>	22 24 70
7	=MEM_B_A<14>	TRUE	MEM_B_CAA<9>	22 24 70
7	=MEM_B_A<13>	TRUE	MEM_B_CAB<0>	23 24 70
7	=MEM_B_CAS_L	TRUE	MEM_B_CAB<1>	23 24 70
7	=MEM_B_WE_L	TRUE	MEM_B_CAB<2>	23 24 70
7	=MEM_B_RAS_L	TRUE	MEM_B_CAB<3>	23 24 70
7	=MEM_B_BA<0>	TRUE	MEM_B_CAB<4>	23 24 70
7	=MEM_B_A<2>	TRUE	MEM_B_CAB<5>	23 24 70
70 63 24 23 7	MEM_B_CAB<6>	TRUE	MEM_B_CAB<6>	7 23 24 63 70
7	=MEM_B_A<10>	TRUE	MEM_B_CAB<7>	23 24 70
7	=MEM_B_A<1>	TRUE	MEM_B_CAB<8>	23 24 70
7	=MEM_B_A<0>	TRUE	MEM_B_CAB<9>	23 24 70
70 63 24 23 22 7	MEM_B_ODT<0>	TRUE	MEM_B_ODT<0>	7 22 23 24 63 70
63 7	TP_LPDDR3_RSVD3	TRUE	TP_LPDDR3_RSVD3	7 63
63 7	TP_LPDDR3_RSVD4	TRUE	TP_LPDDR3_RSVD4	7 63

Pages	Net	MAKE_BASE	Alias	Pages
20	=MEM_A_DQ<0>	TRUE	MEM_A_DQ<9>	7 70
20	=MEM_A_DQ<1>	TRUE	MEM_A_DQ<12>	7 70
20	=MEM_A_DQ<2>	TRUE	MEM_A_DQ<10>	7 70
20	=MEM_A_DQ<3>	TRUE	MEM_A_DQ<11>	7 70
20	=MEM_A_DQ<4>	TRUE	MEM_A_DQ<8>	7 70
20	=MEM_A_DQ<5>	TRUE	MEM_A_DQ<13>	7 70
20	=MEM_A_DQ<6>	TRUE	MEM_A_DQ<14>	7 70
20	=MEM_A_DQ<7>	TRUE	MEM_A_DQ<15>	7 70
20	=MEM_A_DQ<8>	TRUE	MEM_A_DQ<0>	7 70
20	=MEM_A_DQ<9>	TRUE	MEM_A_DQ<1>	7 70
20	=MEM_A_DQ<10>	TRUE	MEM_A_DQ<2>	7 70
20	=MEM_A_DQ<11>	TRUE	MEM_A_DQ<7>	7 70
20	=MEM_A_DQ<12>	TRUE	MEM_A_DQ<4>	7 70
20	=MEM_A_DQ<13>	TRUE	MEM_A_DQ<5>	7 70
20	=MEM_A_DQ<14>	TRUE	MEM_A_DQ<3>	7 70
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70 63 21 7	MEM_A_DQ<33>	TRUE	MEM_A_DQ<33>	7 21 63 70
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21	=MEM_A_DQS_P<4>	TRUE	MEM_A_DQS_P<5>	7 70
21	=MEM_A_DQS_N<4>	TRUE	MEM_A_DQS_N<5>	7 70
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21	=MEM_A_DQS_N<5>	TRUE	MEM_A_DQS_N<4>	7 70
70 63 21 7	MEM_A_DQS_P<6>	TRUE	MEM_A_DQS_P<6>	7 21 63 70
70 63 21 7	MEM_A_DQS_N<6>	TRUE	MEM_A_DQS_N<6>	7 21 63 70
21	=MEM_A_DQS_P<7>	TRUE	MEM_A_DQS_P<7>	7 70
21	=MEM_A_DQS_N<7>	TRUE	MEM_A_DQS_N<7>	7 70

Pages	Net	MAKE_BASE	Alias	Pages
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22	=MEM_B_DQ<8>	TRUE	MEM_B_DQ<0>	7 70
22	=MEM_B_DQ<9>	TRUE	MEM_B_DQ<1>	7 70
22	=MEM_B_DQ<10>	TRUE	MEM_B_DQ<2>	7 70
22	=MEM_B_DQ<11>	TRUE	MEM_B_DQ<7>	7 70
22	=MEM_B_DQ<12>	TRUE	MEM_B_DQ<4>	7 70
22	=MEM_B_DQ<13>	TRUE	MEM_B_DQ<5>	7 70
22	=MEM_B_DQ<14>	TRUE	MEM_B_DQ<6>	7 70
22	=MEM_B_DQ<15>	TRUE	MEM_B_DQ<3>	7 70
22	=MEM_B_DQ<16>	TRUE	MEM_B_DQ<28>	7 70
22	=MEM_B_DQ<17>	TRUE	MEM_B_DQ<29>	7 70
22	=MEM_B_DQ<18>	TRUE	MEM_B_DQ<30>	7 70
22	=MEM_B_DQ<19>	TRUE	MEM_B_DQ<27>	7 70
22	=MEM_B_DQ<20>	TRUE	MEM_B_DQ<24>	7 70
22	=MEM_B_DQ<21>	TRUE	MEM_B_DQ<25>	7 70
22	=MEM_B_DQ<22>	TRUE	MEM_B_DQ<31>	7 70
22	=MEM_B_DQ<23>	TRUE	MEM_B_DQ<26>	7 70
22	=MEM_B_DQ<24>	TRUE	MEM_B_DQ<20>	7 70
22	=MEM_B_DQ<25>	TRUE	MEM_B_DQ<16>	7 70
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22	=MEM_B_DQ<27>	TRUE	MEM_B_DQ<22>	7 70
22	=MEM_B_DQ<28>	TRUE	MEM_B_DQ<21>	7 70
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23	=MEM_B_DQ<35>	TRUE	MEM_B_DQ<43>	7 70
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70 63 23 7	MEM_B_DQ<32>	TRUE	MEM_B_DQ<32>	7 23 63 70
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23	=MEM_B_DQ<63>	TRUE	MEM_B_DQ<54>	7 70
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23	=MEM_B_DQS_N<6>	TRUE	MEM_B_DQS_N<7>	7 70
70 63 23 7	MEM_B_DQS_P<6>	TRUE	MEM_B_DQS_P<6>	7 23 63 70
70 63 23 7	MEM_B_DQS_N<6>	TRUE	MEM_B_DQS_N<6>	7 23 63 70

SYNC_MASTER=J41_MLB SYNC_DATE=08/30/2012

PAGE TITLE

Signal Aliases

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

8 7 6 5 4 3 2 1

Functional Test Points

J3501: AirPort / BT Connector

FUNC_TEST (Need 6 TPs)

Net	Pages
PP3V3_WLAN	29 37 38 39 41
WIFI_EVENT_L	29 37 38
PCIE_AP_R2D_N	29 69
PCIE_AP_R2D_P	29 69
PCIE_CLK100M_AP_N	12 29 69
PCIE_CLK100M_AP_P	12 29 69
PCIE_AP_D2R_P	14 29 69
PCIE_AP_D2R_N	14 29 69
PCIE_WAKE_L	13 29 31
AP_RESET_CONN_L	29
AP_CLKREQ_Q_L	29
USB_BT_CONN_P	29 68
USB_BT_CONN_N	29 68
PP3V3_S4	25 26 27 29 33 36 38 39 58 62 64

(Need to add 8 GND TPs)

J3700: SSD Connector

FUNC_TEST (Need 5 TPs)

Net	Pages
PP3V3_S0SW_SSD_FLT	30
PCIE_SSD_R2D_N<3..0>	30 67
PCIE_SSD_R2D_P<3..0>	30 67
PP3V3_S0	42 43 44 55 56 59 61 8 11 12 13 15 17 18 26 30 35 36 38 39 40 41 62 64 65 74
SSD_RESET_CONN_L	30
SSD_CLKREQ_CONN_L	30
SMC_OOB1_R2D_CONN_L	30
SMC_OOB1_D2R_CONN_L	30
SSD_PCIE_SEL_L	30
SSD_SR_EN_L	15 30
SMC_PWRFAIL_WARN_L	30 37
SSD_PWR_EN	15 30 58 59
PCIE_SSD_D2R_N<3..0>	12 30 67
PCIE_SSD_D2R_P<3..0>	12 30 67
PCIE_CLK100M_SSD_N	12 30 67
PCIE_CLK100M_SSD_P	12 30 67

(Need to add 6 GND TPs)

J4002: Camera Connector

FUNC_TEST

Net	Pages
MIPI_CLK_CONN_N	32 72
MIPI_CLK_CONN_P	32 72
CAM_SENSOR_WAKE_L_CONN	32
MIPI_DATA_CONN_N	32 72
MIPI_DATA_CONN_P	32 72
SMBUS_SMC_1_S0_SDA	14 32 37 40 43 44 69 73
SMBUS_SMC_1_S0_SCL	14 32 37 40 43 44 69 73
I2C_CAM_SCK	31 32
I2C_CAM_SDA	31 32
PP5V_S3RS0_ALSCAM_F (Need TBD TPs)	32

(Need to add TBD GND TPs)

J6100: LPC+SPI Connector

FUNC_TEST

Net	Pages
SPI_ALT_IO2_WP_L	46
SPI_ALT_IO3_HOLD_L	46
LPC_AD<3..0>	14 37 69
SPI_ALT_IO0_MOSI	46
XDP_LPCPLUS_GPIO	15 16
LPCPLUS_RESET_L	69
SMC_TDO	37 38
TP_SMC_TRST_L	
TP_SMC_MD1	
SMC_TX_L	37 38
SPI_ALT_IO1_MISO	46
LPC_FRAME_L	14 37 69
SPIROM_USE_MLB	15 46
PM_CLKRUN_L	13 37
SPI_ALT_CLK	46
SPI_ALT_CS_L	46
LPC_SERIRQ	15 37
LPC_PWRDWN_L	13 37
SMC_TDI	37 38
SMC_TCK	37 38 46
SMC_RESET_L	37 38 46 50
SMC_ROMBOOT	
SMC_RX_L	37 38
SMC_TMS	37 38 46

(Need to add 6 GND TPs)

GND

J6000: Fan Connector

FUNC_TEST

Net	Pages
PP5V_S0	16 17 32 45 51 52 56 58 59 61 62
FAN_RT_TACH	45
FAN_RT_PWM	45

(Need to add 1 GND TP)

J4800: IPD Flex Connector

FUNC_TEST

Net	Pages
SMC_LID	36 37 38 61 65
TPAD_SPI_MISO_R	36
USB_TPAD_P	14 36 68
USB_TPAD_N	14 36 68
TPAD_SPI_CLK_R	36
TPAD_WAKE_L	36
TPAD_SPI_MOSI_R	36
PP3V3_S4_IPD	36
TPAD_SPI_CS_R_L	36
TPAD_SPI_IF_EN_CONN	36
TPAD_SPI_INT_S4_WAKE_L_CONN	36
PP5V_S4_IPD	36
TPAD_USB_IF_EN_CONN	36
SMBUS_SMC_3_SDA	36 37 40 44 73
SMBUS_SMC_3_SCL	36 37 40 44 73
SMC_LSOC_RST_L	36 38
PP3V42_G3H	17 30 35 36 37 38 40 46 49 50 59 61 62 64 65
SMC_ONOFF_L	36 37 38

(Need to add 5 GND TPs)

J7000: DC-In Connector

FUNC_TEST

Net	Pages
PPDCIN_G3H (Need 4 TPs)	49 50 62 64
PP5V_S4RS3 (Need 3 TPs)	32 35 47 49 54 55 58 62

(Need to add 5 GND TPs)

J6404: Speaker Connector

FUNC_TEST

Net	Pages
SPKRAMP_ROUT_P	47 74
SPKRAMP_ROUT_N	47 74

(Need to add 3 GND TPs)

J6950: Battery Connector

FUNC_TEST

Net	Pages
PPVBAT_G3H_CONN (Need 4 TPs)	48 50
SMBUS_SMC_5_G3_SCL	37 40 48 50 73
SMBUS_SMC_5_G3_SDA	37 40 48 50 73
SYS_DETECT_L	48

(Need to add 4 GND TPs near J7050 and 1 for shield)

J8300: Internal DP Connector

FUNC_TEST

Net	Pages
PPHV_S0SW_LCDBKLT (Need 2 TPs)	56 60 62
LED_RETURN_6	56 60
LED_RETURN_5	56 60
LED_RETURN_4	56 60
LED_RETURN_3	56 60
LED_RETURN_2	56 60
LED_RETURN_1	56 60
DP_INT_HPD_CONN	60
I2C_TCON_SDA_R	60
I2C_TCON_SCL_R	60
PP3V3_S0SW_LCD_UF (Need 2 TPs)	60
DP_INT_AUX_CH_C_N	60 67
DP_INT_AUX_CH_C_P	60 67
DP_INT_ML_P<0>	60 67
DP_INT_ML_N<0>	60 67

(Need to add 5 GND TPs)

J7715: KB BKLT Connector

FUNC_TEST

Net	Pages
KBDLED_ANODE	56
KBDLED_FB	56

(Need to add 2 GND TPs)

J1800: XDP Connector

(Only a subset are needed for FCT HVM test fixture)

FUNC_TEST

Net	Pages
XDP_CPU_TCK	6 16 67
XDP_PCH_TCK	12 16 69
XDP_CPU_TDI	6 16 67
XDP_CPU_TDO	6 16 67
XDP_CPUPCH_TRST_L	6 12 16 67
XDP_CPU_TMS	6 16 67
XDP_PCH_TMS	12 16 69
XDP_PCH_TDI	12 16 69
XDP_PCH_TDO	12 16 69
XDP_CPU_PREQ_L	6 16 67
XDP_CPU_PRDY_L	6 16 67
XDP_CPU_VCCST_PWRGD	16
PM_RSMRST_L	13 59
XDP_SYS_PWROK	16
PM_SYSRST_L	13 17 37
CPU_CFG<3>	6 16 67
PP1V05_S0	6 8 11 15 16 17 38 42 51 55 58 59 62 64

(Need to add 2 GND TPs)

Misc Voltages & Control Signals

FUNC_TEST

Net	Pages
PPBUS_G3H	27 41 42 49 50 56 62
PPVIN_SW_TBTBST	62
PPBUS_S5_HS_COMPUTING_ISNS	41 51 52 53 55 62
PPDCIN_G3H	49 50 62 64
PP3V42_G3H	17 30 35 36 37 38 40 46 49 50 59 61 62 64 65
PPVRTC_G3H	8 12 13 17 62
PP3V3_S5	8 11 13 15 16 17 18 28 29 34 42 57 58 59 60 62 74
PP3V3_SUS	8 11 14 18 46 57 58 59 62
PP3V3_S3	15 18 19 33 36 40 41 58 62 62 64 65 74
PP3V3_S0	8 11 12 13 15 17 18 26 30 36 38 39 40 41 42 43 44 45 56 59 61
PP3V3_S0SW_SSD	30 41 62
PP1V5_S0	8 57 58 59 62
PP1V05_S0	6 8 11 15 16 17 38 42 51 55 58 59 62 64
PP15V_TBT	27 28 62
PP3V3_TBTLC	17 18 25 26 62
PP1V05_TBT	26
PPVCC_S0_CPU	8 10 42 52 62
PP1V05_TBTCIO	26 62
PPBUS_S5_HS_OTHER_ISNS	41 54 62
PPDCIN_G3H_ISOL	42 49 50 62
PP3V3_S4	25 26 27 29 33 36 38 39 58 62 64

(Need to add 27 GND TPs)

NO_TEST Nets

NO_TEST MAKE_BASE

CPU/PCH

Pages	Net	Net	Pages
64	NC_PCIE_CLK100M_SDP	NC_PCIE_CLK100M_SDP	64
64	NC_PCIE_CLK100M_SDN	NC_PCIE_CLK100M_SDN	64
64 12	NC_PCIE_CLK100M_FWP	NC_PCIE_CLK100M_FWP	12 64
64 12	NC_PCIE_CLK100M_FWN	NC_PCIE_CLK100M_FWN	12 64
64 14	NC_PCIE_FW_D2RP	NC_PCIE_FW_D2RP	14 64
64 14	NC_PCIE_FW_D2RN	NC_PCIE_FW_D2RN	14 64
64 14	NC_PCIE_FW_R2D_CP	NC_PCIE_FW_R2D_CP	14 64
64 14	NC_PCIE_FW_R2D_CN	NC_PCIE_FW_R2D_CN	14 64
64 14	NC_USB_IRP	NC_USB_IRP	14 64
64 14	NC_USB_IRN	NC_USB_IRN	14 64
64 14	NC_USB_CAMERAP	NC_USB_CAMERAP	14 64
64 14	NC_USB_CAMERAN	NC_USB_CAMERAN	14 64
64 14	NC_USB_SDP	NC_USB_SDP	14 64
64 14	NC_USB_SDN	NC_USB_SDN	14 64
67	DP_INT_ML_C_P<3..1>	NC_INT_ML_CP<3..1>	5
67	DP_INT_ML_C_N<3..1>	NC_INT_ML_CN<3..1>	5
64 12	NC_HDA_SDIN1	NC_HDA_SDIN1	12 64
64 13	NC_PCI_PME_L	NC_PCI_PME_L	13 64
64 14	NC_CLINK_CLK	NC_CLINK_CLK	14 64
64 14	NC_CLINK_DATA	NC_CLINK_DATA	14 64
64 14	NC_CLINK_RESET_L	NC_CLINK_RESET_L	14 64

SMC

Pages	Net	Net	Pages
64 37	NC_SMC_SYS_LED	NC_SMC_SYS_LED	37 64
64	NC_IR_RX_OUT_RC	NC_IR_RX_OUT_RC	64
64	NC_USB_SMCP	NC_USB_SMCP	64
64	NC_USB_SMCN	NC_USB_SMCN	64
64 37	NC_SMC_GFX_OVERTEMP	NC_SMC_GFX_OVERTEMP	37 64
64 37	NC_SMC_GFX_THROTTLE_L	NC_SMC_GFX_THROTTLE_L	37 64
64 37	NC_SMC_FAN_1_CTL	NC_SMC_FAN_1_CTL	37 64
64 37	NC_SMC_FAN_1_TACH	NC_SMC_FAN_1_TACH	37 64
64 37	NC_SMC_FAN_5_CTL	NC_SMC_FAN_5_CTL	37 64
64	NC_ENET_ASF_GPIO	NC_ENET_ASF_GPIO	64
64	NC_SMC_MPM5_LED_PWR	NC_SMC_MPM5_LED_PWR	64
64	NC_SMC_MPM5_LED_CHG	NC_SMC_MPM5_LED_CHG	64
64 37	NC_SMC_T25_EN_L	NC_SMC_T25_EN_L	37 64
64 37	NC_SMC_DP_HPD_L	NC_SMC_DP_HPD_L	37 64
64 37	NC_SMBUS_SMC_4_ASF_SCL	NC_SMBUS_SMC_4_ASF_SCL	37 64
64 37	NC_SMBUS_SMC_4_ASF_SDA	NC_SMBUS_SMC_4_ASF_SDA	37 64
64 37	NC_BDV_BKL_PWM	NC_BDV_BKL_PWM	37 64

TBT

Pages	Net	Net	Pages
71	TBT_B_R2D_C_P<1..0>	NC_TBT_B_R2D_CP<1..0>	25
71	TBT_B_R2D_C_N<1..0>	NC_TBT_B_R2D_CN<1..0>	25
71	TBT_B_D2R_P<1..0>	NC_TBT_B_D2RP<1..0>	25
71	TBT_B_D2R_N<1..0>	NC_TBT_B_D2RN<1..0>	25
64 25	NC_TBT_B_LSTX	NC_TBT_B_LSTX	25 64
71 64	NC_DP_TBTPB_ML_CP<3..1:2>	NC_DP_TBTPB_ML_CP<3..1:2>	64 71
71 64	NC_DP_TBTPB_ML_CN<3..1:2>	NC_DP_TBTPB_ML_CN<3..1:2>	64 71
71 64 25	NC_DP_TBTPB_AUXCH_CP	NC_DP_TBTPB_AUXCH_CP	25 64 71
71 64 25	NC_DP_TBTPB_AUXCH_CN	NC_DP_TBTPB_AUXCH_CN	25 64 71
25	TP_DP_TBTSRC_ML_CP<3>	NC_DP_TBTSRC_ML_CP<3>	
25	TP_DP_TBTSRC_ML_CN<3>	NC_DP_TBTSRC_ML_CN<3>	
25	TP_DP_TBTSRC_ML_CP<2>	NC_DP_TBTSRC_ML_CP<2>	
25	TP_DP_TBTSRC_ML_CN<2>	NC_DP_TBTSRC_ML_CN<2>	
64 25	NC_DP_TBTSRC_ML_CP<1>	NC_DP_TBTSRC_ML_CP<1>	25 64
64 25	NC_DP_TBTSRC_ML_CN<1>	NC_DP_TBTSRC_ML_CN<1>	25 64
25	TP_DP_TBTSRC_ML_CP<0>	NC_DP_TBTSRC_ML_CP<0>	
25	TP_DP_TBTSRC_ML_CN<0>	NC_DP_TBTSRC_ML_CN<0>	
64 25	NC_DP_TBTSRC_AUXCH_CP	NC_DP_TBTSRC_AUXCH_CP	25 64
64 25	NC_DP_TBTSRC_AUXCH_CN	NC_DP_TBTSRC_AUXCH_CN	25 64

Unused nets with offpage

(Nets with offpages not used on this project)

Net	Pages
PCH_BT_UART_D2R	15
PCH_BT_UART_R2D	15
PCH_BT_UART_RTS_L	15
PCH_BT_UART_CTS_L	15
AUD_SPI_CS_L	15
AUD_SPI_CLK	15
AUD_SPI_MISO	15
AUD_SPI_MOSI	15
HDMITBTMUX_LATCH	13
HDD_PWR_EN	15
WOL_EN	14
BT_PWRRST_L	15
HDMITBTMUX_FLAG	13
FW_PWR_EN	
FW_PME_L	
ENET_MEDIA_SENSE	15
LCD_PSR_EN	15
LCD_IRQ_L	15
ODD_PWR_EN_L	13
ENET_LOW_PWR	13
AUD_IP_PERIPHERAL_DET	
AUD_I2C_INT_L	
AP_PCIE_DEV_WAKE	13

SYNC_MASTER=WILL_J43 SYNC_DATE=12/17/2012

PAGE TITLE: Func Test / No Test

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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8	7	6	5	4	3	2	1

Functional Test Points

J9500: LIO Connector

FUNC_TEST

FUNC_TEST	Net	Pages
TRUE	PP3V42_G3H	17 30 35 36 37 38 40 46 49 50 59 61 62 64 62 64 74
TRUE	PP3V3_S0	8 11 12 13 15 17 18 26 30 36 38 39 40 41 42 43 44 45 56 59 61
TRUE	PP1V5_S0SW_AUDIO	58 61
TRUE	SYS_ONEWIRE	37 61
TRUE	SMC_BC_ACOK	37 38 50 61
TRUE	USB_PWR_EN	35 59 61
TRUE	SMBUS_SMC_2_S3_SDA	37 40 61 73
TRUE	SMBUS_SMC_2_S3_SCL	37 40 61 73
TRUE	SPKRAMP_SHDN_L	47 61
TRUE	FINSTACKSNS_ALERT_L	39 61
TRUE	SPKRAMP_INR_N	47 61 74
TRUE	SPKRAMP_INR_P	47 61 74
TRUE	USB_EXTB_N	14 61 68
TRUE	USB_EXTB_P	14 61 68
TRUE	PP5V_S0_ALT_AUD_LDO_EN	61
TRUE	SMC_lID	36 37 38 61 64
TRUE	HDA_SDOUT	12 61 69
TRUE	HDA_BIT_CLK	12 61 69
TRUE	HDA_SDIN0	12 61 69
TRUE	XDP_USB_EXTB_OC_L	14 16 61
TRUE	HDA_RST_L	12 61 69
TRUE	HDA_SYNC	12 61 69
TRUE	USB3_EXTB_D2R_RC_P	61 65 68
TRUE	USB3_EXTB_D2R_RC_N	61 65 68
TRUE	USB3_EXTB_R2D_P	61 65 68
TRUE	USB3_EXTB_R2D_N	61 65 68
TRUE	AUD_PWR_EN	13 59 61

(Need to add 5 GND TPs)

SD Card Aliases

MAKE_BASE

Pages	Net	MAKE_BASE	Net	Pages
68 65 34 14	USB3_SD_D2R_P	TRUE	USB3_SD_D2R_P	14 34 65 68
68 65 34 14	USB3_SD_D2R_N	TRUE	USB3_SD_D2R_N	14 34 65 68
68 65 34 14	USB3_SD_R2D_C_P	TRUE	USB3_SD_R2D_C_P	14 34 65 68
68 65 34 14	USB3_SD_R2D_C_N	TRUE	USB3_SD_R2D_C_N	14 34 65 68
65 39 37 34 15	PP3V3_S0SW_SD		PP3V3_S0SW_SD	15 34 37 39 65

(MAKE_BASE=TRUE on page 45)

Bead Probes

68 61 14	USB3_EXTB_D2R_N	1 TP	BEAD-PROBE	BPA511
68 61 14	USB3_EXTB_D2R_P	1 TP	BEAD-PROBE	BPA510
68 65 61	USB3_EXTB_D2R_RC_N	1 TP	BEAD-PROBE	BPA520
68 65 61	USB3_EXTB_D2R_RC_P	1 TP	BEAD-PROBE	BPA521
68 61 14	USB3_EXTB_R2D_C_N	1 TP	BEAD-PROBE	BPA513
68 61 14	USB3_EXTB_R2D_C_P	1 TP	BEAD-PROBE	BPA512
68 65 61	USB3_EXTB_R2D_N	1 TP	BEAD-PROBE	BPA523
68 65 61	USB3_EXTB_R2D_P	1 TP	BEAD-PROBE	BPA522

SYNC_MASTER=J41_MLB SYNC_DATE=09/13/2012
PAGE TITLE
Project FCT/NC/Aliases
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

J41/J43 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,MEM_TERM	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	TOP,BOTTOM	Y	=50_OHM_SE	=50_OHM_SE			
DEFAULT	ISL2,ISL11	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	ISL3,ISL10	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	ISL4,ISL9	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	*	N	100 MM	100 MM	10 MM	0 MM	0 MM
STANDARD	*	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT

Single-ended Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.310 MM			
27P4_OHM_SE	ISL2,ISL11	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	ISL3,ISL10	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	ISL4,ISL9	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
35_OHM_SE	TOP,BOTTOM	Y	0.195 MM	0.195 MM			
35_OHM_SE	ISL2,ISL11	Y	0.125 MM	0.125 MM			
35_OHM_SE	ISL3,ISL10	Y	0.125 MM	0.125 MM			
35_OHM_SE	ISL4,ISL9	Y	0.125 MM	0.125 MM			
35_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.170 MM			
40_OHM_SE	ISL2,ISL11	Y	0.096 MM	0.096 MM			
40_OHM_SE	ISL3,ISL10	Y	0.096 MM	0.096 MM			
40_OHM_SE	ISL4,ISL9	Y	0.099 MM	0.099 MM			
40_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.135 MM			
45_OHM_SE	ISL2,ISL11	Y	0.075 MM	0.075 MM			
45_OHM_SE	ISL3,ISL10	Y	0.075 MM	0.075 MM			
45_OHM_SE	ISL4,ISL9	Y	0.080 MM	0.080 MM			
45_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.110 MM			
50_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Spacing Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.100 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1x_DIELECTRIC	ISL3,ISL10	0.053 MM	?
1x_DIELECTRIC	ISL4,ISL9	0.050 MM	?
1x_DIELECTRIC	*	0.090 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P075MM	*	0.075 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P075MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	BGA	P070MM_BGA

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P070MM_BGA	*			0.070 MM	5 MM		0.075 MM

Differential Pair Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.165 MM		0.110 MM	0.110 MM
70_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM
70_OHM_DIFF	ISL3,ISL10	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM
70_OHM_DIFF	ISL4,ISL9	Y	0.110 MM	0.110MM		0.095 MM	0.095 MM
70_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
73_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.165 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL2,ISL11	Y	0.106 MM	0.106 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL3,ISL10	Y	0.106 MM	0.106 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL4,ISL9	Y	0.110 MM	0.110 MM		0.150 MM	0.150 MM
73_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	TOP,BOTTOM	Y	0.132 MM	0.132 MM		0.130 MM	0.130 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM
80_OHM_DIFF	ISL3,ISL10	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM
80_OHM_DIFF	ISL4,ISL9	Y	0.088 MM	0.088 MM		0.110 MM	0.110 MM
80_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	TOP,BOTTOM	Y	0.120 MM	0.120 MM		0.150 MM	0.150 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM
85_OHM_DIFF	ISL3,ISL10	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM
85_OHM_DIFF	ISL4,ISL9	Y	0.082 MM	0.082 MM		0.140 MM	0.140 MM
85_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.115 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM
90_OHM_DIFF	ISL3,ISL10	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM
90_OHM_DIFF	ISL4,ISL9	Y	0.076 MM	0.076 MM		0.180 MM	0.180 MM
90_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

SYNC_MASTER=CONSTRAINTS SYNC_DATE=10/24/2012

PAGE TITLE
PCB Rule Definitions

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.100 MM	0.100 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_AGTL	*	=STANDARD	?

Note: CPU_8MIL and CPU_ITP can be converted
back to TABLE_SPACING_RULE
once rdar://10308147 is resolved

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_8MIL	*	*	CPU_8MIL_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_8MIL_2ANY	*	8 MIL	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ITP	*	*	CPU_ITP_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ITP_2ANY	*	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	CPU_COMP	*	CPU_COMP_2SELF
CPU_COMP	*	*	CPU_COMP_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_COMP_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	*	=4x_DIELECTRIC	?
CPU_COMP_2OTHER	*	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	CPU_VCCSENSE	*	CPU_VCCSENSE_2SELF
CPU_VCCSENSE	*	*	CPU_VCCSENSE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_VCCSENSE_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	*	=4x_DIELECTRIC	?
CPU_VCCSENSE_2OTHER	*	=6x_DIELECTRIC	?

PCI-Express Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
CLK_PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

PCIE Clock Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	CLK_PCIE	*	CLK_PCIE_2SELF
CLK_PCIE	*	*	CLK_PCIE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CLK_PCIE_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	*	=4x_DIELECTRIC	?
CLK_PCIE_2OTHER	*	=6x_DIELECTRIC	?

CPU PCIE Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_CPU_TX	PCIE_CPU_TX	*	PCIE_TX2TX
PCIE_CPU_RX	PCIE_CPU_RX	*	PCIE_RX2RX
PCIE_CPU_TX	*_CPU_TX	*	PCIE_TX2OTHERTX
PCIE_CPU_RX	*_CPU_RX	*	PCIE_RX2OTHERRX
PCIE_CPU_TX	*_CPU_RX	*	PCIE_TX2RX
PCIE_CPU_RX	*_CPU_TX	*	PCIE_RX2TX
PCIE_CPU_TX	*_TX	*	PCIE_2OTHERHS
PCIE_CPU_RX	*_TX	*	PCIE_2OTHERHS
PCIE_CPU_TX	*_RX	*	PCIE_2OTHERHS
PCIE_CPU_RX	*_RX	*	PCIE_2OTHERHS
PCIE_CPU_TX	*	*	PCIE_2OTHER
PCIE_CPU_RX	*	*	PCIE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX2OTHERTX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX2OTHERRX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX2RX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_RX2TX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	*	=2.5x_DIELECTRIC	?
PCIE_RX2RX	*	=2.5x_DIELECTRIC	?
PCIE_TX2OTHERTX	*	=4x_DIELECTRIC	?
PCIE_RX2OTHERRX	*	=4x_DIELECTRIC	?
PCIE_TX2RX	*	=6x_DIELECTRIC	?
PCIE_RX2TX	*	=6x_DIELECTRIC	?
PCIE_2OTHERHS	*	=4x_DIELECTRIC	?
PCIE_2OTHER	*	=3x_DIELECTRIC	?

PCH PCIE Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_PCH_TX	PCIE_PCH_TX	*	PCIE_TX2TX
PCIE_PCH_RX	PCIE_PCH_RX	*	PCIE_RX2RX
PCIE_PCH_TX	*_PCH_TX	*	PCIE_TX2OTHERTX
PCIE_PCH_RX	*_PCH_RX	*	PCIE_RX2OTHERRX
PCIE_PCH_TX	*_PCH_RX	*	PCIE_TX2RX
PCIE_PCH_RX	*_PCH_TX	*	PCIE_RX2TX
PCIE_PCH_TX	*_TX	*	PCIE_2OTHERHS
PCIE_PCH_RX	*_TX	*	PCIE_2OTHERHS
PCIE_PCH_TX	*_RX	*	PCIE_2OTHERHS
PCIE_PCH_RX	*_RX	*	PCIE_2OTHERHS
PCIE_PCH_TX	*	*	PCIE_2OTHER
PCIE_PCH_RX	*	*	PCIE_2OTHER

SOURCE: 471984_Chief_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

Note: DisplayPort tables are on Page 113

CPU Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	CPU_PECI	CPU_45S	CPU_COMP	CPU_PECI	6 38
	PM_SYNC	CPU_45S	CPU_AGTL	PM_SYNC	
	PM_MEM_PWRGD	CPU_45S	CPU_AGTL	PM_MEM_PWRGD	
		CPU_45S	CPU_ITP	XDP_DBRESET_L	16 17
		CPU_45S	CPU_ITP	XDP_CPU_PRDY_L	6 16 64
		CPU_45S	CPU_ITP	XDP_CPU_PREQ_L	6 16 64
		CPU_27P4S	CPU_COMP	EDP_COMP	
		CPU_27P4S	CPU_COMP	CPU_PEG_COMP	
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<0>	6
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<1>	6
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2>	6
		CPU_45S	CPU_ITP	CPU_CFG<11..0>	6 16 64
	CPU_CATERR_L	CPU_45S	CPU_AGTL	CPU_CATERR_L	6 37
		CPU_45S	CPU_AGTL	CPU_VCCIO_SEL	
	CPU_PROCHOT_L	CPU_45S	CPU_AGTL	CPU_PROCHOT_L	6 37 38 51
	CPU_PWRGD	CPU_45S	CPU_AGTL	CPU_PWRGD	6
	PM_THRMTRIP_L	CPU_45S	CPU_8MIL	PM_THRMTRIP_L	15 38
	DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_P	
	DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_N	
	DPLL_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DPLL_REF_CLKP	
	DPLL_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DPLL_REF_CLKN	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_P	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_N	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDP_CLK100M_P	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDP_CLK100M_N	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_P	
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_N	
	XDP_TDI	CPU_45S	CPU_ITP	XDP_CPU_TDI	6 16 64
	XDP_TDO	CPU_45S	CPU_ITP	XDP_CPU_TDO	6 16 64
	XDP_TMS	CPU_45S	CPU_ITP	XDP_CPU_TMS	6 16 64
	XDP_TCK	CPU_45S	CPU_ITP	XDP_CPU_TCK	6 16 64
	XDP_TRST_L	CPU_45S	CPU_ITP	XDP_CPUPCH_TRST_L	6 12 16 64
	XDP_BPM_L	CPU_45S	CPU_ITP	XDP_BPM_L<1..0>	6 16
		CPU_45S	CPU_ITP	XDP_BPM_L<7..2>	6 16
		CPU_45S	CPU_ITP	XDP_OBSDATA_B<3..0>	
		CPU_45S	CPU_ITP	CPU_CFG<15..12>	6 16
	(ESB_CPURST_L)	CPU_45S	CPU_ITP	XDP_CPURST_L	16
	CPU_VCCSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_P	8 51
	CPU_VCCSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_N	9 51
	CPU_VCCIOSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_P	
	CPU_VCCIOSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_N	
	CPU_AXG_SENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_P	
	CPU_AXG_SENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_N	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_P	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_N	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	
	CPU_SVIDALERT_L	CPU_45S	CPU_COMP	CPU_VIDALERT_L	8 51
	CPU_SVIDSCLK	CPU_45S	CPU_COMP	CPU_VIDSCLK	8 51
	CPU_SVIDSOUT	CPU_45S	CPU_COMP	CPU_VIDSOUT	8 51
	PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_P<3..0>	12 30
	PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_N<3..0>	12 30
		PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_P<3..0>	30 64
		PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_N<3..0>	30 64
		PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_P<3..0>	
		PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_N<3..0>	
	PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_P<3..0>	12 30 64
	PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_N<3..0>	12 30 64
	PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_P	12 30 64
	PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_N	12 30 64
I212	DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK0_ML_P<3..0>	25
I211	DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK0_ML_N<3..0>	25
I210		DP_80D	DP_TX	DP_TBTSNK0_ML_C_P<3..0>	5 25
I209		DP_80D	DP_TX	DP_TBTSNK0_ML_C_N<3..0>	5 25
I208	DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_P	25
I206	DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_N	25
I207		DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_C_P	13 25
I205		DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_C_N	13 25
I204	DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK1_ML_P<3..0>	25
I203	DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK1_ML_N<3..0>	25
I202		DP_80D	DP_TX	DP_TBTSNK1_ML_C_P<3..0>	5 18 25
I201		DP_80D	DP_TX	DP_TBTSNK1_ML_C_N<3..0>	5 18 25
I200	DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_P	25
I199	DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_N	25
I198		DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_C_P	13 18 25
I197		DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_C_N	13 18 25
	DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_P<3..0>	60 64
	DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_N<3..0>	60 64
		DP_80D	DP_TX	DP_INT_ML_C_P<3..0>	5 60 64
		DP_80D	DP_TX	DP_INT_ML_C_N<3..0>	5 60 64
I215	DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_P	60 64
I214	DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_N	60 64
	DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUXCH_C_P	5 60
	DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUXCH_C_N	5 60
		DP_80D	DP_AUX	DP_INT_AUXCH_P	
		DP_80D	DP_AUX	DP_INT_AUXCH_N	

PCIe SSD

DP

SYNC_MASTER=CONSTRAINTS SYNC_DATE=09/25/2012

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CPU Constraints

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SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ICOMP	*	=4x_DIELECTRIC	?

SOURCE: 471984_Chief_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

UART Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
UART_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
UART	*	=2x_DIELECTRIC	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_PCH_TX	USB3_PCH_TX	*	USB3_TX2TX
USB3_PCH_RX	USB3_PCH_RX	*	USB3_RX2RX
USB3_PCH_TX	*_PCH_TX	*	USB3_TX2OTHERTX
USB3_PCH_RX	*_PCH_RX	*	USB3_RX2OTHERRX
USB3_PCH_TX	*_PCH_RX	*	USB3_TX2RX
USB3_PCH_RX	*_PCH_TX	*	USB3_RX2TX
USB3_PCH_TX	*_TX	*	USB3_2OTHERHS
USB3_PCH_RX	*_TX	*	USB3_2OTHERHS
USB3_PCH_TX	*_RX	*	USB3_2OTHERHS
USB3_PCH_RX	*_RX	*	USB3_2OTHERHS
USB3_PCH_TX	*	*	USB3_2OTHER
USB3_PCH_RX	*	*	USB3_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_TX2OTHERTX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_RX2OTHERRX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_TX2RX	TOP,BOTTOM	=7x_DIELECTRIC	?
USB3_RX2TX	TOP,BOTTOM	=7x_DIELECTRIC	?
USB3_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_TX2TX	*	=2.5x_DIELECTRIC	?
USB3_RX2RX	*	=2.5x_DIELECTRIC	?
USB3_TX2OTHERTX	*	=4x_DIELECTRIC	?
USB3_RX2OTHERRX	*	=4x_DIELECTRIC	?
USB3_TX2RX	*	=6x_DIELECTRIC	?
USB3_RX2TX	*	=6x_DIELECTRIC	?
USB3_2OTHERHS	*	=4x_DIELECTRIC	?
USB3_2OTHER	*	=3x_DIELECTRIC	?

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
PCH_SATA_ICOMP		SATA_ICOMP	PCH_SATAICOMP	
USB_HUB1_UP	USB_80D	USB	USB_HUB_UP_P	
USB_HUB1_UP	USB_80D	USB	USB_HUB_UP_N	
USB_BT	USB_80D	USB	USB_BT_P	14 29
USB_BT	USB_80D	USB	USB_BT_N	14 29
	USB_80D	USB	USB_BT_CONN_P	29 64
	USB_80D	USB	USB_BT_CONN_N	29 64
	USB_80D	USB	USB_BT_WAKE_P	
	USB_80D	USB	USB_BT_WAKE_N	
USB_TPAD	USB_80D	USB	USB_TPAD_P	14 36 64
USB_TPAD	USB_80D	USB	USB_TPAD_N	14 36 64
	USB_80D	USB	USB_TPAD_CONN_P	
	USB_80D	USB	USB_TPAD_CONN_N	
	USB_80D	USB	TPAD_SPI_MOSI_USB_P	
	USB_80D	USB	TPAD_SPI_MISO_USB_N	
USB_TPAD_M	USB_80D	USB	USB_TPAD_M_P	
USB_TPAD_M	USB_80D	USB	USB_TPAD_M_N	
USB_SDCARD	USB_80D	USB	USB_SDCARD_P	
USB_SDCARD	USB_80D	USB	USB_SDCARD_N	

USB Hucopyb nets

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SPI_45S	SPI	TPAD_SPI_MOSI	15 36
	SPI_45S	SPI	TPAD_SPI_MISO	15 36
	SPI_45S	SPI	TPAD_SPI_CLK	15 36

TP SPI nets

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
USB_EXTA	USB_80D	USB	USB_EXTA_P	14 35
USB_EXTA	USB_80D	USB	USB_EXTA_N	14 35
	UART_45S	UART	SMC_DEBUGPRT_TX_L	35 37 38
	UART_45S	UART	SMC_DEBUGPRT_RX_L	35 37 38
USB2_EXTA	USB_80D	USB	USB2_EXTA_MUXED_P	35
USB2_EXTA	USB_80D	USB	USB2_EXTA_MUXED_N	35
USB2_EXTA	USB_80D	USB	USB2_EXTA_MUXED_F_P	35
USB2_EXTA	USB_80D	USB	USB2_EXTA_MUXED_F_N	35
USB3_EXTA_RX	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_P	14 35
USB3_EXTA_RX	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_N	14 35
USB3_EXTA_TX	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_P	35
USB3_EXTA_TX	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_N	35
	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_F_P	
	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_F_N	
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_F_P	
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_F_N	
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_C_P	14 35
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_C_N	14 35

USB EXTA nets (Right USB port)

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
USB_EXTB	USB_80D	USB	USB_EXTB_P	14 61 65
USB_EXTB	USB_80D	USB	USB_EXTB_N	14 61 65
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	USB3_EXTB_D2R_P	14 61 65
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	USB3_EXTB_D2R_N	14 61 65
	USB_80D	USB3_PCH_RX	USB3_EXTB_D2R_RC_P	61 65
	USB_80D	USB3_PCH_RX	USB3_EXTB_D2R_RC_N	61 65
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_P	61 65
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_N	61 65
	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_C_P	14 61 65
	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_C_N	14 61 65

USB EXTB nets (Left USB port)

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
USB3_SD_RX	USB_80D	USB3_PCH_RX	USB3_SD_D2R_P	14 34 65
USB3_SD_RX	USB_80D	USB3_PCH_RX	USB3_SD_D2R_N	14 34 65
USB3_SD_TX	USB_80D	USB3_PCH_TX	USB3_SD_R2D_C_P	14 34 65
USB3_SD_TX	USB_80D	USB3_PCH_TX	USB3_SD_R2D_C_N	14 34 65
	USB_80D	USB3_PCH_RX	USB3_SD_D2R_C_P	34
	USB_80D	USB3_PCH_RX	USB3_SD_D2R_C_N	34
	USB_80D	USB3_PCH_TX	USB3_SD_R2D_P	34
	USB_80D	USB3_PCH_TX	USB3_SD_R2D_N	34
PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS	14
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_PCH_P	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_PCH_N	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK96M_DOT_P	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK96M_DOT_N	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK100M_SATA_P	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK100M_SATA_N	
	CPU_45S	CLK_PCIE	PCH_CLK14P3M_REFCLK	

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REVISION: <E4LABEL>

BRANCH: <BRANCH>

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=3x_DIELECTRIC	?
CLK_LPC	*	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S_R_50S	TOP,BOTTOM	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE		
SMB_45S_R_50S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=4x_DIELECTRIC	?

XDP Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_ITP	*	=2:1_SPACING	?

DisplayPort

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	*	=3x_DIELECTRIC	?
DP_2OTHERHS	*	=4x_DIELECTRIC	?
DP_2OTHER	*	=3x_DIELECTRIC	?
DP_AUX	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
DP_2OTHER	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_AUX	TOP,BOTTOM	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_TX	DP_TX	*	DP_2DP
DP_TX	*_TX	*	DP_2OTHERHS
DP_TX	*_RX	*	DP_2OTHERHS
DP_TX	*	*	DP_2OTHER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 37 64
LPC_FRAME_L	LPC_45S	LPC	LPC_FRAME_L	14 37 64
	LPC_45S	LPC	LPCPLUS_RESET_L	64
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 37
	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS	
	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS_R	
SMBUS_PCH_CLK	SMB_45S_R_50S	SMB	SMBUS_PCH_CLK	14 16 19 40 56
SMBUS_PCH_DATA	SMB_45S_R_50S	SMB	SMBUS_PCH_DATA	14 16 19 40 56
SMBUS_PCH_0_CLK	SMB_45S_R_50S	SMB	SML_PCH_0_CLK	14 40
SMBUS_PCH_0_DATA	SMB_45S_R_50S	SMB	SML_PCH_0_DATA	14 40
SMBUS_SMC_1_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SCL	14 32 37 40 43 44 64 73
SMBUS_SMC_1_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SDA	14 32 37 40 43 44 64 73
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 61 65
	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 61 65
	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST_L	HDA_45S	HDA	HDA_RST_R_L	12
	HDA_45S	HDA	HDA_RST_L	12 61 65
HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	12 61 65
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 61 65
	HDA_45S	HDA	HDA_SDOUT_R	12 17
PM_SUS_CLK	CLK_SLOW_45S	CLK_SLOW	PM_CLK32K_SUSCLK_R	13 38
	CLK_SLOW_45S	CLK_SLOW	SMC_CLK32K	37 38
SPI_CLK	SPI_45S	SPI	SPI_CLK_R	14 46
	SPI_45S	SPI	SPI_CLK	46
SPI_MOSI	SPI_45S	SPI	SPI_MOSI_R	14 46
	SPI_45S	SPI	SPI_MOSI	46
SPI_MISO	SPI_45S	SPI	SPI_MISO	14 46
	SPI_45S	SPI	SPI_MISO_R	46
SPI_CS0	SPI_45S	SPI	SPI_CS0_R_L	14 46
	SPI_45S	SPI	SPI_CS0_L	46
	SPI_45S	SPI	SPI_SMC_CLK	37 46
	SPI_45S	SPI	SPI_SMC_MOSI	37 46
	SPI_45S	SPI	SPI_SMC_MISO	37 46
	SPI_45S	SPI	SPI_SMC_CS_L	37 46
	SPI_45S	SPI	SPI_MLB_CLK	46
	SPI_45S	SPI	SPI_MLB_IO0_MOSI	46
	SPI_45S	SPI	SPI_MLB_IO1_MISO	46
	SPI_45S	SPI	SPI_MLB_CS_L	46
	SPI_45S	SPI	SPI_IO<2>	14 46
	SPI_45S	SPI	SPI_IO2_R	46
	SPI_45S	SPI	SPI_MLB_IO2_WP_L	46
	SPI_45S	SPI	SPI_IO<3>	14 46
	SPI_45S	SPI	SPI_IO3_R	46
	SPI_45S	SPI	SPI_MLB_IO3_HOLD_L	46
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_P	29 64
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_N	29 64
	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_P	14 29
	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_N	14 29
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_P	14 29 64
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_N	14 29 64
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_P	12 29 64
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_N	12 29 64
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_P<3..0>	25
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_N<3..0>	25
	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_P<3..0>	14 25
	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_N<3..0>	14 25
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_P<3..0>	14 25
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_N<3..0>	14 25
	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_P<3..0>	25
	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_N<3..0>	25
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_P	12 25
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_N	12 25
	CLK_PCIE_80D	CLK_PCIE	PEG_CLK100M_P	
	CLK_PCIE_80D	CLK_PCIE	PEG_CLK100M_N	
XDP_TDI	PCH_45S	PCH_ITP	XDP_PCH_TDI	12 16 64
XDP_TDO	PCH_45S	PCH_ITP	XDP_PCH_TDO	12 16 64
XDP_TMS	PCH_45S	PCH_ITP	XDP_PCH_TMS	12 16 64
XDP_TCK	PCH_45S	PCH_ITP	XDP_PCH_TCK	12 16 64
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_P	31 32
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_N	31 32
	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_P	14 32
	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_N	14 32
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_P	14 32
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_N	14 32
	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_P	31 32
	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_N	31 32
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	12 32
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	12 32
	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	31 32
	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	31 32

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTCX1	
SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	17 32
	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP	31 32
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R	32
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP	32
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN	32
	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN	31 32
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	17 25
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	25
SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	17
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	17
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	17
	CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2	34
	CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2_R	34 75
	CLK_25M_45S	CLK_25M	SDSCLK_CLK25M_X1	34

SYNC_MASTER=J41_MLB SYNC_DATE=12/14/2012

PAGE TITLE

PCH Constraints 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DATA2OTHERMEM	*	=8x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=3x_DIELECTRIC	?
MEM_CMD2CMD	*	=3x_DIELECTRIC	?
MEM_CMD2CTRL	*	=3x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=3x_DIELECTRIC	?
MEM_CLK2CLK	*	=6x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	10000
MEM_2GND	*	=2x_DIELECTRIC	10000
MEM_2OTHER	*	=6x_DIELECTRIC	?

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	MEM_TERM	MEM_73D
MEM_40S	MEM_TERM	MEM_50S

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	MEM_*	*	MEM_DATA2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	*	*	MEM_2OTHER
MEM_A_DQS_1	*	*	MEM_2OTHER
MEM_A_DQS_2	*	*	MEM_2OTHER
MEM_A_DQS_3	*	*	MEM_2OTHER
MEM_A_DQS_4	*	*	MEM_2OTHER
MEM_A_DQS_5	*	*	MEM_2OTHER
MEM_A_DQS_6	*	*	MEM_2OTHER
MEM_A_DQS_7	*	*	MEM_2OTHER
MEM_B_DQS_0	*	*	MEM_2OTHER
MEM_B_DQS_1	*	*	MEM_2OTHER
MEM_B_DQS_2	*	*	MEM_2OTHER
MEM_B_DQS_3	*	*	MEM_2OTHER
MEM_B_DQS_4	*	*	MEM_2OTHER
MEM_B_DQS_5	*	*	MEM_2OTHER
MEM_B_DQS_6	*	*	MEM_2OTHER
MEM_B_DQS_7	*	*	MEM_2OTHER
MEM_A_DATA_0	*	*	MEM_2OTHER
MEM_A_DATA_1	*	*	MEM_2OTHER
MEM_A_DATA_2	*	*	MEM_2OTHER
MEM_A_DATA_3	*	*	MEM_2OTHER
MEM_A_DATA_4	*	*	MEM_2OTHER
MEM_A_DATA_5	*	*	MEM_2OTHER
MEM_A_DATA_6	*	*	MEM_2OTHER
MEM_A_DATA_7	*	*	MEM_2OTHER
MEM_B_DATA_0	*	*	MEM_2OTHER
MEM_B_DATA_1	*	*	MEM_2OTHER
MEM_B_DATA_2	*	*	MEM_2OTHER
MEM_B_DATA_3	*	*	MEM_2OTHER
MEM_B_DATA_4	*	*	MEM_2OTHER
MEM_B_DATA_5	*	*	MEM_2OTHER
MEM_B_DATA_6	*	*	MEM_2OTHER
MEM_B_DATA_7	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DATA_0	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_1	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_2	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_3	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_4	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_5	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_6	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_7	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_0	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_1	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_2	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_3	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_4	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_5	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_6	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_7	MEM_*_DATA_*	*	MEM_2OTHERMEM

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK_P<0>	7 20 24
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK_N<0>	7 20 24
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK_P<1>	7 21 24
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK_N<1>	7 21 24
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM_A_CS_L<1..0>	7 20 21 24
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM_A_ODT<0>	7 20 21 24 63
MEM_A_CKE0	MEM_40S	MEM_CMD	MEM_A_CKE<1..0>	7 20 24
MEM_A_CKE1	MEM_40S	MEM_CMD	MEM_A_CKE<3..2>	7 21 24
MEM_A_CMD0	MEM_40S	MEM_CMD	MEM_A_CAA<9..0>	7 20 24 63
MEM_A_CMD1	MEM_40S	MEM_CMD	MEM_A_CAB<9..0>	7 21 24 63
MEM_A_DQ_BYTE0	MEM_40S	MEM_A_DATA_0	MEM_A_DQ<7..0>	7 63
MEM_A_DQ_BYTE1	MEM_40S	MEM_A_DATA_1	MEM_A_DQ<15..8>	7 63
MEM_A_DQ_BYTE2	MEM_40S	MEM_A_DATA_2	MEM_A_DQ<23..16>	7 63
MEM_A_DQ_BYTE3	MEM_40S	MEM_A_DATA_3	MEM_A_DQ<31..24>	7 63
MEM_A_DQ_BYTE4	MEM_40S	MEM_A_DATA_4	MEM_A_DQ<39..32>	7 21 63
MEM_A_DQ_BYTE5	MEM_40S	MEM_A_DATA_5	MEM_A_DQ<47..40>	7 63
MEM_A_DQ_BYTE6	MEM_40S	MEM_A_DATA_6	MEM_A_DQ<55..48>	7 63
MEM_A_DQ_BYTE7	MEM_40S	MEM_A_DATA_7	MEM_A_DQ<63..56>	7 63
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 63
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 63
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 63
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 63
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 63
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 63
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 63
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 63
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 63
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 63
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 63
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 63
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 21 63
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 21 63
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 63
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 63
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK_P<0>	7 22 24
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK_N<0>	7 22 24
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK_P<1>	7 23 24
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK_N<1>	7 23 24
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM_B_CS_L<1..0>	7 22 23 24
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM_B_ODT<0>	7 22 23 24 63
MEM_B_CKE0	MEM_40S	MEM_CMD	MEM_B_CKE<1..0>	7 22 24
MEM_B_CKE1	MEM_40S	MEM_CMD	MEM_B_CKE<3..2>	7 23 24
MEM_B_CMD0	MEM_40S	MEM_CMD	MEM_B_CAA<9..0>	7 22 24 63
MEM_B_CMD1	MEM_40S	MEM_CMD	MEM_B_CAB<9..0>	7 23 24 63
MEM_B_DQ_BYTE0	MEM_40S	MEM_B_DATA_0	MEM_B_DQ<7..0>	7 63
MEM_B_DQ_BYTE1	MEM_40S	MEM_B_DATA_1	MEM_B_DQ<15..8>	7 63
MEM_B_DQ_BYTE2	MEM_40S	MEM_B_DATA_2	MEM_B_DQ<23..16>	7 63
MEM_B_DQ_BYTE3	MEM_40S	MEM_B_DATA_3	MEM_B_DQ<31..24>	7 63
MEM_B_DQ_BYTE4	MEM_40S	MEM_B_DATA_4	MEM_B_DQ<39..32>	7 23 63
MEM_B_DQ_BYTE5	MEM_40S	MEM_B_DATA_5	MEM_B_DQ<47..40>	7 63
MEM_B_DQ_BYTE6	MEM_40S	MEM_B_DATA_6	MEM_B_DQ<55..48>	7 63
MEM_B_DQ_BYTE7	MEM_40S	MEM_B_DATA_7	MEM_B_DQ<63..56>	7 63
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 63
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 63
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 63
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 63
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 63
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 63
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 63
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 63
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 63
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 63
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 63
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 63
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 23 63
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 23 63
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 63
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM_B_DQS_N<7>	7 63
		MEM_PWR	PP1V2_S3	17 19 20 21 22 23 42 53 62
		MEM_PWR	PP0V6_S3_MEM_VREFCA_A	18 19 20 21
		MEM_PWR	PP0V6_S3_MEM_VREFDQ_A	18 19 20 21
		MEM_PWR	PP0V6_S3_MEM_VREFCA_B	18 19 22 23
		MEM_PWR	PP0V6_S3_MEM_VREFDQ_B	18 19 22 23

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Memory Constraints

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DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_TX	TBTDP_TX	*	TBTDP_TX2TX
TBTDP_RX	TBTDP_RX	*	TBTDP_RX2RX
TBTDP_TX	TBTDP_RX	*	TBTDP_TX2RX
TBTDP_RX	TBTDP_TX	*	TBTDP_TX2RX
TBTDP_TX	*_TX	*	TBTDP_2OTHERHS
TBTDP_RX	*_TX	*	TBTDP_2OTHERHS
TBTDP_TX	*_RX	*	TBTDP_2OTHERHS
TBTDP_RX	*_RX	*	TBTDP_2OTHERHS
TBTDP_TX	*	*	TBTDP_2OTHER
TBTDP_RX	*	*	TBTDP_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_TX2TX	TOP,BOTTOM	=6x_DIELECTRIC	?
TBTDP_RX2RX	TOP,BOTTOM	=6x_DIELECTRIC	?
TBTDP_TX2RX	TOP,BOTTOM	=10x_DIELECTRIC	?
TBTDP_2OTHERHS	TOP,BOTTOM	=10x_DIELECTRIC	?
TBTDP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_TX2TX	*	=4x_DIELECTRIC	?
TBTDP_RX2RX	*	=4x_DIELECTRIC	?
TBTDP_TX2RX	*	=6x_DIELECTRIC	?
TBTDP_2OTHERHS	*	=6x_DIELECTRIC	?
TBTDP_2OTHER	*	=4x_DIELECTRIC	?

Thunderbolt/DP Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
TBT_A_R2D	TBTDP_80D	TBTDP_TX	TBT_A_R2D_C_P<1..0>	25 28
TBT_A_R2D	TBTDP_80D	TBTDP_TX	TBT_A_R2D_C_N<1..0>	25 28
	TBTDP_80D	TBTDP_TX	TBT_A_R2D_P<1..0>	28
	TBTDP_80D	TBTDP_TX	TBT_A_R2D_N<1..0>	28
DP_TBTPA_ML1	DP_80D	DP_TX	DP_TBTPA_ML_C_P<1>	25 28
DP_TBTPA_ML1	DP_80D	DP_TX	DP_TBTPA_ML_C_N<1>	25 28
DP_TBTPA_ML3	DP_80D	DP_TX	DP_TBTPA_ML_C_P<3>	25 28
DP_TBTPA_ML3	DP_80D	DP_TX	DP_TBTPA_ML_C_N<3>	25 28
	DP_80D	DP_TX	DP_TBTPA_ML_P<3..1:2>	28
	DP_80D	DP_TX	DP_TBTPA_ML_N<3..1:2>	28
	DP_80D	DP_TX	DP_A_LSX_ML_P<1>	28
	DP_80D	DP_TX	DP_A_LSX_ML_N<1>	28
	TBTDP_80D	TBTDP_RX	TBT_A_D2R_C_P<1..0>	28
	TBTDP_80D	TBTDP_RX	TBT_A_D2R_C_N<1..0>	28
TBT_A_D2R1	TBTDP_80D	TBTDP_RX	TBT_A_D2R_P<1>	25 28
TBT_A_D2R1	TBTDP_80D	TBTDP_RX	TBT_A_D2R_N<1>	25 28
TBT_A_D2R0	TBTDP_80D	TBTDP_RX	TBT_A_D2R_P<0>	25 28
TBT_A_D2R0	TBTDP_80D	TBTDP_RX	TBT_A_D2R_N<0>	25 28
TBT_A_AUXCH	DP_80D	DP_AUX	DP_TBTPA_AUXCH_C_P	25 28
TBT_A_AUXCH	DP_80D	DP_AUX	DP_TBTPA_AUXCH_C_N	25 28
	DP_80D	DP_AUX	DP_TBTPA_AUXCH_P	28
	DP_80D	DP_AUX	DP_TBTPA_AUXCH_N	28
	DP_80D	DP_AUX	DP_A_AUXCH_DDC_P	
	DP_80D	DP_AUX	DP_A_AUXCH_DDC_N	
	TBTDP_80D	TBTDP_RX	TBT_A_D2R1_AUXDDC_P	28
	TBTDP_80D	TBTDP_RX	TBT_A_D2R1_AUXDDC_N	28
TBT_B_R2D	TBTDP_80D	TBTDP_TX	TBT_B_R2D_C_P<1..0>	64
TBT_B_R2D	TBTDP_80D	TBTDP_TX	TBT_B_R2D_C_N<1..0>	64
	TBTDP_80D	TBTDP_TX	TBT_B_R2D_P<1..0>	
	TBTDP_80D	TBTDP_TX	TBT_B_R2D_N<1..0>	
DP_TBTPB_ML	DP_80D	DP_TX	NC_DP_TBTPB_ML_CP<3..1:2>	64
DP_TBTPB_ML	DP_80D	DP_TX	NC_DP_TBTPB_ML_CN<3..1:2>	64
	DP_80D	DP_TX	DP_TBTPB_ML_P<3..1:2>	
	DP_80D	DP_TX	DP_TBTPB_ML_N<3..1:2>	
	DP_80D	DP_TX	DP_B_LSX_ML_P<1>	
	DP_80D	DP_TX	DP_B_LSX_ML_N<1>	
	TBTDP_80D	TBTDP_RX	TBT_B_D2R_C_P<1..0>	
	TBTDP_80D	TBTDP_RX	TBT_B_D2R_C_N<1..0>	
TBT_B_D2R	TBTDP_80D	TBTDP_RX	TBT_B_D2R_P<1..0>	64
TBT_B_D2R	TBTDP_80D	TBTDP_RX	TBT_B_D2R_N<1..0>	64
TBT_B_AUXCH	DP_80D	DP_AUX	NC_DP_TBTPB_AUXCH_CP	25 64
TBT_B_AUXCH	DP_80D	DP_AUX	NC_DP_TBTPB_AUXCH_CN	25 64
	DP_80D	DP_AUX	DP_TBTPB_AUXCH_P	
	DP_80D	DP_AUX	DP_TBTPB_AUXCH_N	
	DP_80D	DP_AUX	DP_B_AUXCH_DDC_P	
	DP_80D	DP_AUX	DP_B_AUXCH_DDC_N	
	TBTDP_80D	TBTDP_RX	TBT_B_D2R1_AUXDDC_P	
	TBTDP_80D	TBTDP_RX	TBT_B_D2R1_AUXDDC_N	

Only used on dual-port hosts.

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	DP_80D	DP_TX	DP_TBTSRC_ML_C_P<3..0>	
	DP_80D	DP_TX	DP_TBTSRC_ML_C_N<3..0>	
	DP_80D	DP_AUX	DP_TBTSRC_AUXCH_C_P	
	DP_80D	DP_AUX	DP_TBTSRC_AUXCH_C_N	
TBT_SPI_CLK	TBT_SPI_45S	TBT_SPI	TBT_SPI_CLK	25
TBT_SPI_MOSI	TBT_SPI_45S	TBT_SPI	TBT_SPI_MOSI	25
TBT_SPI_MISO	TBT_SPI_45S	TBT_SPI	TBT_SPI_MISO	25
TBT_SPI_CS_L	TBT_SPI_45S	TBT_SPI	TBT_SPI_CS_L	25

Only used on hosts supporting Thunderbolt video-in

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Thunderbolt Constraints

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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
I101	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
I102		S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
I103	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
I104	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
I106	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
I108	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
I107	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
I109	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
I110	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
I147	S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	31 32
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	31 32
I127	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32
I128	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32
I129		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 64
I130		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 64
I134	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32
I133	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32
I132		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 64
I131		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 64
I145			S2_MEM_PWR	PP1V35_CAM	31 32
I146			S2_MEM_PWR	PP0V675_CAM_VREF	31 32
I148			S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	32
I149			S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM
2TO1_DIFFPAIR	*	=STANDARD	0.2 MM	0.1 MM	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
SMBUS_SMC_0_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_0_S0_SCL	37 40 60
SMBUS_SMC_0_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_0_S0_SDA	37 40 60
SMBUS_SMC_1_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SCL	14 32 37 40 43 44 64 69
SMBUS_SMC_1_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SDA	14 32 37 40 43 44 64 69
SMBUS_SMC_2_S3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_2_S3_SCL	37 40 61 65
SMBUS_SMC_2_S3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_2_S3_SDA	37 40 61 65
SMBUS_SMC_3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_3_SCL	36 37 40 44 64
SMBUS_SMC_3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_3_SDA	36 37 40 44 64
SMBUS_SMC_5_G3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_5_G3_SCL	37 40 48 50 64
SMBUS_SMC_5_G3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_5_G3_SDA	37 40 48 50 64

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
SENSE_DIFFPAIR			CHGR_CSI_P	50
SENSE_DIFFPAIR	2TO1_DIFFPAIR		CHGR_CSI_N	50
	2TO1_DIFFPAIR		CHGR_CSI_R_P	50
	2TO1_DIFFPAIR		CHGR_CSI_R_N	50
SENSE_DIFFPAIR	2TO1_DIFFPAIR		CHGR_CSO_P	50
SENSE_DIFFPAIR	2TO1_DIFFPAIR		CHGR_CSO_N	50
	2TO1_DIFFPAIR		CHGR_CSO_R_P	43 50
	2TO1_DIFFPAIR		CHGR_CSO_R_N	43 50

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
SENSE_1TO1_P2MM	*	=1TO1_DIFFPAIR	0.200 MM	0.100 MM	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
THERM_1TO1_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
SPKR_DIFFPAIR	*	=1TO1_DIFFPAIR	0.300 MM	0.100 MM	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	CLK_PCIE	*	GND_P2MM
GND	PCIE*	*	GND_P2MM
GND	SATA*	*	GND_P2MM
GND	USB*	*	GND_P2MM
GND	LVDS*	*	GND_P2MM
SB_POWER	CLK_PCIE	*	PWR_P2MM
SB_POWER	SATA*	*	PWR_P2MM
SB_POWER	SATA*	*	PWR_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	10000
PWR_P2MM	*	0.20 MM	10000

J11/J13 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	INLET_THMSNS_D1_P	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	INLET_THMSNS_D1_N	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBTTHMSNS_D2_R_P	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBTTHMSNS_D2_R_N	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBTTHMSNS_D2_P	
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBTTHMSNS_D2_N	
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBT_MLBBOT_THMSNS_P	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBT_MLBBOT_THMSNS_N	44
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	MLBBOT_THMSNS_D3_P	
SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	MLBBOT_THMSNS_D3_N	
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	TBDTHMSNS_D2_P	44
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	TBDTHMSNS_D2_N	44
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUTHMSNS_D2_P	44
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUTHMSNS_D2_N	44
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	CPUVCCIOS0_CS_N	
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	CPUVCCIOS0_CS_P	
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	CPUVR_ISNS1_P	42 52
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	CPUVR_ISNS1_N	42 52
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS2_P	42 52
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS2_N	42 52
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	SENSE_1TO1_45S	SENSE	CPUVR_ISUM_R_P	42
	SENSE_1TO1_45S	SENSE	CPUVR_ISUM_R_N	42
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_CPUDDR_P	42
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_CPUDDR_N	42
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_P3V3S5_N	42
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_P3V3S5_P	42
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_3V3_S0_P	
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_3V3_S0_N	
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_CAMERA_P	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_CAMERA_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_P3V3_S0_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_P3V3_S0_P	41
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_1V05_S0_P	42 55
SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_1V05_S0_N	42 55
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_BMON_GAIN_P	
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_BMON_GAIN_N	
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_N	41 43
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_P	41 43
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER_P	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V2_S3_N	41 53
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V2_S3_P	41 53
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_P	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_SSD_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_SSD_P	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_N	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_P	41
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_PANEL_N	43
SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_PANEL_P	43
I348 SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_GAIN_N	43 44
I349 SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_GAIN_P	43 44
AUD_DIFF	1TO1_DIFFPAIR	AUDIO	SPKRAMP_INR_P	47 61 65
AUD_DIFF	1TO1_DIFFPAIR	AUDIO	SPKRAMP_INR_N	47 61 65
	1TO1_DIFFPAIR	AUDIO	MAX98300_R_P	47
	1TO1_DIFFPAIR	AUDIO	MAX98300_R_N	47
SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT_P	47 64
SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT_N	47 64
		SB_POWER	PP3V3_S5	8 11 13 15 16 17 18 28 29 34 42 57 58 59 60 62 64 62 64 65
		SB_POWER	PP3V3_S0	8 11 12 13 15 17 18 26 30 36 38 39 40 41 42 43 44 45 56 59 61
		GND	GND	

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE		

SD Card Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
I347	SDDATA	SD_45SE		SDCONN_DATA<0..3>	33 34
I346	SDCLK	SD_45SE		SDCONN_CLK	33 34
I348		SD_45SE		SDCONN_WP	33 34
I349		SD_45SE		SDCONN_CMD	33 34
I350		SD_45SE		SDCONN_DETECT_L	33 34
I351		SD_45SE	SPI	SD_SPI_CLK	34
I352		SD_45SE	SPI	SD_SPI_CS_L	34
I353		SD_45SE	SPI	SD_SPI_MOSI	34
I354		SD_45SE	SPI	SD_SPI_MISO	34
I355		CLK_25M_45S		SDSCLK_CLK_25M_X1	
I356		CLK_25M_45S		SDCLK_CLK25M_X2_R	34 69

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<RDAR://COMPONENT/508941> J43 HW EE SCHEMATIC | EVT
<RDAR://COMPONENT/508945> J43 HW EE SCHEMATIC | DVT

Kismet:
afp://kismet.apple.com/Kismet-Projects/J41-J43

Useful Wiki Links:
Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions
Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design

MobileMac HW Radar:
<rdar://component/497591> MobileMac HW | Task
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<rdar://component/497585> MobileMac HW | New Bugs
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<rdar://component/497590> MobileMac HW | Investigation
<rdar://component/497589> MobileMac HW | Architecture

Other Info:
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